VMC Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

Final Step | Part - A | Physics

Introduction to Vector & Forces

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

1. Three non-zero vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ add up to zero. Find which is false?

(A) $(\vec{A}\times\vec{B})\times\vec{C}$ is not zero unless $\vec{B},\vec{C}$ are parallel

(B) $(\vec{A}\times\vec{B})\cdot\vec{C}$ is not zero unless $\vec{B},\vec{C}$ are parallel

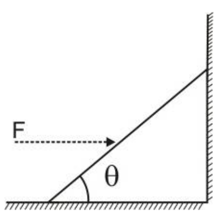
(C) If $\vec{A},\vec{B},\vec{C}$ define a plane, $(\vec{A}\times\vec{B})\times\vec{C}$ is in that plane

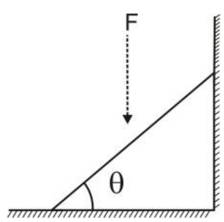
(D) $(\vec{A}\times\vec{B}).\vec{C}=|A||B||C| \Rightarrow C^2=A^2+B^2$

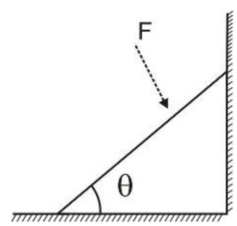
2. When in going east at 10 *kmph*, a train moving with constant velocity appears to be moving exactly 'north-east'. When my velocity is increased to 30 *kmph* east it appears to be moving north. With what speed should i move north so that train appears to be moving exactly south-east?

(A) 30 **(B)** 20 **(C)** 50 **(D)** 10

3. A wedge of mass M rests on a smooth horizontal surface. It is placed against a smooth vertical wall as shown. A force F is applied to the inclined surface (i) horizontally (ii) vertically (iii) $\perp$ to the inclined surface.

(i) 

(ii) 

(iii) 

Let R be normal force between wall and block and N be normal force between ground and the block. Then for the three cases :

(A) (i) $R=F, N=Mg$
(iii) $(F+Mg)\sin\theta=R$
$(F+Mg)\cos\theta=N$
(iii) $R=F\sin\theta,\ F+Mg\cos\theta=N$

(B) (i) $R=F, N=Mg$
(ii) $R=0, N=Mg+F$
(iii) $R=F\sin\theta,$
$N=Mg+F\cos\theta$

(C) $N=Mg$ for each case and $R=F$, 0 and $mg\sin\theta$

(D) None of these

4. It is found that $\left|\vec{A}+\vec{B}\right|=\left|\vec{A}\right|$. This necessarily implies:

(A) $\vec{B}=\vec{0}$ **(B)** $\vec{A}$, $\vec{B}$ are anti-parallel

(C) $\vec{A}$, $\vec{B}$ are perpendicular **(D)** $\vec{A}\cdot\vec{B}\le 0$

5. Let there be two vectors $\vec{a}$ and $\vec{b}$ such that $\vec{a}+\vec{b}$ is in same direction as $\vec{a}-\vec{b}$. Select the correct alternative.

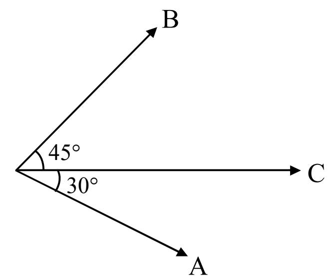
(A) $\vec{a}\times\vec{b}=\vec{0}$ **(B)** $|\vec{a}|>|\vec{b}|$

(C) Both (A) & (B) must be simultaneously true **(D)** $\vec{a}.\vec{b}=0$

6. If $\vec{a}$ denotes a unit vector along an incident light, $\vec{b}$ a unit vector along refracted ray into a medium having refractive index x (relative to first medium) and $\vec{c}$ is a unit vector normal to boundary of two media and directed towards first medium , then law of refraction is

(A) $\vec{a}.\vec{c}=x(\vec{b}.\vec{c})$ **(B)** $\vec{a}\times\vec{c}=x(\vec{b}\times\vec{c})$ **(C)** $\vec{c}\times\vec{a}=x(\vec{b}\times\vec{c})$ **(D)** $x(\vec{a}\times\vec{c})=\vec{b}\times\vec{c}$

7. If $\vec{A}$ and $\vec{B}$ are the components of $\vec{C}$, then :



(A) $A=\dfrac{\sqrt{3}}{2}C$

(B) $B=\dfrac{C}{\sqrt{2}}$

(C) $A=\dfrac{2C}{\sqrt{3}+1}$

(D) $B=\sqrt{2}\left(\sqrt{3}-1\right)C$

***8.** For two vectors $\vec{A}$ and $\vec{B}$, $\left|\vec{A}+\vec{B}\right|=\left|\vec{A}-\vec{B}\right|$ is always true when :

(A) $\left|\vec{A}\right|=\left|\vec{B}\right|\neq 0$ **(B)** $\vec{A}\perp\vec{B}$

(C) $\left|\vec{A}\right|=\left|\vec{B}\right|\neq 0$ and $\vec{A}$ and $\vec{B}$ are parallel or anti parallel **(D)** Either $\left|\vec{A}\right|$ or $\left|\vec{B}\right|$ is zero

***9.** Regarding non-zero vectors, which of the following is a correct statement :

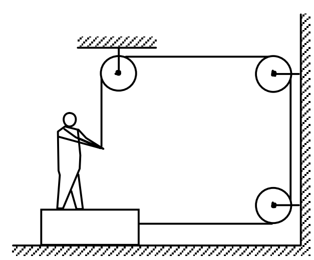
(A) Two equal vectors can never give an addition resultant equal to null – vector.

(B) Three non-coplanar vectors can not give zero vector addition resultant

(C) If $\vec{a}.(\vec{b}\times\vec{c})=0$ and $|\vec{a}|\neq|\vec{b}|\neq|\vec{c}|$ then $\vec{a}+\vec{b}+\vec{c}$ can never be a null vector

(D) If $\vec{a}\times\vec{b}=\vec{0}$ and $|\vec{a}|=|\vec{b}|$, then $\vec{a}+\vec{b}$ can be zero vectors.

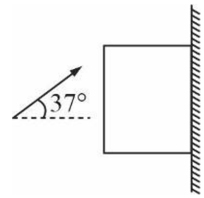
10. The friction coefficient between the board and the floor shown in figure is μ. The maximum force that the man can exert on the rope so that the board does not slip on the floor is : (m is mass of man and M is mass of plank)



(A) $\dfrac{\mu(M+m)g}{(1+\mu)}$ **(B)** $\dfrac{\mu(M-m)g}{(1+\mu)}$

(C) $\mu\dfrac{M}{m}g$ **(D)** None of these

***11.** A block of mass 4 *kg* is acted upon by a 50 *N* force a shown. The friction coefficient between block and wall is μ.



(A) For μ =0.5 block will be at rest

(B) For $\mu=0.2$ block will move down

(C) For $\mu=0.8$ block will move up

(D) Block can never move up for any value of μ

12. Given a parallelogram $ABCD$. If $|\overrightarrow{AB}| = a, |\overrightarrow{AD}| = b \,\&\, |\overrightarrow{AC}| = c$ then $\vec{DB}.\vec{AB}$ has the value

(A) $\dfrac{2a^2+b^2-c^2}{2}$ **(B)** $\dfrac{a^2+3b^2-c^2}{2}$ **(C)** $\dfrac{a^2+b^2+3c^2}{2}$ **(D)** $\dfrac{3a^2+b^2-c^2}{2}$

13. Two forces P and Q are in ratio $P : Q = 1 : 2$. If their addition resultant is at an angle $tan^{-1}\left(\dfrac{\sqrt{3}}{2}\right)$ to vector P, then angle between P and Q is :

(A) $tan^{-1}\left(\dfrac{1}{2}\right)$ **(B)** 45º **(C)** 30º **(D)** 60º

***14.** Consider a set of forces $\vec{F}_1, \vec{F}_2$ and $\vec{F}_3$ acting on a particle of mass 2 kg. Mark the correct options.

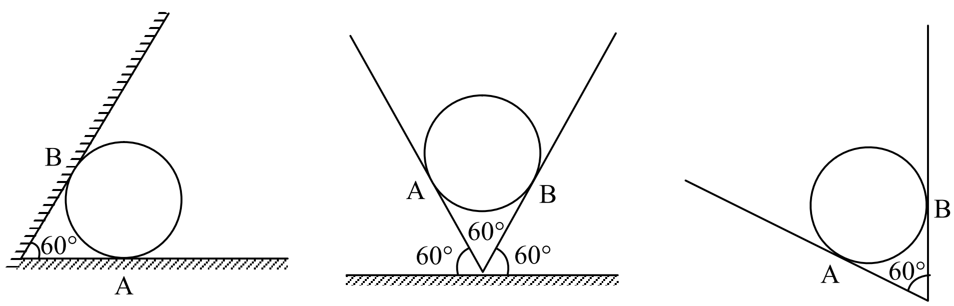
(A) For $|\vec{F}_1| = 2,\ |\vec{F}_2| = 3,\ |\vec{F}_3| = 4$ particle can move with constant velocity

(B) For $|\vec{F}_1| = 1,\ |\vec{F}_2| = 3,\ |\vec{F}_3| = 5$ particle can take an acceleration of $3\,m/s^2$

(C) For $|\vec{F}_1| = 2,\ |\vec{F}_2| = 3,\ |\vec{F}_3| = 4$ particle can take an acceleration of $5\,m/s^2$

(D) For $|\vec{F}_1| = 2,\ |\vec{F}_2| = 3,\ |\vec{F}_3| = 4$ particle can take an acceleration of $2\,m/s^2$

***15.** An iron sphere weighing 10 N rests in a V shaped trough whose sides form an angle $60°$ as shown in figure.



Case I **Case II** **Case III**

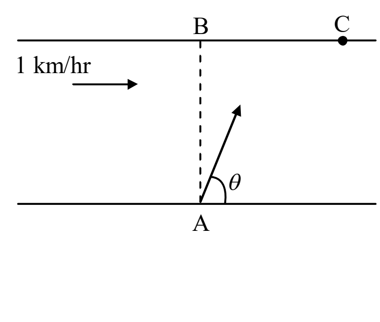
(A) $R_A = 10\,N$ and $R_B = 0$ in case I **(B)** $R_A = 10\,N$ and $R_B = 10\,N$ in case II

(C) $R_A = \dfrac{20}{\sqrt{3}}N$ and $R_B = \left(\dfrac{20}{\sqrt{3}}\right)N$ in case III **(D)** $R_A = 10\,N$ and $R_B = 10\,N$ in all three cases

16. If $\vec{a} = \hat{i} + j + \hat{k},\ \hat{a}.\hat{b} = 1$ and $\vec{a} \times \vec{b} = \hat{j} - \hat{k}$ then $\vec{b}$ is :

(A) $\hat{i} - \hat{j} + \hat{k}$ **(B)** $\hat{i} + \hat{j} - \hat{k}$ **(C)** $\hat{i}$ **(D)** $2\hat{i}$

***17.** A river is flowing with a speed of 1 km/hr. A swimmer wants to go to point 'C' starting from 'A'. He swims with a speed of 5 km/hr, at an angle θ w.r.t. the river. If $AB = BC = 400$ m. Then : $(\sin 53° x 4/5)$



(A) The value of θ is $53°$

(B) Time taken by the man is 6 min

(C) Time taken by the man is 8 min

(D) The value of θ is $45°$

18. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$ which one of the following is correct

(A) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}=\vec{0}$ **(B)** $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}\neq\vec{0}$

(C) $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{a}\times\vec{c}\neq\vec{0}$ **(D)** $\vec{a}\times\vec{b}$, $\vec{b}\times\vec{c}$, $\vec{c}\times\vec{a}$ are mutually perpendicular

***19.** A man is standing on a road and observes that rain is falling at angle $45°$ with the vertical. The man starts running on the road with constant velocity. It appears to him that rain is still falling at angle $45°$ with the vertical, with speed $2\sqrt{2}$ m/s. Motion of the man is in the same vertical plane in which the rain is falling. Then which of the following statement(s) are true :

(A) It is not possible **(B)** Speed of the rain relative to the ground is $2 m/s$

(C) Speed of the man is 4 m/s **(D)** Speed of the rains is $2\sqrt{2}$ m/s

20. If $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$, then $\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$ is equal to :

(A) -1 **(B)** 3 **(C)** 0 **(D)** $-3/2$

21. In a given co-ordinate system, a vector quantity is given as $\vec{A}=3\hat{j}+4\hat{k}$. In another co-ordinate system chosen arbitrarily, $\vec{A}$ cannot be :

(A) $5\hat{i}$ **(B)** $5\hat{j}$ **(C)** $\frac{5}{2}(\hat{i}+\sqrt{3}\hat{j})$ **(D)** $5(\hat{i}+\hat{j})$

***22.** Consider a system of vector $\vec{a}+\vec{b}+\vec{c}+\vec{d}=\vec{0}$ consider two cases

Case 1 : $|\vec{a}|=3$, $|\vec{b}|=5$, $|\vec{c}|=6$ **Case 2 :** $|\vec{a}|=1$, $|\vec{b}|=4$, $|\vec{c}|=7$

Mark the correct alternative(s)

(A) Minimum magnitude of $\vec{d}$ in case 1 is 0 **(B)** Maximum magnitude of $\vec{d}$ in case 1 is 14

(C) Minimum magnitude of $\vec{d}$ in case 2 is 2 **(D)** Maximum magnitude of $\vec{d}$ in case 2 is12

23. Which of the sets given below may represent the magnitudes of three vectors adding to zero ?

(A) 2. 4, 8 **(B)** 4, 8, 16 **(C)** 1, 2, 1 **(D)** 0.5, 1, 2

24. Consider a system of two vector $\vec{a}$ and $\vec{b}$ changing with respect to time $(t \geq 0)$.

$$\vec{a}=8t\hat{i}-t^2\hat{j}; \quad \vec{b}=t^2\hat{i}+2\hat{j}$$

Mark the correct options.

(A) The vectors will become parallel to each other at $t = 4$ s

(B) The vectors will never become perpendicular to each other

(C) The vectors will become perpendicular to each other at $t = 1/4$ s

(D) The vectors will become parallel to each other at $t = 2$ s

25. A vector $\vec{OA}=3\hat{i}$ is rotated by an angle θ about its starting point O in x-z plane in clockwise sense, as seen by an observer located at a point on +y axis. The new vector will be :

(A) $3\cos\theta\,\hat{i}+3\sin\theta\,\hat{j}$ **(B)** $3\cos\theta\,\hat{i}+3\sin\theta\hat{k}$

(C) $3\cos\theta\hat{i}-3\sin\theta\hat{k}$ **(D)** $3\sin\theta\hat{i}-3\cos\theta\hat{k}$

26. Two cars are moving on two mutual perpendicular straight roads. Car a moves along east & towards the crossing with $10\,ms^{-1}$. At any instant it is $1500\,m$ away from the crossing. B at the same instant is $1800\,m$ away from the crossing and is moving towards the crossing with $15\,ms^{-1}$. When do they come closest ?

(A) $109.3\,s$ **(B)** $129.2\,s$ **(C)** $119.3\,s$ **(D)** $99.3\,s$

27. If $\vec{A}\times\vec{B}=\vec{0}$ and $\vec{A}\cdot\vec{B}=-AB$, then angle between $\vec{A}$ and $\vec{B}$ is :

(A) zero **(B)** $\pi/4$ **(C)** $\pi/2$ **(D)** π

28. The angle between vectors $\vec{A}=2\hat{i}+\hat{j}-\hat{k}$ and $\vec{B}=\hat{i}-\hat{k}$ is :

(A) $\frac{\pi}{6}$ **(B)** $\frac{\pi}{3}$ **(C)** $\frac{\pi}{2}$ **(D)** $\frac{\pi}{4}$

29. A unit vector perpendicular to both $\vec{A}=2\hat{i}+3\hat{j}+\hat{k}$ and $\vec{B}=\hat{i}-\hat{k}$ is :

(A) $\frac{1}{\sqrt{42}}\left(4\hat{i}-\hat{j}-5\hat{k}\right)$ **(B)** $\frac{1}{42}\left(4\hat{i}-\hat{j}-5\hat{k}\right)$

(C) $\frac{1}{42}\left(\hat{i}-\hat{j}+5\hat{k}\right)$ **(D)** None of these

30. A particle moving eastwards with $5\,ms^{-1}$. In $10\,s$ the velocity changes to $5\,ms^{-1}$ northwards. The average acceleration in this time is :

(A) $\frac{1}{\sqrt{2}}ms^{-2}$ towards Northeast **(B)** $\frac{1}{2}ms^{-2}$ towards North

(C) $\frac{1}{\sqrt{2}}ms^{-2}$ towards Northwest **(D)** Zero

31. If a vector $2\hat{i}+3\hat{j}+8\hat{k}$ is perpendicular to the vector $4\hat{j}-4\hat{i}+\alpha\hat{k}$ then the value of α is :

(A) $\frac{1}{2}$ **(B)** $\frac{-1}{2}$ **(C)** 1 **(D)** -1

32. A river is flowing from W to E with a speed $5\,m/\text{min}$. A man can swim in still waters at a velocity $10\,m/\text{min}$. In which direction should a man swim to take the shortest path to reach the south bank ?

(A) 30° East of South **(B)** 60° East of North

(C) South **(D)** 30° West of South

33. Find a vector $\vec{x}$ which is perpendicular to both $\vec{A}$ and $\vec{B}$ but has magnitude equal to that of $\vec{B}$.

$\vec{A}=3\hat{i}-2\hat{j}+\hat{k},\ \vec{B}=4\hat{i}+3\hat{j}-2\hat{k}$

(A) $\frac{1}{\sqrt{10}}\left(\hat{i}+10\hat{j}+17\hat{k}\right)$ **(B)** $\frac{1}{\sqrt{10}}\left(\hat{i}-10\hat{j}+17\hat{k}\right)$

(C) $\sqrt{\frac{29}{390}}\left(\hat{i}-10\hat{j}+17\hat{k}\right)$ **(D)** $\sqrt{\frac{29}{390}}\left(\hat{i}+10\hat{j}+17\hat{k}\right)$

34. Rain is falling vertically with $3\,ms^{-1}$ and a man is moving due North with $4\,ms^{-1}$. In which direction he should hold the umbrella to protect himself from rain ?

(A) 37° North of vertical **(B)** 37° South of vertical

(C) 53° North of vertical **(D)** 53° South of vertical

35. A man starts from O moves $500\,m$ turns by 60° and moves $500\,m$ again turns by 60° and moves $500\,m$ and so on. Find the displacement after (i) 5th turns, (ii) 3rd turns :

(A) $500\,m,\ 1000\,m$ **(B)** $500\,m,\ 50\sqrt{3}\,m$ **(C)** $1000\,m,\ 50\sqrt{3}\,m$ **(D)** None of these

36. The acceleration of a particle as seen from two frames S_1 and S_2 has equal magnitude $5\,ms^{-2}$.

(A) The frames must be at rest with respect to each other
(B) The frames may be moving with respect to each other but neither should be moving with respect to the particles.
(C) The acceleration of S_2 with respect to S_1 be 0 or $10\,ms^{-2}$
(D) The acceleration of S_2 with respect to S_1 lies between 0 and $10\,ms^{-2}$

37. A man running on a horizontal road at $8\,ms^{-1}$, finds rain falling vertically. If he increases his speed to $12\,ms^{-1}$, he finds that drops make 30° angle with respect to the vertical. Find the velocity of rain with respect to the road.

(A) $4\sqrt{7}\,ms^{-1}$ **(B)** $8\sqrt{2}\,ms^{-1}$ **(C)** $7\sqrt{3}\,ms^{-1}$ **(D)** $8\,ms^{-1}$

38. Which of the following cannot be in equilibrium ?
(A) $10\,N, 10\,N, 5\,N$ **(B)** $5\,N, 7\,N, 9\,N$ **(C)** $8\,N, 4\,N, 13\,N$ **(D)** $9\,N, 6\,N, 5\,N$

39. A steamer is moving due east with $36\,km/h$. To a man in the steamer the wind appears to blow at $18\,km/h$ due north. Find the velocity of the wind.

(A) $5\sqrt{5}\,ms^{-1}\tan^{-1}\frac{1}{2}$ North of east **(B)** $5\,ms^{-1}\tan^{-1}2$ North of east

(C) $5\sqrt{5}\,ms^{-1}\tan^{-1}2$ North of east **(D)** $5\,ms^{-1}\tan^{-1}\frac{1}{2}$ North of east

40. A force $6\hat{i}+3\hat{j}+\hat{k}$ Newton displaces a particles from $A(0, 3, 2)$ to $B(5, 1, 6)$. Find the work done.
(A) $10\,J$ **(B)** $22\,J$ **(C)** $28\,J$ **(D)** $41\,J$

41. Wind is blowing due *NE* with $18\sqrt{2}\,km\ h^{-1}$ and steamer is heading due east with $18\,km\ h^{-1}$. In which direction is the flag on the mast fluttering ?
(A) North West **(B)** North **(C)** South West **(D)** South

42. A man goes $100\,m$ North then $100\,m$ East and then $20\,m$ North and then $100\sqrt{2}\,m$ South West. Find the displacement.
(A) 20 *m* West **(B)** 20 *m* East **(C)** 20 *m* North **(D)** 20 *m* South

43. A river flows $3\,km\ h^{-1}$ and a man is capable of swimming $2\,km\ h^{-1}$. He wishes to cross it with minimum drift. At what angle with river will he swim?

(A) $\sin^{-1}\left(\frac{2}{3}\right)$ **(B)** $\cos^{-1}\left(\frac{2}{3}\right)$ **(C)** $\tan^{-1}\left(\frac{2}{3}\right)$ **(D)** $\cot^{-1}\left(\frac{2}{3}\right)$

44. A pilot is to fly an aircraft with velocity *v* in still air. Wind is blowing due south with velocity *u*. Find the time for a round journey *A* to *B* and back along east-west direction. (*A* and *B* are *l* distance away).

(A) $\frac{l}{\sqrt{v^2-u^2}}$ **(B)** $\frac{2l}{\sqrt{v^2-u^2}}$ **(C)** $\frac{2l}{v}$ **(D)** $\frac{2l}{\sqrt{v^2+u^2}}$

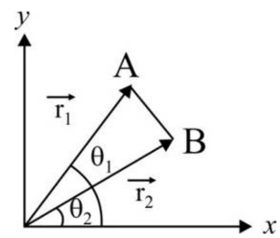
45. If $\left|\vec{A}+\vec{B}\right| = \left|\vec{A}-\vec{B}\right|$ then angle between the non-zero vectors $\vec{A}$ and $\vec{B}$ is :

(A) 0 **(B)** $\frac{\pi}{3}$ **(C)** $\frac{\pi}{2}$ **(D)** $\frac{\pi}{4}$

INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

46. A particle moves in *xy* plane from point A, position vector $\vec{r}_1$, to point *B*, position vector $\vec{r}_2$, if the magnitudes of these vectors are, respectively, $r_1 = 3$ *and* $r_2 = 4$ and the angles they make with the x-axis are $\theta_1 = 75°$ *and* $\theta_2 = 15°$, respectively then the square of the magnitude of the displacement vector *i.e.*, $|\overrightarrow{AB}|^2$ is ……



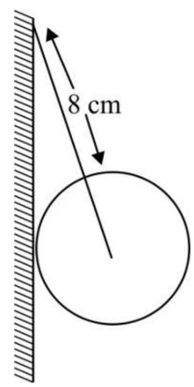
47. Given $\vec{A} = \hat{i} + 2\hat{j} - 3\hat{k}$. When a vector $\vec{B}$ is added to $\vec{A}$, we get a unit vector along X-axis. Then, $\vec{B}$ is $x\hat{j} + y\hat{k}$. The sum $x + y$ is …….

48. Given that *A* = *B* = *C*. If $\vec{A} + \vec{B} = \vec{C}$, then the angle between $\vec{A}$ and $\vec{C}$ is θ_1. If $\vec{A} + \vec{B} + \vec{C} = \vec{0}$, then the angle between $\vec{A}$ *and* $\vec{C}$ is θ_2. The ratio $\dfrac{\theta_2}{\theta_1}$ is ……..

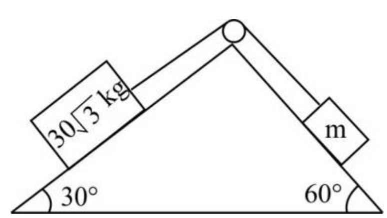
49. The magnitude of the *X* and *Y* components of $\vec{P}$ are 7 and 6 respectively. Also, the magnitudes of the *X* and *Y* components of $\vec{P} + \vec{Q}$ are 11 and 9 respectively. The magnitude of $\vec{Q}$ is ………

50. The addition resultant of two vectors $\vec{A}$ *and* $\vec{B}$ is perpendicular to $\vec{A}$. Magnitude of addition resultant $\vec{R}$ is equal to half of magnitude of $\vec{B}$. The angle between $\vec{A}$ and $\vec{B}$ in degree is ……….

51. Uniform sphere of weight 240*N* and radius 5 cm is being held by a string from centre of sphere as shown in the figure. The tension in the string is …………*N*

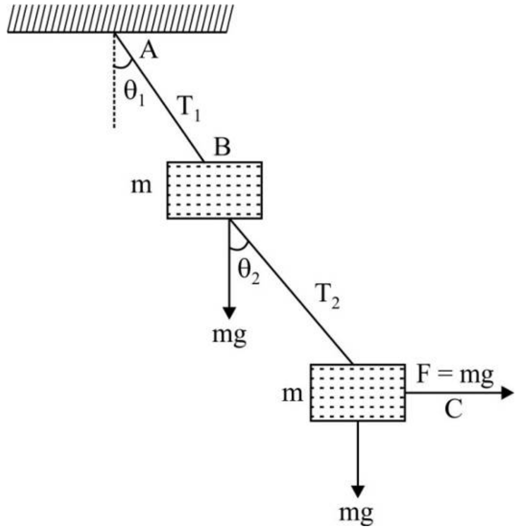


52. Two blocks are kept on smooth wedge which is fixed on a horizontal ground. If the system is in equilibrium, then *m* is …… kg.



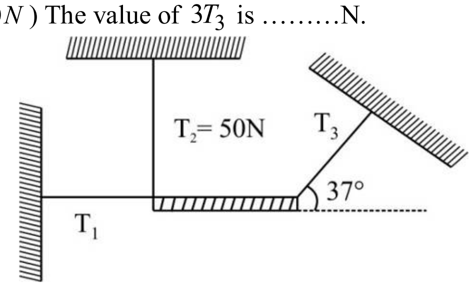
53. A person moves 30m north, then 20m east, then $30\sqrt{2}\,m$ south west. His displacement from the original position is …..... in meters towards west.

54. The blocks B and C in the figure have mass $m = \sqrt{5}$kg each. The strings AB and BC are light, having tensions T_1 and T_2 respectively. The system is in equilibrium with a constant horizontal force mg acting on C. Vertical mg shown is weight of block, then $T_1 =$ ....N. Take $g = 10 m/s^2$.

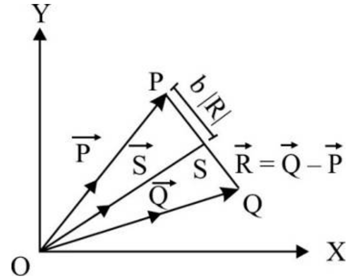


55. Find the magnitude and the direction of the addition resultant of the two vectors $\vec{Q}$ *and* $\vec{P}$ of magnitudes 25 and 40 respectively making an angle of 120° between them. (cos 120°= – 1/2). The addition resultant of $\vec{Q}$ *and* $\vec{P}$ makes an angle of $\sin^{-1}\left(\frac{5\sqrt{n}}{14}\right)$ with $\vec{P}$. Find value of *n*.

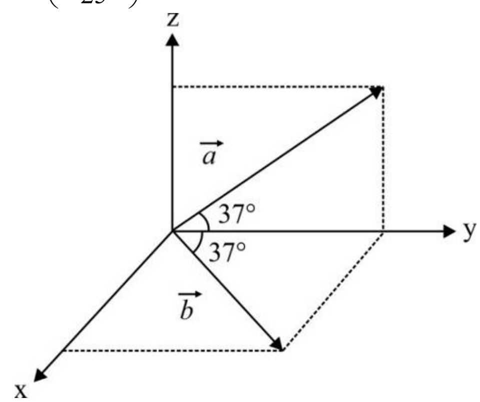
56. In the given situation a uniform rod of mass 10 kg is in equilibrium in horizontal position. (Take $g = 10m/s^2, T_2 = 50N$) The value of $3T_3$ is ....N.



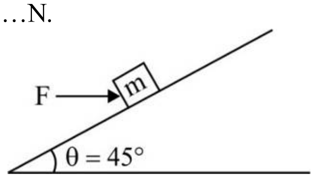
57. Three vectors $\vec{P}, \vec{Q}$ *and* $\vec{R}$ are shown in the figure. Let S be any point on the vector R. The distance between the point P and S is $b|R|$. The general relation among vectors $\vec{P}, \vec{Q}$ *and* $\vec{S}$ is $\vec{S} = (n-b)\vec{p} + b\vec{Q}$. Find value of *n*.



58. Figure shows two vectors $\vec{a}$ (in y-z plane) and $\vec{b}$ (in x- y plane) such that $|\vec{a}|=|\vec{b}|=5$ units. The angle between $\vec{a}$ and $\vec{b}$ is $\theta=\cos^{-1}\left(\frac{10+n}{25}\right)$. Find value of n.



59. A block of weight 210N is placed on a fixed inclined plane. The inclined surface of angle of inclination $\theta=45°$ has coefficient of friction $\mu=0.4$ such that $\mu<\tan\theta$. The minimum horizontal force F needed to keep the block stationary is ....N.



60. Three cars A, B and C are moving at constant velocity. The velocity of car A as seen from car B is $5\hat{i}-2\hat{j}\ m/s$. The velocity of car B as seen from car C is $3\hat{i}+4\hat{j}\ m/s$. The velocity of car A as seen from car $C(in\ m/s)$ is $n\hat{i}+2\hat{j}$. Find value of n.

Kinematics of a Particle

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

***1.** A spring with one end attached to a mass and the other to a rigid support is stretched and released.

(A) Magnitude of acceleration, when just released is maximum
(B) Magnitude of acceleration, when at equilibrium position, is maximum
(C) Speed is maximum when mass is at equilibrium position
(D) Magnitude of displacement is always maximum whenever speed is minimum.

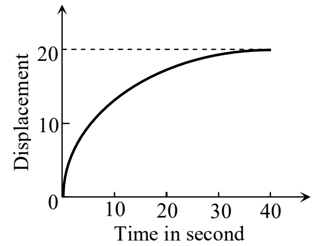
2. A particle moving in a straight line covers half the distance with speed of 3 m/s. The other half of the distance covered in two equal time intervals with speed of 4.5 m/s and 7.5 m/s respectively. The average speed of the particle during this motion is :

(A) 4.0 m/s **(B)** 5.0 m/s **(C)** 5.5 m/s **(D)** 4.8 m/s

***3.** A ball is bouncing elastically with a speed 1 m/s between walls of a railway compartment of size 10 m in a direction perpendicular to walls. The train is moving at a constant velocity of $10\,m/s$ parallel to the direction of motion of the ball. As seen from the ground,

(A) The direction of motion of the ball changes every 10 seconds.
(B) Speed of ball changes every 10 seconds
(C) Average speed of ball over any 20 second interval is fixed
(D) The acceleration of ball is the same as from the train

4. The displacement of a particle as a function of time is shown in the figure. The figure shows that



(A) The particle starts with certain velocity but the motion is retarded and finally the particle stops
(B) The velocity of the particle is constant throughout
(C) The acceleration of the particle is constant throughout
(D) The particle starts with constant velocity, then motion is accelerated and finally the particle moves with another constant velocity

5. A projectile is fired vertically upwards with an initial velocity u. After an interval of T seconds a second projectile is fired vertically upwards, also with initial velocity u.

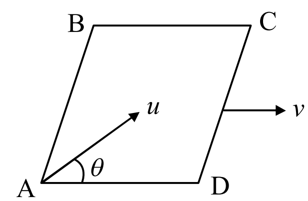
(A) They meet at time $t = \frac{u}{g}$ and at a height $\frac{u^2}{2g} + \frac{gT^2}{8}$

(B) They meet at time $t = \frac{u}{g} + \frac{T}{2}$ and at a height $\frac{u^2}{2g} + \frac{gT^2}{8}$

(C) They meet at time $t = \frac{u}{g} + \frac{T}{2}$ and at a height $\frac{u^2}{2g} - \frac{gT^2}{8}$

(D) They never meet

6. A smooth square platform $ABCD$ is moving towards right with a uniform speed v. At what angle θ must a particle be projected from A with speed u so that it strikes the point B :



(A) $sin^{-1}\left(\frac{u}{v}\right)$ **(B)** $cos^{-1}\left(\frac{v}{u}\right)$ **(C)** $cos^{-1}\left(\frac{u}{v}\right)$ **(D)** $sin^{-1}\left(\frac{v}{u}\right)$

7. A stone is thrown vertically upward. On its way up it passes point A with speed of v, and point B, $3m$ higher than A, with speed $v/2$. The maximum height reached by stone above point B is :

(A) 1 m **(B)** 2 m **(C)** 3 m **(D)** 5 m

8. A Person can swim in still water at the rate of 3 km/hr. He wants to cross the river so that path traveled by swimmer is minimum. If river is flowing at the rate of 5 km/hr and width of the river is 120 m. Then :

(A) Swimmer must swim making angle 127° with flow of river
(B) Swimmer must swim perpendicular to flow to river
(C) Length of possible shortest path is 200 m
(D) Length f possible shortest path is 120 m

9. From a tower of height H, a particle is thrown vertically upwards with a speed u. The time taken by the particle to hit the ground is n times that taken by it to reach the highest point of its path. The relation between H, u and n is :

(A) $2\,gH = n^2u^2$ **(B)** $gH = (n-2)^2 u^2$

(C) $2gH = nu^2(n-2)^2$ **(D)** $gH = (n-2)^2 u^2$

***10.** Let be the instantaneous velocity and acceleration of a particle moving in a plane. The rate of change of speed dv/dt of the particle is equal to :

(A) $|\vec{a}|$ **(B)** $\frac{\vec{v}\,.\,\vec{a}}{v}$

(C) The component of $\vec{a}$ parallel to $\vec{v}$ **(D)** The component of $\vec{a}$ perpendicular to $\vec{v}$

11. Initial acceleration of a particle moving in a straight line is a_0 and initial velocity is zero. The acceleration reduces continuity to half in every t_0 seconds. The terminal velocity of the particle is :

(A) $a_0 t_0 \ell n(2)$ **(B)** $\dfrac{a_0 t_0}{\ell n(2)}$ **(C)** $a_0 t_0$ **(D)** $\dfrac{a_0 t_0}{2}$

***12.** Pick the correct statements :

(A) Average speed of a particle in a given time is never less than the magnitude of the average velocity

(B) It is possible to have a situation in which $\left|\dfrac{d\vec{v}}{dt}\right| \neq 0$ but $\dfrac{d}{dt}|\vec{v}| = 0$

(C) The average velocity of a particle is zero in a time interval. It is possible that the instantaneous velocity is never zero in the interval

(D) The average velocity of a particle moving on a straight line is zero in a time interval. It is possible that the instantaneous velocity is never zero in the interval. (Infinite accelerations are not allowed).

13. Acceleration of a particle which is at rest at $x = 0$ is $\vec{a} = (4 - 2x)\hat{i}$. Select the correct alternative(s).

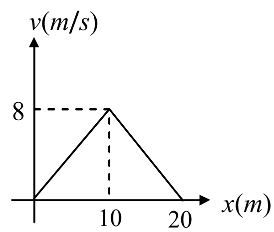
(A) Particle further comes to rest at $x = 4$ **(B)** Particle oscillates about $x = 2$

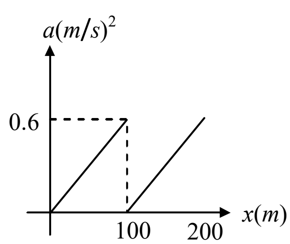
(C) Maximum speed of particle is 4 units **(D)** All of the above

14. A particle starts from rest and traverses a distance ℓ with uniform acceleration, then moves uniformly over a further distance 2ℓ and finally comes to rest after moving a further distance 3ℓ under uniform retardation. Assuming entire motion to be rectilinear motion the ratio of average speed over the journey to the maximum speed on its way is :

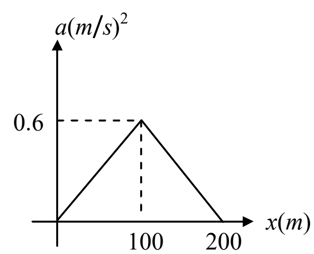
(A) 1/5 **(B)** 2/5 **(C)** 3/5 **(D)** 4/5

15. The *v-s* graph for a car in a race on a straight road is given. Identify the correct *a-x* graph :

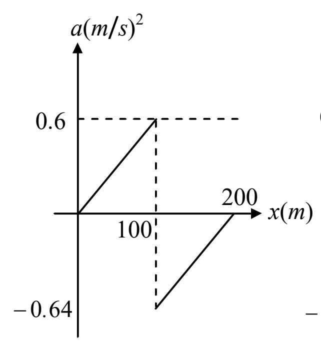




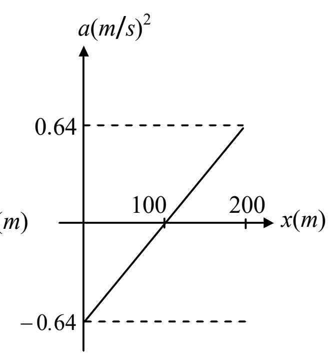
(A)



(B)

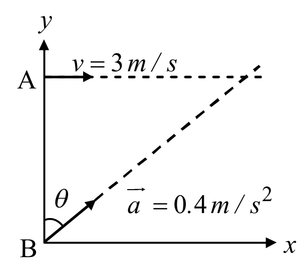


(C)



(D)

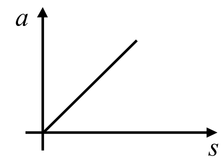
16. Particle A moves along the line $y = 30\ m$ with a constant velocity $\vec{v}$ and directed parallel to the positive x-axis. Particle B starts at the origin with zero speed and constant acceleration $\vec{a}$ at the same time at the instant that particle A passes the y-axis. The square of the time at which they colloide is :

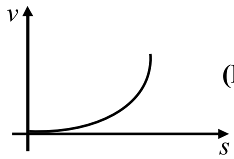


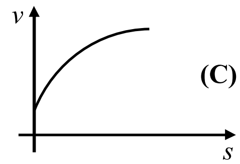
(A) 1 *min* **(B)** 2 *min*

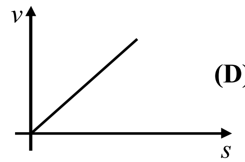
(C) 2.5 *min* **(D)** 5 *min*

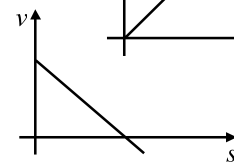
17. Acceleration (a)-displacement (s) graph of a particle moving in a straight line is as shown in the figure. The initial velocity of the particle is zero. The v-s graph of the particle would be :



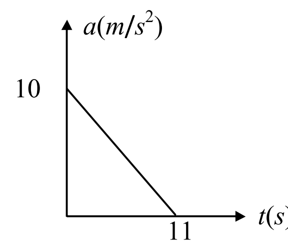
(A)


(B)


(C)


(D)


18. A particle starting from rest undergoes a rectilinear motion with acceleration a. The variation of a with time t is shown in figure. The maximum velocity attained by the particle during the motion is :



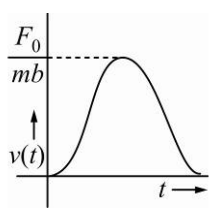
(A) $55\ m/s$ **(B)** $500\ m/s$
(C) $110\ m/s$ **(D)** $650\ m/s$

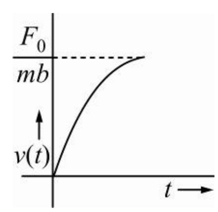
19. A particle has an initial velocity of 9 m/s due east and a constant acceleration of 2 m/s^2 due west. The distance covered by the particle in the fifth second of its motion is :
(A) 0 **(B)** $0.5\ m$ **(C)** $2\ m$ **(D)** None of these

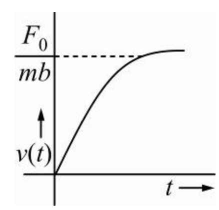
20. A ball of weight W is thrown upward with a velocity v. If air exerts an average resisting force F, the velocity with which the ball returns back to the thrower is :

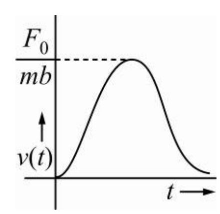
(A) $v\sqrt{\dfrac{W}{W+F}}$ **(B)** $v\sqrt{\dfrac{W}{W-F}}$ **(C)** $v\sqrt{\dfrac{W-F}{W+F}}$ **(D)** $v\sqrt{\dfrac{W+F}{W}}$

21. A particle of mass m is at rest at the origin at time $t = 0$. It is subjected to a force $F(t) = F_0 e^{-bt}$ in the x-direction. Its speed $v(t)$ is depicted by which of the following curves?

(A)


(B)


(C)


(D)


22. Two balls are dropped to the ground from different heights. One ball is dropped 2s after the other but they both strike the ground at the same time. If the first ball takes 5s to reach the ground, then the difference in initial heights is $\left(g = 10\ ms^{-2}\right)$

(A) $20\ m$ **(B)** $80\ m$ **(C)** $170\ m$ **(D)** $40\ m$

***23.** A point moves in a straight line so that its displacement x at time t is given by $x^2 = 1 + t^2$. Its acceleration at any time t is :

(A) $\dfrac{1}{x^3}$ **(B)** $\dfrac{-t}{x^3}$ **(C)** $\dfrac{1}{x} - \dfrac{t^2}{x^3}$ **(D)** $\dfrac{1}{x} - \dfrac{1}{x^2}$

24. A particle of unit mass undergoes one-dimensional motion such that its velocity varies according to $v(x) = \beta x^{-2n}$, where β and n are constants and x is the position of the particle. The acceleration of the particle as a function of x, is given by :

(A) $-2\beta^2 x^{-2n+1}$ **(B)** $-2n\beta^2 e^{-4n+1}$ **(C)** $-2n\beta^2 x^{-2n-1}$ **(D)** $-2n\beta^2 x^{-4n-1}$

25. A stone falls freely under gravity. It covers distance h_1, h_2 *and* h_3 in the first 5 seconds, the next 5 seconds and the next 5 seconds respectively. The relation between h_1, h_2 *and* h_3 is :

(A) $h_2 = 3h_1$ and $h_3 = 3h_2$ **(B)** $h_1 = h_2 = h_3$

(C) $h_1 = 2h_2 = 3h_3$ **(D)** $h_1 = \frac{h_2}{3} = \frac{h_3}{5}$

26. The motion of a particle along a straight line is described by equation $x = 8 + 12t - t^3$ where x is in metre and t in second. The retardation of the particle when its velocity becomes zero is :

(A) $24\,m\,s^{-2}$ **(B)** zero **(C)** $6\,m\,s^{-2}$ **(D)** $12\,m\,s^{-2}$

27. A particle moves a distance x in time t according to equation $x = (t+5)^{-1}$. The acceleration of particle is proportional to :

(A) $(\text{velocity})^{3/2}$ **(B)** $(\text{distance})^2$ **(C)** $(\text{distance})^{-2}$ **(D)** $(\text{velocity})^{2/3}$

28. A ball is dropped from a high rise platform at $t = 0$ starting from rest. After 6 seconds another ball is thrown downwards from the same platform with a speed v. The two balls meet at $t = 18\,s$. What is the value of v ?

(A) $75\,m/s$ **(B)** $55\,m/s$ **(C)** $40\,m/s$ **(D)** $60\,m/s$

29. A particle starts its motion from rest under the action of a constant force. If the distance covered in first 10 seconds is S_1 and that covered in the first 20 seconds is S_2, then :

(A) $S_2 = 3S_1$ **(B)** $S_2 = 4S_1$ **(C)** $S_2 = S_1$ **(D)** $S_2 = 2S_1$

30. A bus is moving with a speed of $10\,ms^{-1}$ on a straight road. A scooterist wishes to overtake the bus in $100\,s$. If the bus is at a distance of $1\,km$ from the scooterist, with what speed should the scooterist chase the bus ?

(A) $40\,ms^{-1}$ **(B)** $25\,ms^{-1}$ **(C)** $10\,ms^{-1}$ **(D)** $20\,ms^{-1}$

31. A particle moving along x-axis has acceleration f, at time t, given by $f = f_0\left(1 - \frac{t}{T}\right)$, where f_0 and T are constants. The particle at $t = 0$ has zero velocity. In the time interval between $t = 0$ and the instant when $f = 0$, the particle's velocity $v(x)$ is :

(A) $\frac{1}{2}f_0T^2$ **(B)** f_0T^2 **(C)** $\frac{1}{2}f_0T$ **(D)** f_0T

32. A car moves from X to Y with a uniform speed v_u and returns to Y with a uniform speed v_d. The average speed for this round trip is :

(A) $\sqrt{v_u v_d}$ **(B)** $\frac{v_d v_u}{v_d + v_u}$ **(C)** $\frac{v_4 + v_d}{2}$ **(D)** $\frac{2v_d v_u}{v_d + v_u}$

33. The position x of a particle with respect to time t along x-axis is given by $x = 9t^2 - t^3$ where x is in metres and t in seconds. What will be the position of this particle when it achieves maximum speed along the $+x$ direction ?

(A) $54\,m$ **(B)** $81\,m$ **(C)** $24\,m$ **(D)** $32\,m$

34. A car runs at a constant speed on a circular track of radius $100\,m$, taking 62.8 seconds for every circular lap. The average velocity and average speed for each circular lap respectively is :

(A) $10\,m/s, 0$ **(B)** $0, 0$ **(C)** $0, 10\,m/s$ **(D)** $10\,m/s, 10\,m/s$

35. The displacement x of a particle varies with time t as $x = ae^{-\alpha t} + be^{-\beta t}$, where a, b, α and β are positive constants. The velocity of the particle will :

(A) be independent of β **(B)** drop to zero when $\alpha = \beta$

(C) go on decreasing with time **(D)** go on increasing with time

36. If a ball is thrown vertically upwards with speed u, the distance covered during the last t seconds of its ascent is :

(A) ut **(B)** $\frac{1}{2}gt^2$ **(C)** $ut - \frac{1}{2}gt^2$ **(D)** $(u + gt)t$

37. A rubber ball is dropped from a height of $5\,m$ on a plane. On bouncing it rises to $1.8\,m$. The ball loses its velocity on bouncing by a factor of :

(A) $\frac{3}{5}$ **(B)** $\frac{2}{5}$ **(C)** $\frac{16}{25}$ **(D)** $\frac{9}{25}$

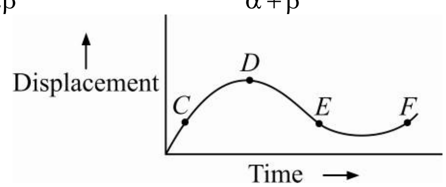
38. A body dropped from a height h with initial velocity zero, strikes the ground with a velocity $3\,m/s$. Another body of same mass dropped from the same height h with an initial velocity of $4\,m/s$. The final velocity of second mass, with which it strikes the ground is :

(A) $5\,m/s$ **(B)** $12\,m/s$ **(C)** $3\,m/s$ **(D)** $4\,m/s$

39. A car accelerates from rest at constant rate α for some time after which it decelerations at a constant rate β and comes to rest. If total time elapsed is t, then maximum velocity acquired by car will be :

(A) $\frac{(\alpha^2 - \beta^2)t}{\alpha\beta}$ **(B)** $\frac{(\alpha^2 + \beta^2)t}{\alpha\beta}$ **(C)** $\frac{(\alpha + \beta)t}{\alpha\beta}$ **(D)** $\frac{\alpha\beta t}{\alpha + \beta}$

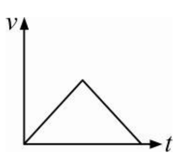
40. The displacement-time graph of a moving particle is shown below. The instantaneous velocity of the particle is negative at the point :

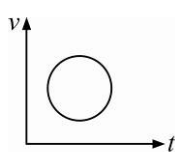


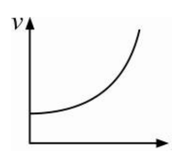
(A) E **(B)** F

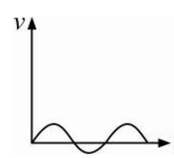
(C) C **(D)** D

41. Which of the following curve does not represent motion in one dimension ?

(A)


(B)


(C)


(D)


42. What will be the ratio of the distance moved by a freely falling body from rest in 4^{th} and 5^{th} seconds of journey ?

(A) $4:5$ **(B)** $7:9$ **(C)** $16:25$ **(D)** $1:1$

43. A car is moving along a straight road with a uniform acceleration. It passes through two points P and Q separated by a distance with velocity 30 *km/h* and 40 *km/h* respectively. The velocity of the car midway between P and Q is :

(A) $33.3\,km/h$ **(B)** $20\sqrt{2}\,km/h$ **(C)** $25\sqrt{2}\,km/h$ **(D)** $35\,km/h$

44. Two balls A and B of same masses are thrown from the top of a building. A, thrown upward with velocity V and B, thrown downward with velocity V, then :

(A) Velocity of A is more than B at the ground **(B)** Velocity of B is more than A at the ground

(C) Both A and B strike the ground with same velocity **(D)** None of these

45. Speed of two identical cars and u and $4u$ at a specific instant. The ratio of the respective distances in which the two cars are stopped from that instant is :

(A) 1 : 1 **(B)** 1 : 4 **(C)** 1 : 8 **(D)** 1 : 16

46. A car moving with a speed of 50 *km/h* can be stopped by brakes after at least 6 *m*. If the same car is moving at a speed of 100 *km/h*, the minimum stopping distance is :

(A) 12 *m* **(B)** 18 *m* **(C)** 24 *m* **(D)** 6 *m*

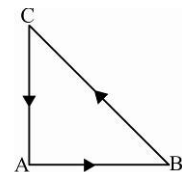
47. Three forces start acting simultaneously on a particle moving with velocity $\vec{v}$. These forces are represented in magnitude and direction by the three sides of a triangle *ABC* (as shown). The particle will now move with velocity.

(A) $\vec{v}$ remaining unchanged

(B) less than $\vec{v}$

(C) greater than $\vec{v}$

(D) $|\vec{v}|$ in the direction of the largest force *BC*



48. A ball is released from the top of a tower of height h metre. It takes T second to reach the ground. What is the position of the ball at $T/3$ second ?

(A) $\frac{(8h)}{9}$ metre from the ground **(B)** $\frac{(7h)}{9}$ metre from the ground

(C) $\frac{h}{9}$ metre from the ground **(D)** $\frac{(17h)}{18}$ metre from the ground

49. A car traveling with a speed of 60 *km/h* can stop within a distance of 20 *m*. If the car is going twice as fact, *i.e*., 120 *km/h*, the stopping distance will be :

(A) 60 *m* **(B)** 40 *m* **(C)** 20 *m* **(D)** 80 *m*

50. A car starting from rest accelerates at the rate f through a distance S, then continues at constant speed for time t and then decelerates at the rate $f/2$ to come to rest. If the total distance traversed is 15 S, then :

(A) $S = \frac{1}{4}ft^2$ **(B)** $S = \frac{1}{2}ft^2$ **(C)** $S = \frac{1}{6}ft^2$ **(D)** $S = \frac{1}{72}ft^2$

51. The relation between time t and distance x is $t = ax^2 + bx$, where a and b are constants. The acceleration is :

(A) $2av^2$ **(B)** $-2av^3$ **(C)** $2bv^3$ **(D)** $-2abv^2$

52. A parachutist after bailing out falls 50 *m* without friction. When his parachute opens, it decelerates at 2 m/s^2. He reaches the ground with a speed of 3 *m/s*. At what height did he bail out ?

(A) 111 *m* **(B)** 293 *m* **(C)** 182 *m* **(D)** 91 *m*

53. A bullet fired into a fixed target loses half of its velocity after penetrating 3 *cm*. How much further it will penetrate before coming to rest assuming that it faces constant resistance to motion ?

(A) 1.0 *cm* **(B)** 1.5 *cm* **(C)** 2.0 *cm* **(D)** 3.0 *cm*

54. A particle located at $x = 0$ at time $t = 0$ starts moving along the positive x-direction with a velocity v that varies as $v = \alpha\sqrt{x}$. The displacement of the particle varies with time as:

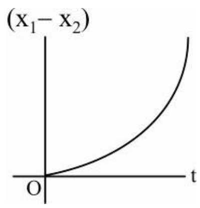
(A) $t^{1/2}$ **(B)** t^3 **(C)** t^2 **(D)** t

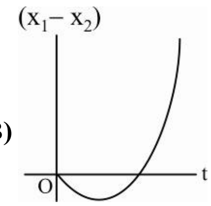
55. The velocity of a particle is $v = v_0 + gt + ft^2$. If its position is $x = 0$ at $t = 0$, then its displacement after unit time ($t = 1$) is :

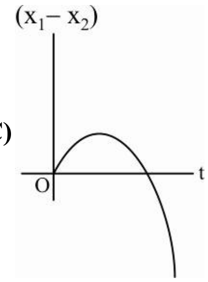
(A) $v_0 + (g/2) + (f/3)$ **(B)** $v_0 + g + f$

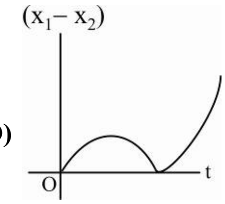
(C) $v_0 + (g/2) + f$ **(D)** $v_0 + 2g + 3f$

56. A body is at rest at $x = 0$. At $t = 0$, it starts moving in the positive x-direction with a constant acceleration. At the same instant another body passes through $x = 0$ moving in the positive x-direction with a constant speed. The position of the first body is given by $x_1(t)$ after time t and that of the second body by $x_2(t)$ after the same time interval. Which of the following graphs correctly describes $(x_1 - x_2)$ as a function of time t ?

(A) 

(B) 

(C) 

(D) 

57. An object moving with a speed of 6.25 *m/s*, is decelerated at a rate given by :

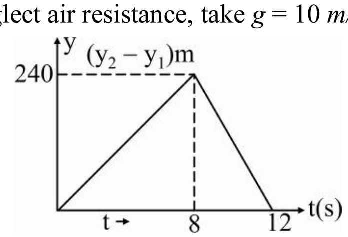
$\frac{dv}{dt} = -2.5\sqrt{v}$ where v is instantaneous speed. The time taken by the object, to come to rest, would be :

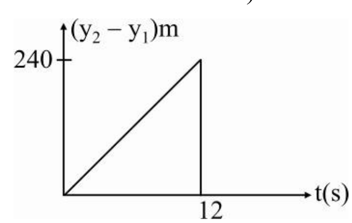
(A) 1 *s* **(B)** 2 *s* **(C)** 4 *s* **(D)** 8 *s*

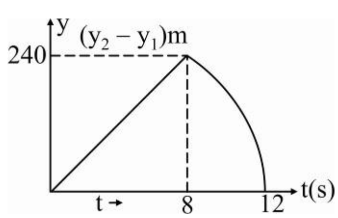
58. From a tower of height H, a particle is thrown vertically upwards with a speed u. The time taken by the particle, to hit the ground, is n times that taken by it to reach the highest point of its path. The relation between H, u and n is :

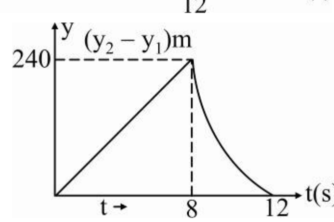
(A) $2gH = nu^2(n-2)$ **(B)** $gH = (n-2)u^2$ **(C)** $2gH = n^2u^2$ **(D)** $gH = (n-2)2u^2$

59. Two stones are through up simultaneously from the edge of a cliff 240 *m* high with initial speed of 10 *m/s* and 40 *m/s* respectively. Which of the following graphs best represents the time variation of relative position of the second stone with respect to the first ? (Assume stones do not rebound after hitting the ground and neglect air resistance, take $g = 10\ m/s^2$). (The figures are schematic and not drawn to scale.)

(A) 

(B) 

(C) 

(D) 

INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

60 The speed of a train increases at constant rate α from zero to v and then remains constant for an interval and finally decreases to zero at constant rate β. The total distance travelled by the train is l. The time taken to complete the journey is t then time t is minimum when $v = \sqrt{\frac{x\alpha\beta}{(\alpha+\beta)}}$. Find the value of x.

61. Starting from rest a particle is first accelerated for time t_1 with constant acceleration a_1 and then stops in time t_2 with constant retardation a_2. Let v_1 be the average velocity in this case and s_1 the total displacement. In the second case, it is accelerated for the same time t_1 with constant acceleration $2a_1$ and comes to rest with constant retardation a_2 in time t_3. If v_2 is the average velocity in this case and s_2 the total displacement, then $\mathrm{v}_2 = \mathrm{m}\ \mathrm{v}_1$. Find the value of m

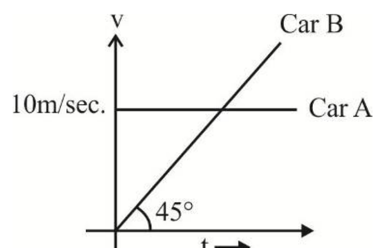
62. The velocity of a particle is at any time related to the distance travelled by the particle by the relation v = ax + b, where a is a positive and b is $\le \frac{a}{2}$. The displacement of particle at time t is $\mathrm{x} = \frac{\mathrm{b}}{\mathrm{a}}\left(\mathrm{e}^{\mathrm{at}} - \mathrm{k}\right)$. Find the value of k.

63. Starting from rest, a particle moving in a straight line is accelerated with an acceleration $a = \left(16 - t^2\right)^{1/2}\ m/\sec^2$ for $0 \le t \le 4$ second and $a = 2m/\sec^2$ for $t > 4$ second. The velocity of the particle at t = 6 second is $\mathrm{p} \times 10^{-2}\left(\pi + 1\right)\mathrm{m/sec}$. Find the value of p.

64. Two cars start off to race with velocities u and u^1 and move with accelerations a and a^1, the result being a dead hit. The length of the course is $\frac{p\left(u'-u\right)\left(u'a - ua'\right)}{\left(a - a'\right)^2}$. Find the value of p.

65. A railway track runs parallel to a road until a turn brings the road to railway crossing. A cyclist rides along the road everyday at a constant speed 20 km/hr. He normally meets a train that travels in same direction at the crossing. The speed of the train in (in km/hr) is 30 N. Find the value of N.

66. Initially car A is 10.5m ahead of car B. Both starting moving at time t = 0 in the same direction along a straight line. The velocity time graph of two cars is shown in figure. The time in second when the car B will catch the car A will be.



67. A burglar's car had started with an acceleration of 2 m/sec^2. A police vigilant party came after 5 second and continued to close the burglar's car with a uniform velocity of 20 m/sec. Find the time taken in which the police van will overtake the burglar's car.

68. An elevator, in which a man is standing is moving upward with a constant acceleration of 1m/sec^2. At some instant when speed of elevator is 10 m/sec, the man drops a coin from a height of 2m. The time taken by the coin to reach the floor is $p \times 10^{-2}$ second. Find the value of p. $\left(g = 10m/\sec^2\right)$

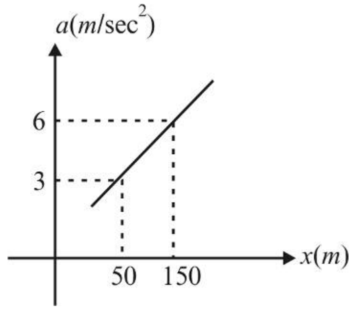
69. Men are running in a line along a road with velocity 9 km/hr behind one another at equal interval of 20m. Cyclist are also riding along the same line in the same direction at 18 km/hr at equal intervals of 30 m. Find the speed in km/hr with which an observer must travel along the road in opposite direction so that whenever he meets a runner he also meet a cyclist

70. A particle P is initially at a distance d = 16 m from a fixed point O. The particle P moves with a velocity $\vec{V} = 5\widehat{PO} + 3\hat{i}$. Where $\widehat{PO}$ is a unit vector from P to O at any time t. Initially $\widehat{PO}$ is perpendicular to $\hat{i}$. Find the time in seconds after which point P meets point O.

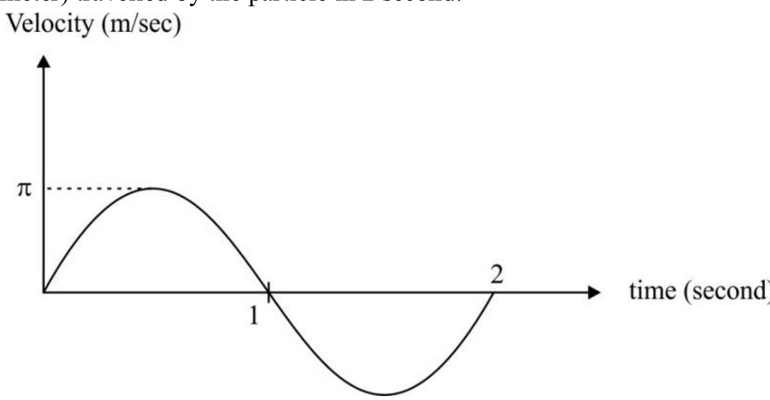
71. The acceleration of the particle varies with time as $a = 2t + |t-2|$. Find the speed (in m/sec) of the particle at

$t = 9$ sec. If the particle was moving in x-axis with a speed of 4 m/sec.

72. Referring to a-x graph. The velocity $V = \sqrt{p}$ meter/second when the displacement of the particle is $100m$. Assume initial velocity as zero. Find the value of p.



73. If a particle is moving on a straight line, then its velocity-time graph is sinusoidal as shown in the figure. Find distance (in meter) travelled by the particle in 2 second.



74. Two friends start bikes from one corner of a square field of edge L towards the diagonally opposite corner in the same time t. They both start from the same place and take different routes. One travels along the diagonal with constant acceleration a, and the other accelerates momentarily and then travels along the edge of the field with constant speed v. The relationship between a and v is $a = \frac{v^2}{\sqrt{k}L}$. Find the value of K

75. A train is targeted to run from Delhi to Pune at an average speed of 80 kph but due to repairs of track looses 2 hrs in the first part of the journey. If then accelerates at a rate of 20 kph^2 till the speed reaches 100 kph. Its speed is now maintained till the end of the journey. If the train now reaches station in time, find the distance from when it started accelerating?

76. Two cars travelling towards each other on a straight road at velocity 10 m/s and 12 m/s respectively. When they are 150m apart, both drivers apply their brakes and each car decelerates at $2m/s^2$ until it stops. How far apart will they be when they have both come to a stop?

Motion in Two Dimensions

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

1. A particle starts traveling on a circle with constant tangential acceleration. The angle between velocity vector and acceleration vector, at the moment when particle completes half the circular track, is :

(A) $tan^{-1}(2\pi)$ **(B)** $tan^{-1}(\pi)$ **(C)** $tan^{-1}(3\pi)$ **(D)** zero

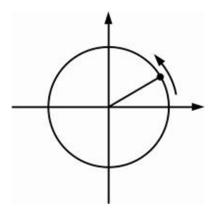
2. In a two dimensional motion, instantaneous speed v_0 is a positive constant. Then which of the following are necessarily true?

(A) The acceleration of the particle is zero
(B) The acceleration of the particle is bounded
(C) The acceleration of the particle is necessarily in the plane of motion
(D) The particle must be undergoing a uniform circular motion

***3.** Two particles are projected in air with speed v_0 at angles θ_1 and θ_2 (both acute) to the horizontal, respectively. If the height reached by the first particle is greater than that of the second, then tick the right choices.

(A) Angle of projection : $q_1 > q_2$ **(B)** Time of flight : $T_1 > T_2$
(C) Horizontal range : $R_1 > R_2$ **(D)** Total energy : $U_1 > U_2$

***4.** A particle moving in a circle centered at the origin in anticlockwise sense as shown. The position of the particle is given as $\vec{r} = R\left(cos\ \omega t\ \hat{i} + sin\ \omega t\ \hat{j}\right)$ where, ω is constant. Mark the correct statement.

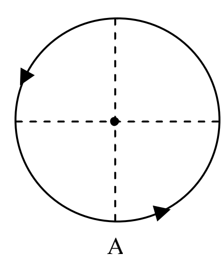


(A) The particle has a constant acceleration
(B) The particle has a variable acceleration
(C) The acceleration of the particle changes according to the rate of $\left|\frac{d\vec{a}}{dt}\right| = R\omega^3$
(D) The speed of the particle changes according to the rate of $\frac{d|\vec{v}|}{dt} = 0$

5. A ball is projected from origin with speed 20 m/s at an angle $30°$ with x-axis. The x-coordinate of the ball at the instant when the velocity of the ball becomes perpendicular to the velocity of projection will be :

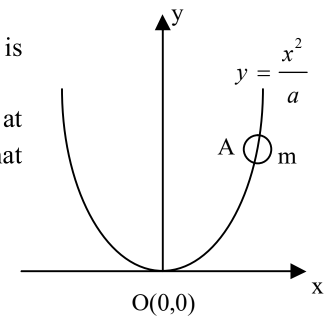
(A) $40\sqrt{3}m$ **(B)** 40 m **(C)** $20\sqrt{3}m$ **(D)** 20 m

6. A particle is describing uniform circular motion in the anti-clockwise sense such that its time period of revolution is T. At $t = 0$ the particle is observed to be at A. If θ_1 be the angle between acceleration at $t = \frac{T}{4}$ and average velocity in the time interval 0 to $\frac{T}{4}$ and θ_2 be the angle between acceleration at $t = \frac{T}{4}$ and the change in velocity in the time interval 0 to $\frac{T}{4}$, then :



(A) $\theta_1 = 135°, \theta_2 = 45°$ **(B)** $\theta_1 = 135°, \theta_2 = 135°$
(C) $\theta_1 = 45°, \theta_2 = 135°$ **(D)** $\theta_1 = 45°, \theta_2 = 45°$

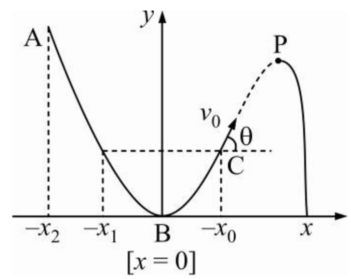
7. In the given figure, a smooth parabolic wire track lies in the vertical plane ($x - y$ plane). The shape of track is defined by the equation $y = \left(\frac{x^2}{a}\right)$ (where a is constant). A ring of mass m which can slide freely on the wire track, is placed at the position A(a, a). The track is rotated with constant angular speed ω such that there is no relative slipping between the ring and the track then ω is equal to :



(A) $\sqrt{\frac{g}{a}}$ **(B)** $\sqrt{\frac{g}{2a}}$

(C) $\sqrt{\frac{2g}{a}}$ **(D)** $(2)^{\frac{1}{4}}\left(\frac{g}{a}\right)^{\frac{1}{2}}$

8. A particle slides down a frictionless parabolic $\left(y = x^2\right)$ track $\left(A - B - C\right)$ starting from rest at point A (in figure). Point B is at the vertex of parabola and point C is at a height less than that of point A. After C, the particle moves freely in air as a projectile. If the particle reaches highest point at P, then :



(A) KE at P = KE at B
(B) Height at P = Height at A
(C) Total energy at P = total energy at A
(D) Time of travel from A to B = Time of travel from B to P.

9. A particle is thrown with a speed u at angle θ to the horizontal. When the particle makes an angle ϕ with the horizontal, its speed changes to v.
(A) $v = u \cos\theta$ **(B)** $v = u \cos\theta \,.\, \cos\phi$ **(C)** $v = \cos\theta \,.\, \sec\phi$ **(D)** $v = u \sec\theta \,.\, \cos\phi$

***10.** For a particle performing uniform circular motion, choose the correct statement(s) from the following:
(A) Magnitude of particle velocity (speed) remains constant
(B) Particle velocity remains directed perpendicular to radius vector
(C) Direction of acceleration keeps changing as particle moves
(D) Angular momentum is constant in magnitude but direction keeps changing

***11.** A cart moves with a constant speed along a horizontal circular path. From the cart, a particle is thrown up vertically with respect to the cart.
(A) The particle will land somewhere on the circular path
(B) The particle will land outside the circular path
(C) The particle will follow an elliptical path
(D) The particle will follow a parabolic path

***12.** Two particles are projected from the same point with the same speed, at different angles θ_1 and θ_2 to the horizontal. They have the same horizontal range. Their times of flight are t_1 and t_2 respectively :

(A) $\theta_1 + \theta_2 = 90°$ **(B)** $\frac{t_1}{t_2} = \tan\theta_1$ **(C)** $\frac{t_1}{t_2} = \tan\theta_2$ **(D)** $\frac{t_1}{\sin\theta_1} = \frac{t_2}{\sin\theta_2}$

13. A particle is projected with a speed u. After 2 seconds of projection it is found to be marking an angle of $45°$ with the horizontal and $0°$ after 3 sec :

(A) Angle of projection is $\tan^{-1}(3)$ **(B)** Angle of projection is $\tan^{-1}(1/3)$

(C) Speed of projection $= 30\sqrt{2}\ m/s$ **(D)** Speed of projection is 30 m/s

14. A ladder placed on a smooth floor slips. If at a given instant the velocity with which the ladder is slipping is, v_1 and the velocity of that part of ladder which is touching the wall is v_2, then the velocity of the centre of the ladder at that instant is :

(A) v_1 **(B)** v_2 **(C)** $\frac{v_1 + v_2}{2}$ **(D)** $\frac{\sqrt{v_1^2 + v_2^2}}{2}$

15. A very broad elevator is going up vertically with a constant acceleration of 2 m/s^2. At the instant when its velocity is 4 m/s a ball is projected from the floor of the lift with a speed of 4 m/s relative to the floor at an elevation of $30°$. The time taken by the ball to return the floor is : ($g = 10\ m/s^2$)

(A) 0.5 s **(B)** 0.33 s **(C)** 0.25 s **(D)** 1s

16. The horizontal range and maximum height attained by a projectile are R and H respectively. If a constant horizontal acceleration $a = g/4$ is imparted to the projectile due to wind, then its horizontal range and maximum height will be :

(A) $(R + H), \frac{H}{2}$ **(B)** $\left(R + \frac{H}{2}\right), 2H$ **(C)** $(R+2H), H$ **(D)** $(R+H), H$

17. For a particle moving along a circular path the radial acceleration a_r is proportional to t^2 (square of time). If a_z is tangential acceleration which of the following is independent of time :

(A) $a_r \cdot a_z$ **(B)** a_z **(C)** a_r / a_z **(D)** $\frac{(a_r)^2}{a_z}$

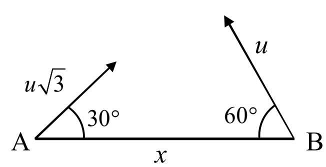
18. With what minimum speed must a particle be projected from origin so that it is able to pass through a given point
(30 m, 40 m). Take $g = 10\ m/s^2$:

(A) 60 m/s **(B)** 30 m/s **(C)** 50 m/s **(D)** 40 m/s

19. A projectile is thrown with a velocity of $10\sqrt{2}\ m/s$ at an angle of $45°$ with horizontal. The interval between the moments when speed is $\sqrt{125}\ m/s$ is : ($g = 10\ m/s^2$)

(A) 1.0 s **(B)** 1.5 s **(C)** 2.0 s **(D)** 0.5 s

20. Two particles A and B are separated by a horizontal distance x. They are projected at the same instant towards each other with speeds $u\sqrt{3}$ and u at angle of projections $30°$ and $60°$ respectively figure. The time after which the horizontal distance between them becomes zero is :



(A) $\frac{x}{u}$ **(B)** $\frac{x}{2u}$ **(C)** $\frac{2x}{u}$ **(D)** $\frac{4x}{u}$

21. A projectile moves from the ground such that its horizontal displacement is $x = Kt$ and vertical displacement is $y = Kt(1 - \alpha t)$, where K and α are constants and t is time. Find out total time of flight (T) and maximum height attained (Y_{max}):

(A) $T = \alpha, Y_{max} = \frac{K}{2\alpha}$ **(B)** $T = \frac{1}{\alpha}, Y_{max} = \frac{2K}{\alpha}$

(C) $T = \frac{1}{\alpha}, Y_{max} = \frac{K}{6\alpha}$ **(D)** $T = \frac{1}{\alpha}, Y_{max} = \frac{K}{4\alpha}$

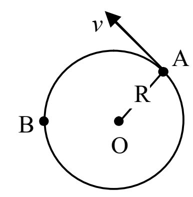
22. A projectile is given in an initial velocity of $(i+2j)\,m/s,$ where i is along the ground and j is along the vertical. If $g = 10\ m/s^2$, the equation of its trajectory is :

(A) $y = x - 5x^2$ **(B)** $y = 2x - 5x^2$ **(C)** $4y = 2x - 5x^2$ **(D)** $4y = 2x - 25x^2$

23. A projectile is aimed at a mark on a horizontal plane through the point of projection and falls 6 *m* short when its elevation is 30° but overshoot the mark by 9 *m* when its elevation is 45°. The angle of elevation of projectile to hit the target on the horizontal plane :

(A) $sin^{-1}\left[\frac{3\sqrt{3}+4}{5}\right]$ **(B)** $cos^{-1}\left[\frac{3\sqrt{3}+4}{5}\right]$ **(C)** $\frac{1}{2}cos^{-1}\left[\frac{3\sqrt{3}+4}{10}\right]$ **(D)** $\frac{1}{2}sin^{-1}\left[\frac{3\sqrt{3}+4}{10}\right]$

24. In uniform circular motion where B is fixed $\frac{\omega_{AO}}{\omega_{AB}} = \frac{\text{ang. velocity of A wrt. O}}{\text{ang. velocity of A wrt.B}}$



(A) $\frac{1}{2}$ **(B)** 2

(C) 1 **(D)** None of these

25. A heavy particle is projected from a point on the horizontal at an angle 60° with the horizontal, with a speed of 10 *m/s*. Then the radius of curvature of its path at the instant of its crossing the same horizontal is ($g = 10\ m/s^2$)

(A) Infinite **(B)** 10 *m* **(C)** 11.54 *m* **(D)** 20 *m*

26. If vectors $\vec{A} = \cos\omega t\ \hat{i} + \sin\omega t\ \hat{j}$ *and* $B = \cos\frac{\omega t}{2}\hat{i} + \sin\frac{\omega t}{2}\hat{j}$ are functions of time, then the value of t at which they are orthogonal to each other is :

(A) $t = \frac{\pi}{\omega}$ **(B)** $t = 0$ **(C)** $t = \frac{\pi}{4\omega}$ **(D)** $t = \frac{\pi}{2\omega}$

27. The position vector of a particle $\vec{R}$ as a function of time is given by $\vec{R} = 4\sin(2\pi)\hat{i} + 4\cos(2\pi t)\hat{j}$. Where R is in meters, t is in seconds and $\hat{i}$ and $\hat{j}$ denotes unit vectors along x- and y-directions, respectively. Which one of the following statements is wrong for the motion of particle ?

(A) Magnitude of the velocity of particle is 8 meter/second

(B) Path of the particle is a circle of radius 4 meter

(C) Acceleration vector is along $-\vec{R}$

(D) Magnitude of acceleration vector is $\frac{v^2}{R}$, where v is the velocity of particle

28. A projectile is filled from the surface of the earth with a velocity of $5\,ms^{-1}$ and angle θ with the horizontal. Another projectile fired from another planet with a velocity of $3\,ms^{-1}$ at the same angle follows a trajectory of the projectile fired from the earth. The value of the acceleration due to gravity on the planet is (in ms^{-2}) is:

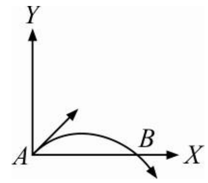
(A) 3.5 **(B)** 5.9 **(C)** 16.3 **(D)** 110.8

29. A particle is moving such that its position coordinates (x, y) are $(2m, 3m)$ at time $t = 0$, $(6m, 7m)$ at time $t = 2s$ and $(13m, 14m)$ at time $(t = 5s)$ is :

Average velocity vector $(\vec{v}_{av})$ from $t = 0$ to $t = 5s$ is :

(A) $\frac{1}{5}(13\hat{i} + 14\hat{j})$ **(B)** $\frac{7}{3}(\hat{i} + \hat{j})$ **(C)** $2(\hat{i} + \hat{j})$ **(D)** $\frac{11}{5}(\hat{i} + \hat{j})$

30. The velocity of a projectile at the initial point A is $\left(2\hat{i}+3\hat{j}\right)m/s$. It's velocity (in m/s) at point B is :



(A) $2\hat{i}-3\hat{j}$ **(B)** $2\hat{i}+3\hat{j}$
(C) $-2\hat{i}-3\hat{j}$ **(D)** $-2\hat{i}+3\hat{j}$

31. A particle has initial velocity $\left(2\hat{i}+3\hat{j}\right)$ and acceleration $\left(0.3\hat{i}+0.2\hat{j}\right)$. The magnitude of velocity after 10 seconds will be :

(A) $9\sqrt{2}$ units **(B)** $5\sqrt{2}$ units **(C)** 5 units **(D)** 9 units

32. A particle moves in a circle of radius $5\,cm$ with constant speed and time period $0.2\pi\,s$. The acceleration of the particle is :

(A) $15\,m/s^2$ **(B)** $25\,m/s^2$ **(C)** $36\,m/s^2$ **(D)** $5\,m/s^2$

33. A missile is fired for maximum range with an initial velocity of $20\,m/s$. If $g=10\,m/s^2$, the range of the missile is :

(A) $40\,m$ **(B)** $50\,m$ **(C)** $60\,m$ **(D)** $20\,m$

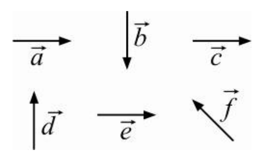
34. A body is moving with velocity $30\,m/s$ towards east. After 10 seconds its velocity becomes $40\,m/s$ towards north. The average acceleration of the body is :

(A) $1\,m/s^2$ **(B)** $7\,m/s^2$ **(C)** $\sqrt{7}\,m/s^2$ **(D)** $5\,m/s^2$

35. A projectile is fired at an angle 45° with the horizontal. Elevation angle of the projectile at its highest point as seen from the point of projection, is :

(A) 45° **(B)** 60° **(C)** $\tan^{-1}\frac{1}{2}$ **(D)** $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

36. Six vectors, $\vec{a}$ through $\vec{f}$ have the magnitudes and directions indication in the figure. Which of the following statements is true ?



(A) $\vec{b}+\vec{c}=\vec{f}$ **(B)** $\vec{d}+\vec{c}=\vec{f}$
(C) $\vec{d}+\vec{e}=\vec{f}$ **(D)** $\vec{b}+\vec{e}=\vec{f}$

37. A particle of mass m is projected with velocity v making and angle 45° with the horizontal. When the particle lands on the level ground the magnitude of the change in its momentum will be :

(A) $\Delta P=-mu\,sin\,45°\,\hat{j}-mu\,cos\,45°\,\hat{i}$ **(B)** $\Delta P=-mu\,sin\,45°\,\hat{j}$
(C) $mu\,cos\,45°\,\hat{i}$ **(D)** 0

38. A tube of length L is filled completely with an incompressible liquid of mass M and closed at both the ends. The tube is then rotated in a horizontal plane about one of its ends with a uniform angular velocity ω. The force exerted by the liquid at the other end is :

(A) $\frac{ML^2\omega^2}{2}$ **(B)** $\frac{ML\omega^2}{2}$ **(C)** $\frac{ML^2\omega}{2}$ **(D)** $ML\omega^2$

39. For angles of projection of a projectile at angle $\left(45°-\theta\right)$ and $\left(45°+\theta\right)$, the horizontal range described by the projectile are in the ratio of :

(A) $2:1$ **(B)** $1:1$ **(C)** $2:3$ **(D)** $1:2$

40. Two boys are standing at the ends A and B of a ground where $AB = a$. The boy at B starts running in a direction perpendicular to AB with velocity v_1. The boy at A starts running simultaneously with velocity v and catches the other in a time t, where t is :

(A) $\dfrac{a}{\sqrt{v^2 + v_1^2}}$ **(B)** $\dfrac{a}{v + v_1}$ **(C)** $\dfrac{a}{v - v_1}$ **(D)** $\sqrt{\dfrac{a^2}{v^2 - v_1^2}}$

41. An object of mass 3 *kg* is at rest. Now a force of $\vec{F} = 6t^2\hat{i} + 4t\,\hat{j}$ is applied on the object then velocity of object at $t = 3\sec$ is :

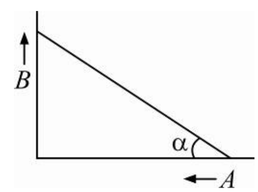
(A) $18\hat{i} + 3\hat{j}$ **(B)** $18\hat{i} + 6\hat{j}$ **(C)** $3\hat{i} + 18\hat{j}$ **(D)** $18\hat{i} + 4\hat{j}$

42. What is the value of linear velocity, if $\vec{r} = 3\hat{i} - 4\hat{j} + \hat{k}$ and $\vec{\omega} = 5\hat{i} - 6\hat{j} + 6\hat{k}$?

(A) $4\hat{i} - 13\hat{j} + 6\hat{k}$ **(B)** $18\hat{i} + 13\hat{j} - 2\hat{k}$ **(C)** $6\hat{i} + 2\hat{j} - 3\hat{k}$ **(D)** $6\hat{i} - 2\hat{j} + 8\hat{k}$

43. Two particles A and B are connected by a rigid rod AB. The rod sides along perpendicular rails as shown here. The velocity of A to the left is $10\,m/s$. What is the velocity of B when angle $a = 60°$?

(A) $10\,m/s$ **(B)** $9.8\,m/s$ **(C)** $5.8\,m/s$ **(D)** $17.3\,m/s$



44. A ball of mass 0.25 *kg* attached to the end of a string of length $1.96\,m$ is moving in a horizontal circle. The string will break if the tension is more than 25 *N*. What is the maximum speed with which the ball can be moved ?

(A) $5\,m/s$ **(B)** $3\,m/s$ **(C)** $14\,m/s$ **(D)** $3.92\,m/s$

45. The angular speed of flywheel making 120 revolutions/minute is :

(A) $4\pi\ rad/s$ **(B)** $4\pi^2\ rad/s$ **(C)** $\pi\ rad/s$ **(D)** $2\pi\ rad/s$

46. A ball is projected with kinetic energy E at an angle of 45° to the horizontal. At the highest point during its flight, its kinetic energy will be :

(A) Zero **(B)** $\dfrac{E}{2}$ **(C)** $\dfrac{E}{\sqrt{2}}$ **(D)** E

47. In a projectile motion, velocity at maximum height is :

(A) $\dfrac{u\cos\theta}{2}$ **(B)** $u\cos\theta$ **(C)** $\dfrac{u\sin\theta}{2}$ **(D)** None of these

48. The coordinates of a moving particle at any time t are given by $x = \alpha t^3$ and $y = \beta t^3$. The speed of the particle at time t is given by :

(A) $\sqrt{\alpha^2 + \beta^2}$ **(B)** $3t\sqrt{\alpha^2 + \beta^2}$ **(C)** $3t^2\sqrt{\alpha^2 + \beta^2}$ **(D)** $t^2\sqrt{\alpha^2 + \beta^2}$

49. A boy playing on the roof of a 10 *m* high building throws a ball with a speed of 10 *m/s* at an angle of 30° with the horizontal. How far from the throwing point will the ball be at the height of 10 *m* from the ground ? ($g = 10\ m/s^2$)

(A) 8.66 *m* **(B)** 5.20 *m* **(C)** 4.33 *m* **(D)** 2.60 *m*

50. Which of the following statements is false for a particle moving in a circle with a constant angular speed ?

(A) The acceleration vector points to the centre of the circle
(B) The acceleration vector is tangent to the circle
(C) The velocity vector is tangent to the circle
(D) The velocity and acceleration vectors are perpendicular to each other

51. A ball is thrown from a point with a speed v_0 at an elevation of angle θ. From the same point and at the same instant, a person starts running with a constant speed $\frac{v_0}{2}$ to catch the ball. Will the person be able to catch the ball ? If yes, what should be the angle of projection θ ?

(A) No **(B)** Yes, 30° **(C)** Yes, 60° **(D)** Yes, 45°

52. For a given velocity, a projectile has the same range R for two angles of projection. If t_1 and t_2 are the times of flight in the two cases then :

(A) $t_1 t_2 \propto R^2$ **(B)** $t_1 t_2 \propto R$ **(C)** $t_1 t_2 \propto \frac{1}{R}$ **(D)** $t_1 t_2 \propto \frac{1}{R^2}$

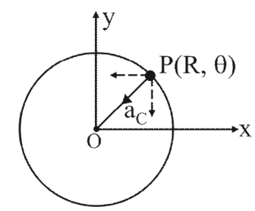
53. A particle has an initial velocity $3\hat{i} + 4\hat{j}$ and an acceleration of $0.4\hat{i} + 0.3\hat{j}$. Its speed after 10 s is :

(A) 10 units **(B)** $7\sqrt{2}$ units **(C)** 7 units **(D)** 8.5 units

54. A particle is moving with velocity $\vec{v} = K(y\hat{i} + x\hat{j})$, where K is constant. The general equation for its path is:

(A) $y = x^2$ + constant **(B)** $y^2 = x$ + constant **(C)** xy = constant **(D)** $y^2 = x^2$ + constant

55. For a particle in uniform circular motion the acceleration $\vec{a}$ at a point $P(R, \theta)$ on the circle of radius R is : (Here θ is measured from the x-axis)

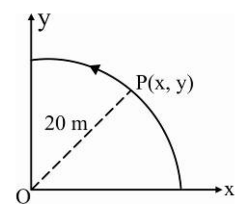


(A) $-\frac{v^2}{R}\cos\theta\,\hat{i} + \frac{v^2}{R}\sin\theta\,\hat{j}$

(B) $-\frac{v^2}{R}\sin\theta\,\hat{i} + \frac{v^2}{R}\cos\theta\,\hat{j}$

(C) $-\frac{v^2}{R}\cos\theta\,\hat{i} - \frac{v^2}{R}\sin\theta\,\hat{j}$ **(D)** $\frac{v^2}{R}\hat{i} + \frac{v^2}{R}\hat{j}$

56. A point P moves in counter-clockwise direction on a circular path as shown in the figure. The movement of 'P' is such that it sweeps out a length $s = t^3 + 5$, where s is in metres and t is in seconds. The radius of the path is 20 m. The acceleration of 'P' when $t = 2$ s is nearly.



(A) 13 m/s^2 **(B)** 12 m/s^2

(C) 7.2 m/s^2 **(D)** 14 m/s^2

57. A water fountain on the ground sprinkles water all around it. If the speed of water coming out of the fountains is v, the total area around the fountain that gets wet is :

(A) $\pi\frac{v^2}{g}$ **(B)** $\pi\frac{v^4}{g^2}$ **(C)** $\frac{\pi}{2}\frac{v^4}{g^2}$ **(D)** $\pi\frac{v^2}{g^2}$

58. A boy can throw a stone up to a maximum height of 10 m. The maximum horizontal distance that the boy can throw the same stone up to will be :

(A) $20\sqrt{2}$ m **(B)** 10 m **(C)** $10\sqrt{2}$ m **(D)** 20 m

59. Two cars of masses m_1 and m_2 are moving in circles of radii r_1 and r_2, respectively. Their speeds are such that they make complete circles in the same time t. The ratio of their centripetal acceleration is :

(A) $m_1 r_1 : m_2 r_2$ **(B)** $m_1 : m_2$ **(C)** $r_1 : r_2$ **(D)** 1 : 1

60. A projectile is given an initial velocity of $(\hat{i} + 2\hat{j})\,m/s$, where $\hat{i}$ is along the ground and $\hat{j}$ is along the vertical. If $g = 10$ m/s^2, the equation of its trajectory is :

(A) $y = 2x - 5x^2$ **(B)** $y = x - 5x^2$ **(C)** $4y = 2x - 5x^2$ **(D)** $y = 2x + 5x^2$

INTEGER ANSWER TYPE QUESTIONS

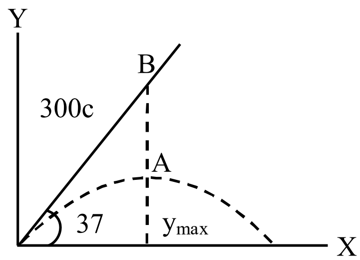
The Answers to the following questions are positive integers of 1/2/3 digits or zero.

61. A projectile is thrown with a velocity of 20 m/s, at an angle of 60º with the horizontal. If t is time(in sec) after which the velocity vector will make an angle of 45º with the horizontal (in upward direction), then integral value of 100 t is (take $g = 10 m/s^2$):

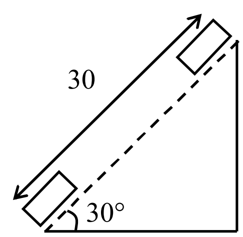
62. An aeroplane was flying horizontally with a velocity of 720 km/h at an altitude of 490 m. When it is just vertically above the target a bomb is dropped from it. How far(in km) horizontally it missed the target?

63. A man standing on a road has to hold his umbrella at 30º with the vertical to keep the rain away. He thrown the umbrella and starts running at 10 km/h. He finds that rain drop are hitting his head vertically. Find the speed(in km/hr) of rain w.r.t. road:

64 A ball A is projected from origin with an initial velocity v_0 = 700 cm/s, in a direction 37º above the horizontal as shown in fig. Another ball B 300 cm from origin on a line 37º above the horizontal is released from rest at the instant A starts. Then how far(in cm) will B have fallen when it is hit by A:

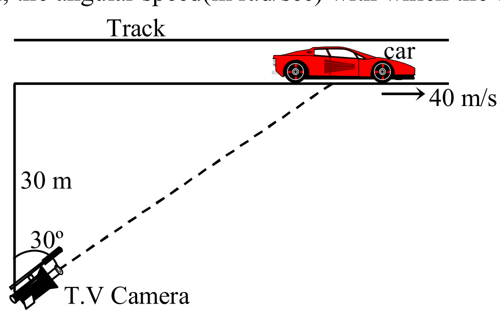


65. Two guns are pointed at each other one upwards at an angle of elevation of 30º and other at the same angle of depression, the muzzle being 30 m apart. If the charges leave the gun with velocities of 350 m/s and 300 m/s respectively. If t is the time (in millisec) when they will meet, integral value of t is:



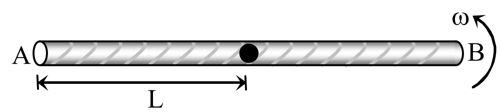
66. A point moves along a circle with velocity v = at where a = 0.5 m/sec^2. Then the total acceleration(in cm/sec^2) of the point at the moment when it covered $(1/10)^{th}$ of the circle after beginning of motion:

67. A racing car is travelling along a track at a constant speed of 40 m/s. A T.V. camera men is recording the event from a distance of 30m directly away from the track as shown in figure. In order to keep the car under view in the position shown, the angular speed(in rad/sec) with which the camera should be rotated, is:



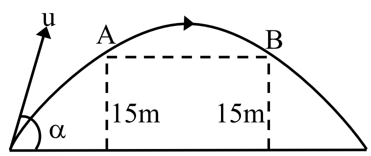
68. A stone is thrown horizontally with a velocity of 10m/sec. Find the find the greatest integer value of radius of curvature(in meter) of it's trajectory at the end of 3 sec after motion began?

69. A long horizontal rod has a bead which can slide its length and initially placed at a distance L= 1m from one end A of the rod. The rod is set in angular motion about A with constant angular acceleration $\alpha = 0.02$ rad/sec. If the coefficient of friction between the rod and the bead is μ=0.32, and gravity is neglected, then the time (in sec.) after which the bead starts slipping is:



70. A ball is projected upwards from the top of tower with a velocity 50 m/s making an angle 30^0 with the horizontal. The height of the tower is 70m. After how many seconds from the instant of throwing will the ball reach the ground:

71. A golfer standing on level ground hits a ball with a velocity of u = 52 m/s at an angle α above the horizontal. If $\tan\alpha$ = 5/12, then the time(in sec) for which the ball is at least 15m above the ground (i.e. between A and B) will be (take g = 10 m/s^2).



72. A shell is fired from a point O at an angle of 60º with a speed of 40 m/s & it strikes a horizontal plane through O, at a point A. The gun is fired a second time with the same angle of elevation but a different speed v. If it hits the target which starts to rise vertically from A with a constant speed 9 $\sqrt{3}$ m/s at the same instant as the shell is fired, find v(in m/sec). (Take g = 10 m/s^2)

73 A batsman hits the ball at a height 4.0 ft from the ground at projection angle of 45° and the horizontal range is 350 ft. Ball falls on left boundary line, where a 24 ft height fence is situated at a distance(in ft.) of 320 ft. if ball passes the fence *h* (in feet) height above it then 100*h* is:

74. A man running on a horizontal road at 8 km/h finds the rain falling vertically. He increases his speed to 12 km/h and finds that the drops are making 30º with vertical. If the speed (in km/h) of the rain with respect to the road is *v*, 10*v* is:

75. A pilot is taking his plane towards north with a velocity of 100 km/h. At that place the wind is blowing with a speed of 60 km/h from east to west. If the plane will be at distance d (in km) from the starting point then integral value of 10d is:

Dynamics of a Particle

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

1. A block of mass m is placed on a surface with a vertical cross-section given by $y = x^3/6$. If the coefficient of friction is 0.5, the maximum height above the ground at which the block can be placed without slipping is :

(A) $\frac{1}{6}m$ **(B)** $\frac{2}{3}m$ **(C)** $\frac{1}{3}m$ **(D)** $\frac{1}{2}m$

2. Two billiard balls A and B, each of mass $50g$ and moving in opposite directions with speed of $5\,m\,s^{-1}$ each, collide and rebound with the same speed. If the collision lasts for 10^{-3} which of the following statements are true?

(A) The impulse imparted to each ball is $0.25\,kg\,m\,s^{-1}$ and the force on each ball is 250 N

(B) The impulse imparted to each ball is $0.25\,kg\,m\,s^{-1}$ and the force exerted on each ball is $25 \times 10^{-5}\,N$

(C) The impulse imparted to each ball is $0.5\,Ns$

(D) The impulse and the force on each ball are equal in magnitude and opposite in direction

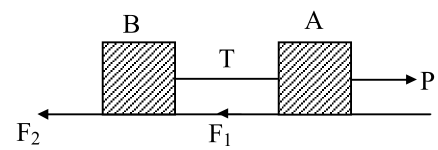
***3.** Two blocks A and B of the same mass are joined by a light string and placed on a horizontal surface. An external horizontal force P acts on A. The tension in the string is T. The forces of static friction acting on A and B are F_1 and F_2 respectively. The limiting value of F_1 and F_2 is F_0. As P is gradually increased.

(A) For $P < F_0,\ T = 0$

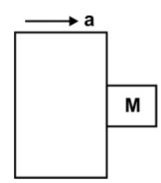
(B) For $F_0 < P < 2F_0,\ T = P - F_0$

(C) For $P > 2F_0,\ T = P/2$

(D) None of the above



4. A rough vertical board has an acceleration 'a' along the horizontal so that a block of mass m pressing against it, does not fall. The coefficient of friction between the block and board should be at least



(A) $= g/a$ **(B)** $> g/a$

(C) $< g/a$ **(D)** $> a/g$

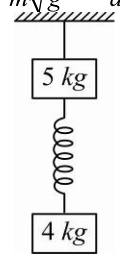
***5.** A simple pendulum with a bob of mass m is suspended from the roof of a car moving with a horizontal acceleration a. The bob is at rest with respect to the car, then :

(A) The string makes an angle of $tan^{-1}(a/g)$ with the vertical

(B) The string makes an angle of $tan^{-1}\left(1 - \frac{a}{g}\right)$ with the vertical

(C) The tension in the string is $m\sqrt{a^2 + g^2}$ **(D)** The tension in the string is $m\sqrt{g^2 - a^2}$

6. Two block of mass $4\,kg$ and $5\,kg$ attached a light spring are suspended by a string as shown in figure. Find acceleration of block $5\,kg$ and $4\,kg$. Just after the string is cut :



(A) $8\,m/s^2,\ 10\,m/s^2$ **(B)** $10\,m/s^2,\ 10\,m/s^2$

(C) $18\,m/s^2,\ 0$ **(D)** $8\,m/s^2,\ 0$

***7.** A block of weight W is suspended from a spring balance. The lower surface of the block rests on a weighing machine. The spring balance reads W_1 and the weighing machine reads W_2. (W, W_1, W_2 are in the same unit.)

(A) $W = W_1 + W_2$ if the system is at rest

(B) $W > W_1 + W_2$ if the system moves down with some acceleration

(C) $W_1 > W_2$ if the system moves up with some acceleration

(D) No relation between W_1 and W_2 can be obtained with the given description of the system

***8.** The motion of a particle of mass m is given by $x = 0$ for $t < 0s$, $x(t) = A\, sin 4p\, t$ for $0 < t < (1/4)s\, (A > 0)$ and $x = 0$ for $t > (1/4)s$:

(A) The force at $t = (1/8)s$ on the particle is $-16\pi^2 A\, m$

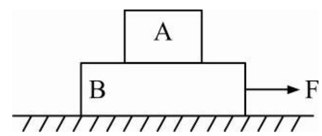
(B) The particle is acted upon by on impulse of magnitude $4\pi^2 A\, m$ at $t = 0s$ and $t = (1/4)s$

(C) The particle is not acted upon by any force

(D) The particle is not acted upon by a constant force

(E) There is no impulse acting on the particle

***9.** In fig. the co-efficient of friction between the floor and the body B is 0.1. The co-efficient of friction between the bodies B and A is 0.2. A force F is applied as shown on B. the mass of A is $m/2$ and of B is m. Which of the following statements are true ?



(A) The bodies will move together if $F = 0.25mg$.

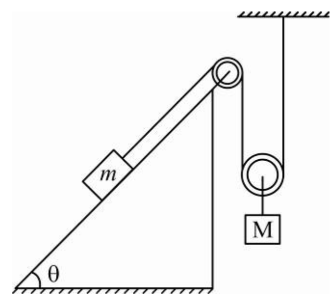
(B) The body A will slip with respect to B if $F = 0.5mg$.

(C) The bodies will move together if $F = 0.5mg$.

(D) The bodies will be at rest if $F = 0.1mg$.

(E) The maximum value of F for which the two bodies will move together is $0.45mg$.

10. Friction co-efficient between block and inclined surface is μ_s. This maximum value of M for which system will remain in equilibrium.



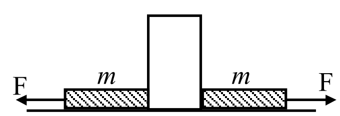
(A) $m(sin\theta + \mu_s cos\theta)$

(B) $2m\, sin\theta$

(C) $2m(sin\theta + \mu_s cos\theta)$

(D) $m\, sin\theta$

11. Figure shows a heavy block kept on a frictionless surface and being pulled by two ropes of equal mass m. At $t = 0$, the force on the left rope is withdrawn but the force on the right end continues to act. Let F_1 and F_2 be the magnitudes of the forces by the right rope and the left rope on the block respectively.



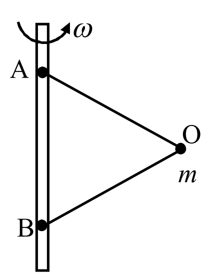
(A) $F_1 = F_2 = F$ for $t < 0$

(B) $F_1 = F_2 = F + mg$ for $t < 0$

(C) $F_1 = F, F_2 = F$ for $t > 0$

(D) $F_1 < F, F_2 = F$ for $t > 0$

12. A particle O, of mass m is attached to a vertical rod with two inextensible strings AO and BO of equal lengths ℓ. The distance between the points of suspension on the vertical rod is also ℓ. If the setup rotates with angular frequency ω, then :



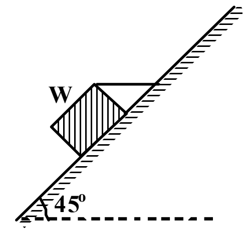
(A) Tension in thread BO is greater

(B) Tension in thread AO is greater

(C) Tension in the two threads are equal

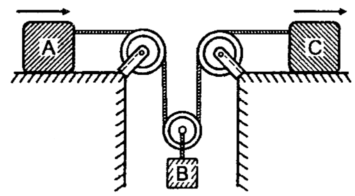
(D) Tension in AO or BO is greater according as ω is anticlockwise or clockwise

13. A rectangular block weighing 150 N, is lying on a rough inclined plane with inclination angle 45° as shown in the figure. The block is tied up by a horizontal string which has a tension of 50 N to keep the block just in equilibrium, then the coefficient of friction between the block and the inclined surface is :



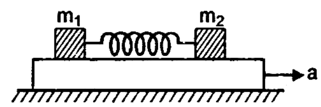
(A) Zero **(B)** 0.33
(C) 0.5 **(D)** 0.7

14. Blocks A and C start from rest and move to the right with acceleration $a_A = 12t\ m/s^2$ and $a_c = 3\ m/s^2$. Here t is in seconds. The time when block B again comes to rest is :



(A) $2s$ **(B)** $1\ s$
(C) $3/2\ s$ **(D)** $1/2\ s$

15. Two blocks of masses m_1 and m_2 are connected with a massless spring and placed over a plank moving with an acceleration 'a' as shown in figure. The coefficient of friction between the blocks and platform is μ.

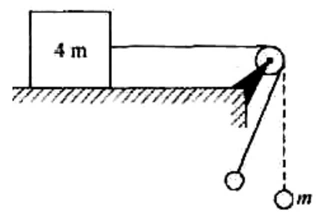


(A) Spring will be stretched if $a > \mu g$
(B) Spring will be compressed if $a < \mu g$
(C) Spring will neither be compressed nor be stretched for $a < \mu g$
(D) Spring will be in its natural length under all conditions

***16.** A block of mass m is placed on a rough horizontal surface. The coefficient of friction between them is μ. An external horizontal force is applied to the block and its magnitude is gradually increased. The force exerted by the block on the surface is R.

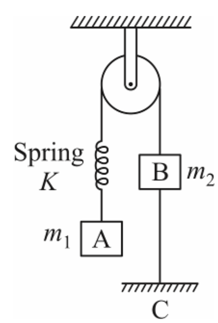
(A) The magnitude of R will gradually increase **(B)** $R \le mg\sqrt{\mu^2+1}$
(C) The angle made by R with the vertical will gradually increase
(D) The angle made by R with the vertical $\le tan^{-1}\mu$

17. Two bodies of mass m and 4 m are attached by a string shown in the figure. The body of mass m hanging from a string of length l is executing simple harmonic motion with amplitude A while other body is at rest on the surface. The minimum coefficient of friction between the mass 4 m and the horizontal surface must be :



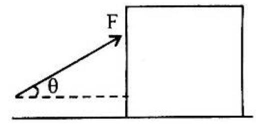
(A) $\frac{1}{4}\left(1-\frac{A^2}{l^2}\right)$ **(B)** $\frac{1}{4}\left(1+\frac{A^2}{l^2}\right)$ **(C)** $\frac{1}{4}\frac{A}{l}cos\,\theta$ **(D)** $\frac{1}{4}$

18. In the system shown in figure, $m_1, > m_2$ system is held at rest by thread BC. Just after the thread BC is burnt:



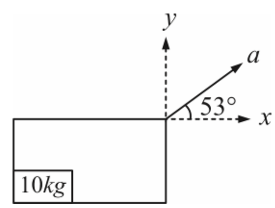
(A) Acceleration of m_2 will be upwards
(B) Magnitude of acceleration of both blocks will be equal to $\left(\frac{m_1-m_2}{m_1+m_2}\right)g$
(C) Acceleration of m_x will be equal to zero
(D) Magnitudes of acceleration of two blocks will be non-zero and unequal

19. Pushing force that makes an angle θ with the horizontal is applied on a block of weight W placed on a horizontal table. If the angle of friction be λ, the magnitude of force required to move the body is equal to



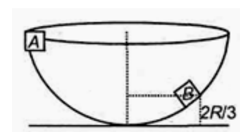
(A) $\dfrac{W\cos\lambda}{\cos(\theta-\lambda)}$ **(B)** $\dfrac{W\sin\lambda}{\cos(\theta-\lambda)}$ **(C)** $\dfrac{W\tan\lambda}{\cos(\theta-\lambda)}$ **(D)** $\dfrac{W\sin\lambda}{g\sin(\theta-\lambda)}$

20. A block of mass 10 kg is placed in a box as shown in figure. Box is moving with constant acceleration of 5 m/s^2 at an angle of 53° from x axis (horizontal direction). Force exerted by box on block in y-direction (vertical direction) will be : (g = 10 m/s^2, $\tan 53° = 4/3$)



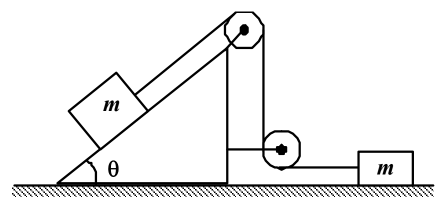
(A) 140N **(B)** 40N
(C) 50N **(D)** 150N

21. A particle of mass m is released from rest at point A along the inside surface of a smooth hemispherical bowl of radius R. The speed at B which is at a height $h = \dfrac{2R}{3}$ from the lowest point is :



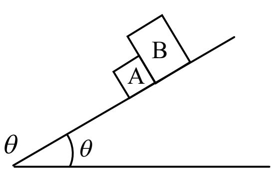
(A) $\sqrt{2gR}$ **(B)** $\sqrt{\dfrac{4gR}{3}}$ **(C)** $\sqrt{gR}$ **(D)** $\sqrt{\dfrac{2}{3}gR}$

22. For the system shown in the figure, the pulleys are light and frictionless. The tension in the string will be :



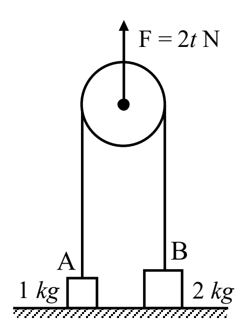
(A) $\dfrac{2}{3}mg\sin\theta$ **(B)** $\dfrac{3}{2}mg\sin\theta$
(C) $\dfrac{1}{2}mg\sin\theta$ **(D)** $2\,mg\sin\theta$

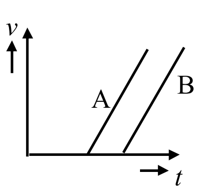
***23.** The two blocks A and B of equal mass are initially in contact. When released from rest on the inclined plane, they slide down the incline. The coefficients of friction between the inclined plane and A and B are μ_1 and μ_2 respectively.

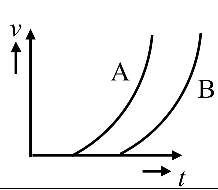


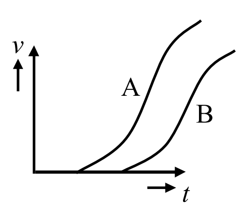
(A) If $\mu_1 > \mu_2$, the blocks will always remain in contact
(B) If $\mu_1 < \mu_2$, the blocks will slide down with different accelerations
(C) If $\mu_1 > \mu_2$, the blocks will have a common acceleration $\dfrac{1}{2}(\mu_1+\mu_2)g\sin\theta$
(D) If $\mu_1 < \mu_2$, the blocks will have a common acceleration $\dfrac{\mu_1\mu_2 g}{\mu_1+\mu_2}\sin\theta$

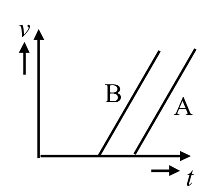
24. Two blocks A and B of masses 1 kg and 2 kg respectively are placed on a smooth horizontal surface. They are connected by a massless inextensible string going over a pulley as shown. The pulley is being acted upon by a vertical force of magnitude varying with time as $F = 2t\ N$. Which of the following represent the velocity time variation of A and B.



(A)


(B)


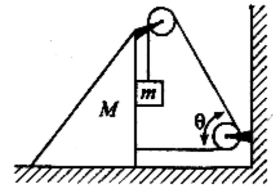
(C)


(D)


25. The force required to just move a body up the inclined plane is double the forcerequired to just prevent the body from sliding down the plane. The coefficient of friction is u. The inclination θ of the plane is:

(A) $tan^{-1}(\mu)$ **(B)** $tan^{-1}(\mu/2)$ **(C)** $tan^{-1}(2\mu)$ **(D)** $tan^{-1}(3\mu)$

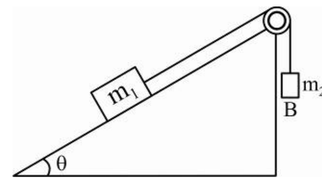
26. *M* and *m* are connected as shown in figure. If *v* and *u* denote the horizontal veloeity of Mand vertical veloeity component of *m* respectively then the ratio of *u*/*v* is:



(A) $1 + cos\ \theta$ **(B)** $1 + cos^2\ \theta$

(C) $1 - cos\ \theta$ **(D)** None of these

***27.** Mass m_1 moves on a slope making an angle θ with the horizontal and is attached to mass m_2 by a string passing over a frictionless pulley as shown in fig. The co-efficient to friction between m_1 and the sloping surface is μ . Which of the following statements are true ?



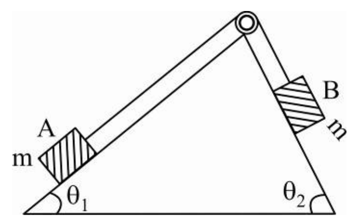
(A) If $m_2 > m_1\ sin\theta$, the body will move up the plane

(B) If $m_2 < m_1\ (sin\theta + \mu\ cos\theta)$, the body will move up the plane

(C) If $m_2 < m_1\ (sin\theta + \mu\ cos\theta)$, the body will move down the plane

(D) If $m_2 < m_1\ (sin\theta - \mu\ cos\theta)$, the body will move down the plane

***28.** In figure a body *A* of mass *m* slides on plane inclined at angle θ_1 on the horizontal and μ_1 is the coefficient of friction between and the plane. *A* is connected by a light string passing over a frictionless pulley to another body *B*, also of mass *m*, sliding on a frictionless plane inclined at angle θ_2 to the horizontal. Which of the following statements are true ?



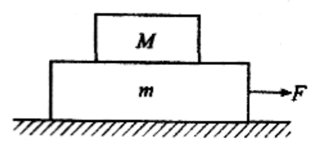
(A) *A* will never move up the plane

(B) *A* will never move up the plane when $\mu = \dfrac{sin\theta_2 - sin\theta_1}{cos\theta_1}$

(C) For *A* to move up the plane, θ_2 must always be greater than θ_1

(D) *B* will always slide down with constant speed

29. A light string fixed at one end to a clamp on ground passes over a fixed pulley and hangs at the other side. It makes an angle of 30° with the ground. A monkey of mass 5 *kg* climbs up the rope. The clamp can tolerate a vertical force of 40 *N* only. The maximum acceleration in upward direction with which the monkey can climb safely is : (Neglect friction and take $g = 10\ m/s^2$)

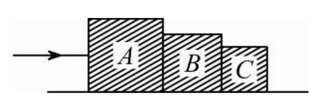


(A) $2m/s^2$ **(B)** $4m/s^2$ **(C)** $6\ m/s^2$ **(D)** $8m/s^2$

30. Two stones of masses *m* and 2*m* are whirled in horizontal circles, the heavier one in a radius $\frac{r}{2}$ and the lighter one in radius *r*. The tangential speed of lighter stone is *n* times that of the value of heavier stone when they experience same centripetal forces. The value of *n* is :

(A) 4 **(B)** 1 **(C)** 2 **(D)** 3

31. Three blocks *A*, *B* and *C*, of masses 4 *kg*, 2 *kg* and 1 *kg* respectively, are in contact on a frictionless surface, as shown. If a force of 14 *N* is applied on the 4 *kg* block, then the contact force between *A* and *B* is :



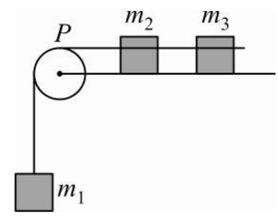
(A) 4 **(B)** 1 **(C)** 2 **(D)** 3

32. A block *A* of mass m_1 rests on a horizontal table. A light string connected to it passes over a frictionless at the edge of table and from its other end another block *B* of the mass m_2 is suspended. The coefficient of kinetic friction between the block and the table is μ_k. When the block *A* is sliding on the table, the tension in the string is :

(A) $\dfrac{m_1 m_2 (1+\mu_k) g}{(m_1+m_2)}$ **(B)** $\dfrac{m_1 m_2 (1-\mu_k) g}{(m_1+m_2)}$

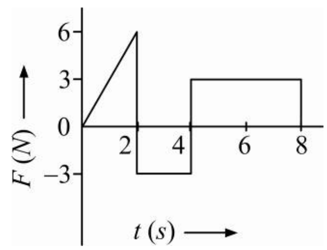
(C) $\dfrac{(m_2+\mu_k m_1) g}{(m_1+m_2)}$ **(D)** $\dfrac{(m_2-\mu_k m_1) g}{(m_1+m_2)}$

33. A system consists of three masses m_1, m_2 and m_3 connected by a string passing over a pulley *P*. The mass m_1 hangs freely and m_2 and m_3 are on a rough horizontal table (the coefficient of friction $= \mu$). The pulley is frictionless and of negligible mass. The downward acceleration of mass m_1 is (Assume $m_1 = m_2 = m_3 = m$) :



(A) $\dfrac{g(1-g\mu)}{9}$ **(B)** $\dfrac{2g\mu}{3}$ **(C)** $\dfrac{g(1-2\mu)}{3}$ **(D)** $\dfrac{g(1-2\mu)}{2}$

34. The force *F* acting on a particle of mass *m* is indicated by the force-time graph shown below. The change in moment of the particle over the time interval from zero to 8 *s* is :



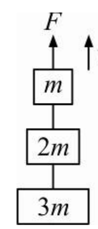
(A) $24\,N\,s$ **(B)** $20\,N\,s$

(C) $12\,N\,s$ **(D)** $6\,N\,s$

35. A balloon with mass *m* is descending down with an acceleration *a* (where $a < g$). How much mass should be removed from it so that it starts moving up with an acceleration *a* ?

(A) $\dfrac{2ma}{g+a}$ **(B)** $\dfrac{2ma}{g-a}$ **(C)** $\dfrac{ma}{g+a}$ **(D)** $\dfrac{ma}{g-a}$

36. Three blocks with masses $m, 2m$ and $3m$ are connected by strings, as shown in figure. After an upward force *F* is applied on block *m*, the masses move upward at constant speed *v*. What is the net force on the block of mass $2m$? (*g* is the acceleration due to gravity)



(A) $3mg$ **(B)** $6mg$ **(C)** zero **(D)** $2mg$

37. An explosion breaks a rock into three parts in a horizontal plane. Two of them go off at right angles to each other. The first part of mass 1 *kg* moves with a speed of $12\,ms^{-1}$ and the second part of mass 2 *kg* moves with $8\,ms^{-1}$ speed. If the third part files off with $4\,ms^{-1}$ speed, then its mass is :

(A) $7\,kg$ **(B)** $17\,kg$ **(C)** $3\,kg$ **(D)** $5\,kg$

38. The upper half of an inclined plane of inclination θ is perfectly smooth while lower half is rough. A block starting from rest at the top of the plane will again come to rest at the bottom, if the coefficient of friction between the block and lower half of the plane is given by :

(A) $\mu = 2\tan\theta$ **(B)** $\mu = \tan\theta$ **(C)** $\mu = \dfrac{1}{\tan\theta}$ **(D)** $\mu = \dfrac{2}{\tan\theta}$

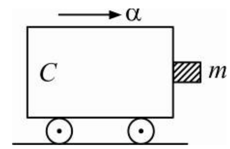
39. A person holding a rifle (mass of person and rifle together is $100 kg$) stands on a smooth surface and fires 10 shots horizontally, in $5 s$. Each bullet has a mass of $10 g$ with a muzzle velocity of $800 ms^{-1}$. The final velocity acquired by the person and the average force expected on the person are :

(A) $-0.08 ms^{-1}, 16 N$ **(B)** $-0.8 ms^{-1}, 8 N$ **(C)** $-1.6 ms^{-1}, 16 N$ **(D)** $-1.6 ms^{-1}, 8 N$

40. A person of mass 60 kg is inside a life of mass $940 kg$ and presses the button on control penal, The lift starts moving upwards with an acceleration $1.0 m/s^2$. If $g = 10 ms^{-2}$, the tension in the supporting cable is :

(A) $8600 N$ **(B)** $9680 N$ **(C)** $11000 N$ **(D)** $1200 N$

41. A block of mass m is in contact with the cart C as shown in the figure. The coefficient of static between the block and the cart is μ. The acceleration α of the cart that will prevent the block from falling satisfies :



(A) $\alpha > \frac{mg}{\mu}$ **(B)** $\alpha > \frac{g}{\mu m}$ **(C)** $\alpha \geq \frac{g}{\mu}$ **(D)** $\alpha < \frac{g}{\mu}$

42. A roller coaster is designed such that riders experience "weightlessness" as they go round the top of a hill of radius of curvature $20 m$. The speed of the car at the top of the hill is between :

(A) $16 m/s$ and $17 m/s$ **(B)** $13 m/s$ and $14 m/s$

(C) $14 m/s$ and $15 m/s$ **(D)** $15 m/s$ and $16 m/s$

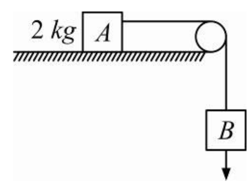
43. Sand is being dropped on a conveyer belt at the rate of $M\ kg/s$. The force necessary to keep the belt moving with a constant velocity of $v\ m/s$ will be :

(A) $\frac{Mv}{2}$ Newton **(B)** zero **(C)** Mv Newton **(D)** 2 Mv Newton

44. A 0.5 kg ball moving with a speed of $12 m/s$ strikes a hard wall at an angle of 30° with the wall. It is reflected with the same speed at the same angle. If ball is in contact with the wall for 0.25 seconds, the average force acting on the wall is :

(A) 96 N **(B)** 48 N **(C)** 24 N **(D)** 12 N

45. The coefficient of static friction, μ_x, between block A of mass 2 kg and the table as shown in the figure is 2.0. What would be the maximum mass value of block B, so that the two blocks do not move? The string and the pulley are assumed to the smooth and massless. $(g = 10 m/s^2)$



(A) $2.0 kg$ **(B)** $4.0 kg$ **(C)** $0.2 kg$ **(D)** $0.4 kg$

46. A man weighs $80 kg$. He stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of $5 m/s^2$. What would be the reading on the scale? $(g = 10 m/s^2)$

(A) zero **(B)** 400 N **(C)** 800 N **(D)** 1200 N

47. A monkey of mass $20 kg$ is holding a vertical rope. The rope will not break when a mass of $25 kg$ is suspended from it but will break of the mass exceeds $25 kg$. What is the maximum acceleration with which the monkey can climb up along which is connected to lift is :

(A) $5 m/s^2$ **(B)** $10 m/s^2$ **(C)** $25 m/s^2$ **(D)** $2.5 m/s^2$

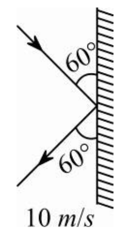
48. 250 N force is required to raise 75 kg mass from a pulley. If rope is pulled $12\,m$ then the load is lifted to $3\,m$, the efficiency of pulley system will be :

(A) 25% **(B)** 33.3% **(C)** 75% **(D)** 90%

49. On the horizontal surface of a truck a block of mass $1\,kg$ is placed $(\mu = 0.6)$ and truck a block of mass 1 kg is placed $(\mu = 0.6)$ and truck is moving with acceleration $5\,m/\sec^2$ then the frictional force on the block will be :

(A) 5 N **(B)** 6 N **(C)** 5.88 N **(D)** 8 N

50. A body of mass $3\,kg$ hits a wall at an angle of 60° and return at the same angle. The impact time was $0.2\,\sec$. The force exerted on the wall :



(A) $150\sqrt{3}\,N$ **(B)** $50\sqrt{3}\,N$

(C) $100\,N$ **(D)** $75\sqrt{3}\,N$

51. A bullet is fired from a gun. The force on the bullet is given by $F = 600 - 2\times10^5 t$ where, F is in Newton and t in seconds. The force on the bullet becomes zero as soon as it leaves the barrel. What is the average impulse imparted to the bullet ?

(A) 9 N-s **(B)** zero **(C)** 1.8 N-s **(D)** 0.9 N-s

52. In a rocket, fuel burns at the rate of $1\,kg/s$. This fuel is ejected from the rocket with a velocity of $60\,km/s$. This exerts a force on the rocket equal to :

(A) 6000 N **(B)** 60000 N **(C)** 60 N **(D)** 600 N

53. A $600\,kg$ rocket is set for a vertical fringe. If the exhaust speed is $1000\,ms^{-1}$, the mass of the gas ejected per second to supply the thrust needed to overcome the weight of rocket is :

(A) $117.6\,kgs^{-1}$ **(B)** $58.6\,kgs^{-1}$ **(C)** $6\,kgs^{-1}$ **(D)** $76.4\,kgs^{-1}$

54. A body of mass $5\,kg$ explodes at rest into three fragments with equal masses in the ratio 1 : 1 : 3. The fragments with equal masses fly in mutually perpendicular directions with speeds of $21\,m/s$. The velocity of heaviest fragment in m/s will be :

(A) $7\sqrt{2}$ **(B)** $5\sqrt{2}$ **(C)** $3\sqrt{2}$ **(D)** $\sqrt{2}$

55. When forces F_1, F_2 and F_3 are acting on a particle of mass m such that F_2 and F_3 are mutually perpendicular, then the particle remains stationary. If the force F_1 is now removed, then the acceleration of the particle is:

(A) F_1/m **(B)** F_2F_3/mF_1 **(C)** $(F_2 - F_3)/m$ **(D)** F_2/m

56. A lift is moving down with acceleration a. A man in the lift drops a ball inside the lift. The acceleration of the ball as observed by the man in the lift and a man standing on the ground are, respectively:

(A) g, g **(B)** $g-a, g-a$ **(C)** $g-a, g$ **(D)** a, g

57. A light string passing over a smooth light pulley connects two blocks of masses m_1 and m_2 (vertically). If the acceleration of the system is $g/8$, then the ratio of the masses is :

(A) $8:1$ **(B)** $9:7$ **(C)** $4:3$ **(D)** $5:3$

58. Two forces are such that sum of their magnitudes is 18 N and their resultant, of magnitudes 12 N, is perpendicular to the smaller force. Then the magnitudes of the forces are :

(A) $12\,N, 6\,N$ **(B)** $13\,N, 5\,N$ **(C)** $10\,N, 8\,N$ **(D)** $16\,N, 2\,N$

59. The minimum velocity (in m/s) with which a car driver must traverse a flat curve of radius 150 m and coefficient of friction 0.6 to avoid skidding is :

(A) 60 **(B)** 30 **(C)** 15 **(D)** 25

60. A light spring balance hangs from the hook of the other light spring balance and a block of mass M kilogram hangs from the former one. Which of the following statements about the scale reading is true ?

(A) Both the scales read $M/2$ kilogram each

(B) Both the scales read M kilogram each

(C) The scale of the lower one reads M kilogram and of the upper one zero

(D) The reading of the two scales can be anything but the sum of the reading will be M kilogram.

61. A block of mass M is pulled along a horizontal frictionless surface by a rope of mass m. If a force P is applied at the free end of the rope, the force exerted by the rope on the block is :

(A) $\frac{PM}{M+m}$ **(B)** $\frac{Pm}{M+m}$ **(C)** $\frac{PM}{M-m}$ **(D)** p

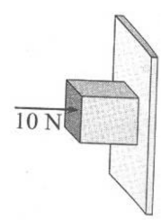
62. A spring balance is attached to the ceiling of a lift. A man hangs his bag on the spring and the spring reads 49 N, when the lift is stationary. If the lift moves downward with an acceleration of $5\,m/s^2$, the reading of the spring balance will be :

(A) 49 N **(B)** 24 N **(C)** 74 N **(D)** 15 N

63. A rocket with a lift-off mass $3.5\times10^4\ kg$ is blasted upward with an initial acceleration of $10\,m/s^2$. Then the initial thrust of blast is :

(A) $1.75\times10^5\,N$ **(B)** $3.5\times10^5\,N$ **(C)** $7.0\times10^5\,N$ **(D)** $1.40\times10^5\,N$

64. A horizontal force of $10N$ is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is :



(A) $2N$ **(B)** $20N$

(C) $50N$ **(D)** $100N$

65. A marble block of mass 2 kg lying on ice when given a velocity of $6\,m/s$ is stopped by friction in 10s. Then the coefficient of friction is :

(A) 0.01 **(B)** 0.02 **(C)** 0.03 **(D)** 0.06

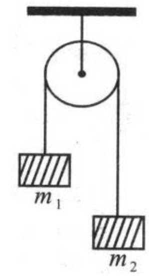
66. A car is moving on a circular path of radius 500 m with a speed of 30 m/s. If the speed is increased at the rate of $2\,m/s^2$, the resultant acceleration of the car at this moment is :

(A) $2\,m/s^2$ **(B)** $2.5\,m/s^2$ **(C)** $2.7\,m/s^2$ **(D)** $4\,m/s^2$

67. A machine gun fires a bullet of mass $40\,g$ with a velocity of $1200\,m/s$. The man holding it can exert a maximum force of 144 N on the gun. How many bullets can he fire per second at the most ?

(A) two **(B)** four **(C)** one **(D)** three

68. Two masses $m_1 = 5\,kg$ and $m_2 = 4.8\,kg$ tied to a string are hanging over a light frictionless pulley. What is the acceleration of the masses when the system is left free to move ? $\left(g = 9.8\,m/s^2\right)$

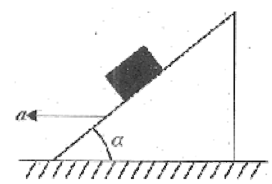


(A) $5\,m/s^2$ **(B)** $9.8\,m/s^2$
(C) $0.2\,m/s^2$ **(D)** $4.8\,m/s^2$

69. A block rests on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of static friction between the block and the plane is 0.8. If the frictional force on the block is 10 *N*, the mass of the block (in *kg*) is : (Take $g = 10\,m/s^2$)

(A) 1.6 **(B)** 4.0 **(C)** 2.0 **(D)** 2.5

70. A block is kept on a frictionless inclined surface with angle of inclination α. The incline is given an acceleration *a* to keep the block stationary.
Then *a* is equal to :



(A) $g \tan\alpha$ **(B)** g
(C) $g \operatorname{cosec}\alpha$ **(D)** $g/\tan\alpha$

71. A particle of mass $0.3\,kg$ is subjected to a force $F = -kx$ with $k = 15\,N/m$. What will be its initial acceleration if it is released from a point $20\,cm$ away from the origin ?

(A) $5\,m/s^2$ **(B)** $10\,m/s^2$ **(C)** $3\,m/s^2$ **(D)** $15\,m/s^2$

72. A smooth block is released from rest on a 45° incline and then slides a distance *d*. The time taken to slide is *n* times as much to slide on a rough inclined than on a smooth incline. The coefficient of friction is :

(A) $\mu_s = \sqrt{1-\frac{1}{n^2}}$ **(B)** $\mu_s = 1-\frac{1}{n^2}$ **(C)** $\mu_k = \sqrt{1-\frac{1}{n^2}}$ **(D)** $\mu_k = 1-\frac{1}{n^2}$

73. Consider a car moving on a straight road with a speed of $100\,m/s$. The distance at which the car can be stopped is : $[\mu_k = 0.5]$

(A) $400\,m$ **(B)** $100\,m$ **(C)** $1000\,m$ **(D)** $800\,m$

(**Note :** It should be minimum distance in which the car can be stopped.)

74. The upper half of an inclined plane of inclination θ is perfectly smooth while the lower half is rough. A body starting from rest at top comes back to rest at the bottom if the coefficient of friction for the lower half is given by :

(A) $\mu = \sin\theta$ **(B)** $\mu = \cot\theta$ **(C)** $\mu = 2\cos\theta$ **(D)** $\mu = 2\tan\theta$

75. A player caught a cricket ball of mass 150 *g* moving at a rate of $20\,m/s$. If the catching process is completed in 0.1*s*, the force of the blow exerted by the ball on the hand of the player is :

(A) $30\,N$ **(B)** $300\,N$ **(C)** $150\,N$ **(D)** $3\,N$

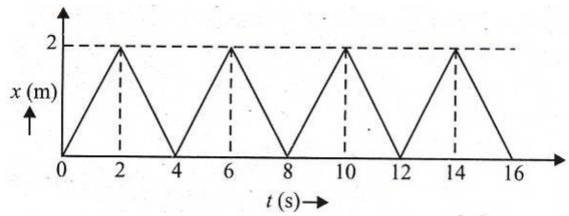
76. A ball of mass $0.2\,kg$ is thrown vertically upwards by applying a force by hand. If the hand moves $0.2\,m$ while applying the force and the ball goes up to $2\,m$ height further, find the magnitude of the force. Consider $g = 10\,m/s^2$.

(A) $20\,N$ **(B)** $22\,N$ **(C)** $4\,N$ **(D)** $16\,N$

77. A body of mass $m = 3.513\,kg$ is moving along the x-axis with a speed of $5.00\,m/s$. The magnitude of its momentum is recorded as :

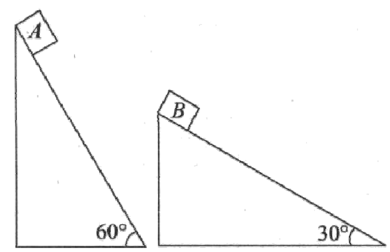
(A) 17.6 *kg-m/s* **(B)** 17.565 *kg-m/s* **(C)** 17.56 *kg-m/s* **(D)** 17.57 *kg-m/s*

78. The figure shows the position-time $(x-t)$ graph of one-dimensional motion of a body of mass 0.4 *kg*. The magnitude of each impulse is :



(A) $0.4\,Ns$ **(B)** $0.8\,Ns$ **(C)** $1.6\,Ns$ **(D)** $0.2\,Ns$

79. Two fixed frictionless inclined planes making angles 30° and 60° with the vertical are shows in the figure. Two blocks A and B are placed on the two planes. What is the relative vertical acceleration of A with respect to B ?



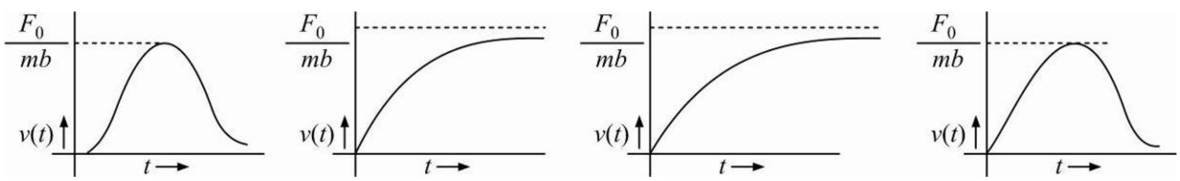
(A) $4.9\,m/s^2$ in horizontal direction

(B) $9.8\,m/s^2$ in vertical direction

(C) Zero

(D) $4.9\,m/s^2$ in vertical direction

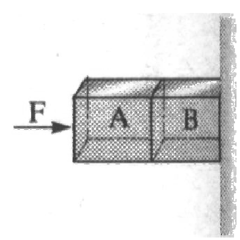
80. A particle of mass m is at rest at the origin at time $t = 0$. It is subjected to a force $F(t) = F_0 e^{-bt}$ in the x direction. Its speed $v(t)$ is depicted by which of the following curves ?



81. A block of mass m is placed on a surface with a vertical cross section given by $y = x^3/6$. If the coefficient of friction is 0.5, the maximum height above the ground at which the block can be placed without slipping is:

(A) $\frac{1}{3}m$ **(B)** $\frac{1}{2}m$ **(C)** $\frac{1}{6}m$ **(D)** $\frac{2}{3}m$

82. Given in the figure are two blocks A and B of weight 20 N and 100 N, respectively. These are being pressed against a wall by a force F as shown. If the coefficient of friction between the blocks is 0.1 and between block B and the wall is 0.15, the frictional force applied by the wall on block B is :



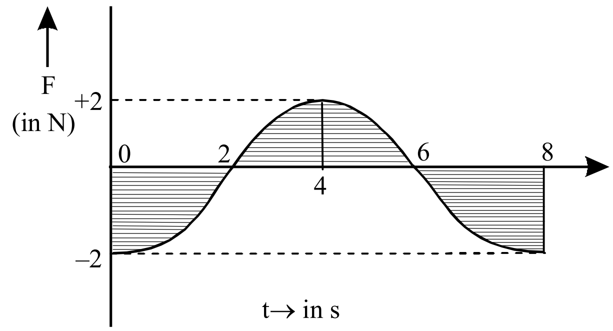
(A) $100\,N$ **(B)** $80\,N$

(C) $120\,N$ **(D)** $150\,N$

INTEGER ANSWER TYPE QUESTIONS

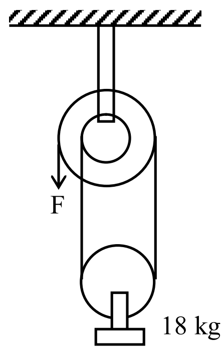
The Answers to the following questions are positive integers of 1/2/3 digits or zero.

83. A force – time (Sine curve) graph for the motion of a body is shown in figure. Change in linear momentum between 0 and 8 s is:

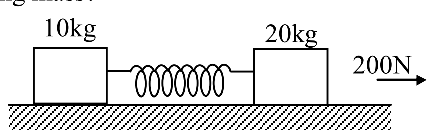


84. A particle is acted upon by two mutually perpendicular forces of 3N and 4N. In order that the particle remains stationary, the magnitude of the third force in N that should be applied is:

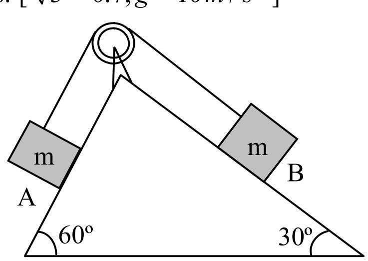
85. In the figure at the free end a force F is applied to keep the suspended mass of 18 kg at rest . The value of F in N is:



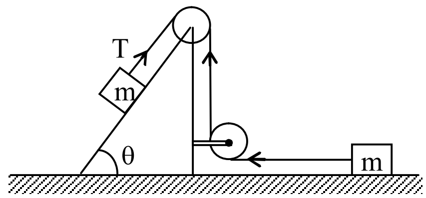
86. The masses of 10 kg and 20 kg respectively are connected by massless spring as shown in the figure. A force of 200 N acts on the 20kg mass. At the instant shown, the 10 kg mass has acceleration of 12 m/s^2. What is the acceleration in m/s^2 of 20 kg mass?



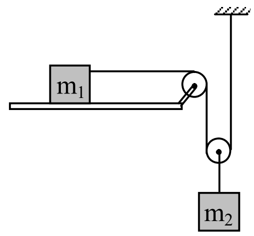
87. Two blocks each of mass *m* are resting on a frictionless inclined plane as shown in figure. Then, magnitude of acceleration in cm/s^2 of blocks: [$\sqrt{3} = 0.7, g = 10\, m/s^2$]



88. For the system shown in the figure, the pulleys are light and frictionless. The tension in N in the string will be: [$m = 4kg, \theta = 30°, g = 10\,ms^2$]

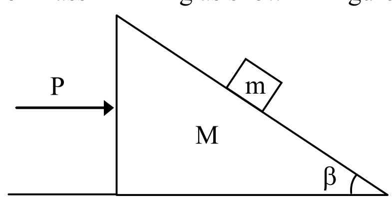


89. If the surface is smooth, the acceleration in m/s^2 of the block m_2 will be: [$m_1 = 1kg,\ m_2 = 4kg$]

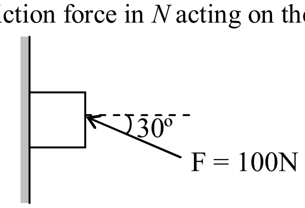


90. A man goes up in a uniformly accelerating lift. He returns downward with the lift accelerating at the same rate. The ratio of apparent weighs in the two cases is 3 : 1. The acceleration in m/s^2 of the lift is:

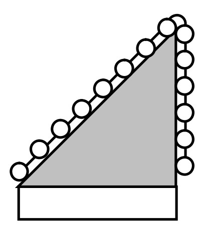
91. Two wooden blocks are moving on a smooth horizontal surface such that the mass m = 1 kg remains stationary with respect to block of mass M = 4 kg as shown in figure. The magnitude of force P in N is:



92. A force of 100 N is applied on a block of mass 3 kg as shown in figure. The coefficient of friction between the surface and block is 1/4. The friction force in N acting on the block is:



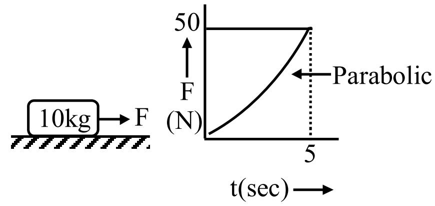
93. Sixteen beads in a string are placed on a smooth inclined plane of inclination $\sin^{-1}$ (1/3) such that some of them lie along the incline whereas the rest hang over the top of the plane. If acceleration at first bead is g/2, the arrangement of beads is that:



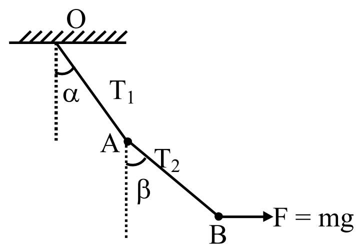
94. An empty plastic box of mass m = 2 kg is found to accelerate up at the rate of g/6 when placed deep inside water. How much sand in gms should be put inside the box so that it may accelerate down at the rate of g/6?

95. A spring toy of weight 1 kg rests on a weighing machine. The toy suddenly jumps and the balance reads 11 N. The acceleration in m/s^2 of the toy just on jumping up is(g = 10 m/s^2):

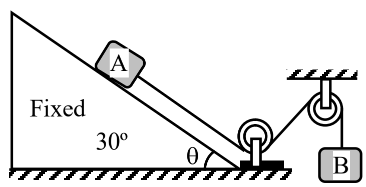
96 A force F is applied to the initially stationary cart. The variation of force with time is shown in the figure. The greatest integer value of speed in m/s of cart at t = 5 sec is:



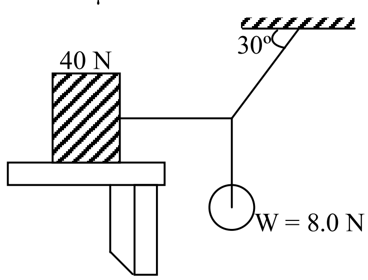
97. Two particles A and B, each of mass m, are kept stationary by applying a horizontal force F = mg on particle B as shown in figure. Then the ratio $10\ T_1^2/T_2^2$



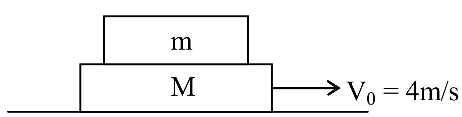
98. Two blocks A and B of equal mass m = 2kg are connected through a massless string and arranged as shown in figure. Friction is absent everywhere. When the system is released from rest, tension (in *N*) in the string is ($g = 10\ m/s^2$)



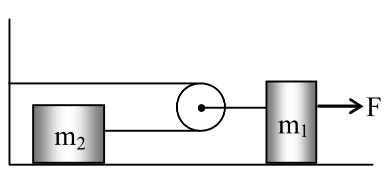
99. The system shown in figure is just on the verge of slipping. The coefficient of static friction between the block and table top is μ, the value of 100μ is:



100. A stationary body of mass m is slowly lowered onto a massive platform of mass M (M>>m) moving at a speed V_0 = 4 m/s as shown in fig. How far (in meter) will the body slide along the platform? (μ = 0.2 and g = 10 m/s^2)



101. In the given figure, calculate the acceleration (in m/s^2) of the block m_2, if the coefficient of friction between the blocks and the surface is 0.2. Take m_1 = 2 kg, m_2 = 6 kg, F = 93 N, g = 10 m/s^2, the pulley is smooth and weightless.



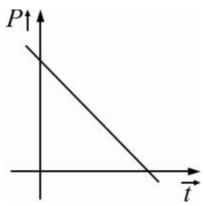
Energy & Momentum

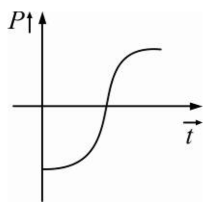
CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

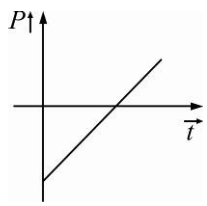
1. The density of a non-uniform rod of length 1 m is given by $\rho(x) = \alpha\left(1 + bx^2\right)$ where a and b are constants $o \le x \le 1$. The centre of mass of the rod will be at :

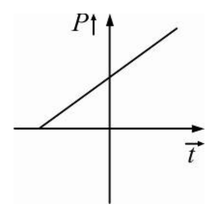
(A) $\dfrac{3(2+b)}{4(3+b)}$ **(B)** $\dfrac{4(2+b)}{3(3+b)}$ **(C)** $\dfrac{3(3+b)}{4(2+b)}$ **(D)** $\dfrac{4(3+b)}{3(2+b)}$

2. A particle is projected with speed u in air at an angle θ with the horizontal. The graph showing the variation of instantaneous power due to gravity P with time t will be :

(A) 

(B) 

(C) 

(D) 

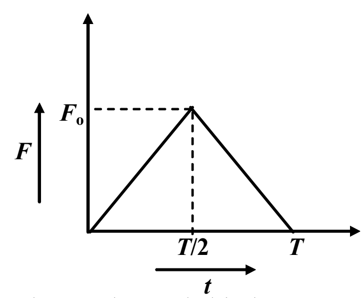
3. A body of mass M moving with a speed u has a 'head on' collision with a body of mass m initially at rest. If $M >> m$, the speed of the body of mass m after collision, will be nearly.

(A) um/M **(B)** uM/m **(C)** $u/2$ **(D)** $2u$

4. A massive ball moving with speed v collides with a stationary tiny ball having a mass very much smaller than the mass of the first ball. The collision is elastic. Immediately after the impact, the second ball will move with a speed approximately equal to :

(A) v **(B)** $2v$ **(C)** $v/2$ **(D)** ∞

5. A particle of mass m moving with velocity u makes an elastic one-dimensional collision with a stationary particle of mass m. They are in contact for a very brief time T. Their force of interaction increases from zero to F_o linearly in time $T/2$, and decreases linearly to zero in further time $T/2$. The magnitude of F_o is :

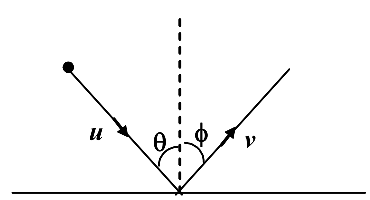


(A) mu/T **(B)** $2mu/T$
(C) $mu/2T$ **(D)** None of these

6. Two identical spheres move in opposite directions with the speeds v_1 and v_2 and pass behind an opaque screen, where they may either cross without touching (Event 1) or make an elastic head-on collision (Event 2) :

(A) We can never make out which event has occurred
(B) We cannot make out which event has occurred only if $v_1 = v_2$
(C) We can always make out which event has occurred
(D) We can make out which event has occurred only $v_1 = v_2$

7. A particle strikes a horizontal frictionless floor with a speed u, at an angle θ with the vertical, and rebounds with a speed v, at an angle ϕ with the vertical. The coefficient of restitution between the particle and the floor is e. The magnitude of v is :

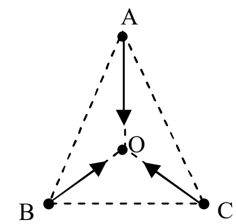


(A) eu **(B)** $(1-e)u$
(C) $u\sqrt{sin^2\theta + e^2 cos^2\theta}$ **(D)** $u\sqrt{e^2 sin^2\theta + cos^2\theta}$

8. The first ball of mass m moving with velocity u collides head on with the second ball of mass m at rest. If the coefficient of restitution is e, then the ratio of the velocities of the first and the second ball after the collision is :

(A) $\frac{1-e}{1+e}$ **(B)** $\frac{1+e}{1-e}$ **(C)** $\frac{1+e}{2}$ **(D)** $\frac{1-e}{2}$

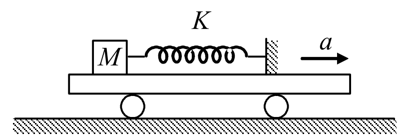
9. Three particles A, B and C of equal mass move with equal speed v along the medians of an equilateral triangle. They collide at the centroid O of the triangle. After collision A comes to rest while B retraces its path will speed v. The velocity of C is then :



(A) v, direction $\overrightarrow{OA}$ **(B)** $2v$, direction $\overrightarrow{OA}$

(C) $2v$, direction $\overrightarrow{OB}$ **(D)** v direction $\overrightarrow{BO}$

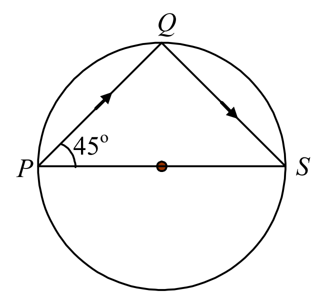
10. A block of mass M is attached with a spring of spring constant K. The whole arrangement is placed on a vehicle as shown in the figure. If the vehicle starts moving towards right with an acceleration a (there is no friction anywhere), then :



(A) Maximum elongation in the spring is $\frac{Ma}{K}$ **(B)** Maximum elongation in the spring is $\frac{2Ma}{K}$

(C) Maximum compression in the spring is $\frac{2ma}{K}$ **(D)** None of these

11. A body is fired from point P and strikes at Q inside a smooth circular wall as shown in the figure. It rebounds to point S (diametrically opposite to P), then :



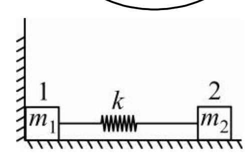
(A) The coefficient of restitution is zero

(B) The coefficient of restitution is 1

(C) Kinetic energy is not conserved

(D) The coefficient of restitution is $\frac{1}{\sqrt{3}}$

12. Two bars connected by a weightless spring of stiffness k rest on a smooth horizontal plane as shown in figure. Bar 2 is shifted a small distance x to the left and then released. The velocity of the centre of inertia of the system after bar 1 breaks off the wall is :



(A) $\frac{x\sqrt{m_2 k}}{m_1+m_2}$ **(B)** $\frac{kx}{m_1+m_2}$ **(C)** zero **(D)** $\frac{\sqrt{m_1 k}\,x}{m_1+m_2}$

13. Two equal masses are attached to the two ends of a spring of spring constant k. The masses are pulled out symmetrically to stretch the spring by a length x over its natural length. The work done by the spring on each mass is :

(A) $\frac{1}{2}kx^2$ **(B)** $-\frac{1}{2}kx^2$ **(C)** $\frac{1}{4}kx^2$ **(D)** $-\frac{1}{4}kx^2$

14. Two particles of masses m and $2m$ start moving from origin with same constant speed along x and y axis respectively. The center of mass of system of particles will move along the line :

(A) $3y = x$ **(B)** $y = 3x$ **(C)** $y = 2x$ **(D)** $y = 2x/3$

15. A particle of mass m moving with velocity v_o collides elastically with another stationary particle of same mass m. The velocity of the first particle as observed from the reference frame of center of mass of the two particle system :

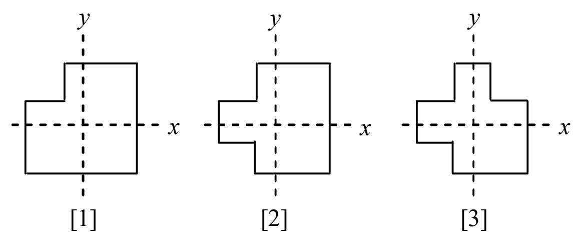
(A) Remains constant

(B) Changes by v_o in the direction of motion of center of mass

(C) Changes by v_o in opposite direction to the motion of center of mass

(D) Changes by $v_o/2$ in the direction of motion of center of mass.

16. A machinist starts with three identical square places but cuts one corner form one of them, two corners from the second, and three corners from the third. Rank the three according to the x-coordinate of their centre of mass, from smallest to largest:



(A) 3, 1, 2 **(B)** 1, 3, 2

(C) 3, 2, 1 **(D)** 1 and 3 tie, then 2

17. A smooth sphere is moving on horizontal surface with velocity vector $2\hat{i} + 2\hat{j}$ immediately before it hits a vertical wall. The wall is parallel to $\hat{j}$ vector and the coefficient of restitution between the sphere and the wall is $e = 1/2$. The velocity vector of the sphere after it hits the wall is:

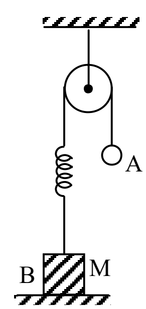
(A) $\hat{i} - \hat{j}$ **(B)** $-\hat{i} + 2\hat{j}$ **(C)** $-\hat{i} - \hat{j}$ **(D)** $2\hat{i} - \hat{j}$

18. A rocket of initial mass 5000 kg ejects gas at a constant rate of 60 kg/s with a relative speed of 2050 m/s. Acceleration of the rocket 15 second after it is blasted off from the surface of earth will be ($g = 10$ m/s^2)

(A) 10 m/s^2 **(B)** 20 m/s^2

(C) 30 m/s^2 **(D)** 40 m/s^2

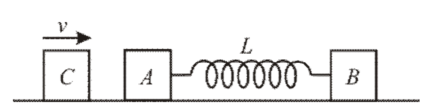
19. In the figure, the ball A is released from rest when the spring is at its natural (unscratched) length. For the block B of mass M to leave contact with the ground at some stage, the minimum mass of A must be :



(A) $2M$ **(B)** M

(C) $M/2$ **(D)** A function of M and the force constant of the spring

***20.** Two blocks A and B, each of mass m, are connected by a massless spring of natural length L and spring constant K. The blocks are initially resting on a smooth horizontal floor with the spring at its natural length, as shown in the figure. A third identical block C, also of mass m, moves on the floor with a speed v along the line joining A and B, and collides elastically with A. Then



(A) The kinetic energy of the AB system, at maximum compression of the spring, is zero

(B) The kinetic energy of the AB system, at maximum compression of the spring, is $mv^2/4$

(C) The maximum compression of the spring is $v\sqrt{(m/K)}$

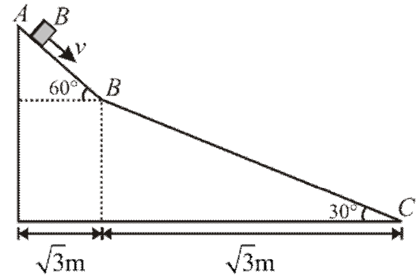
(D) The maximum compression of the spring is $v\sqrt{(m/2K)}$

21. A ball impinges directly upon another ball at rest and is itself brought to rest by the impact. If two-third of initial kinetic energy is lost in the collision, then the coefficient of restitution is :

(A) $\frac{1}{\sqrt{2}}$ **(B)** $\frac{1}{2}$ **(C)** $\frac{1}{3}$ **(D)** Zero

PARAGRAPH FOR QUESTIONS 22 - 23

A small block of mass M moves on a frictionless surface of an inclined plane, as shown in the figure. The angle of the incline suddenly changes from $60°$ to $30°$ at point B. The block is initially at rest at A. Assume that collisions between the block and the incline are totally inelastic.



22. The speed of the block at point B immediately after it strikes the second incline is :

(A) $\sqrt{60}$ *m/s* **(B)** $\sqrt{45}$ *m/s* **(C)** $\sqrt{30}$ *m/s* **(D)** $\sqrt{15}$ *m/s*

23. The speed of the block at point C immediately before it leaves the second incline is :

(A) $\sqrt{120}$ *m/s* **(B)** $\sqrt{105}$ *m/s* **(C)** $\sqrt{90}$ *m/s* **(D)** $\sqrt{75}$ *m/s*

24. If collision between the block and the incline is completely elastic, then the vertical (upward) component of the velocity of the block at point B, immediately after it strikes the second incline is :

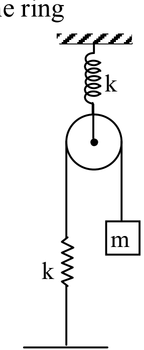
(A) $\sqrt{30}$ *m/s* **(B)** $\sqrt{15}$ *m/s* **(C)** 0 **(D)** $-\sqrt{15}$ *m/s*

25. A heavy ring of mass m is clamped on the periphery of a light circular disc. A small particle having equal mass is clamped at the centre of the disc. The system is rotated in such a way that the centre moves in a circle of radius r with a uniform speed v. We conclude that an external force.

(A) $\frac{mv^2}{r}$ must be acting on the central particle **(B)** $\frac{2mv^2}{r}$ must be acting on the central particle

(C) $\frac{2mv^2}{r}$ must be acting on the system **(D)** $\frac{2mv^2}{r}$ must be acting on the ring

26. A system is released from rest as shown in figure. Kinetic energy of mass m when its moves distance x in downward direction is (initially both spring are unstretched and all spring are massless)



(A) 10 mgx **(B)** $\frac{10mgx - kx^2}{10}$

(C) $mgx - kx^2$ **(D)** $mgx - \frac{kx^2}{2}$

27. In a reference frame K, two particles move along the x-axis, one of mass m_1 with velocity $\vec{v}_1$ and the other of mass m_2 with velocity $\vec{v}_2$. There exists another reference frame K' in which the total kinetic energy of the two masses m_1 and m_2 will be the minimum. The instantaneous velocity $\vec{v}$ of the reference frame K' to the reference frame K will be:

(A) $\vec{v} = \vec{v}_1 + \vec{v}_2$ **(B)** $\vec{v} = \vec{v}_1$ **(C)** $\vec{v} = \vec{v}_2$ **(D)** $v = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$

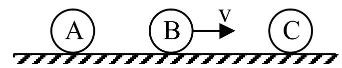
28. A railway flat car has an artillery gun installed on it. The combined system has a mass M and moves with a velocity V. The barrel of the gun makes an angle α with the horizontal. A shell of mass *m* leaves the barrel at a speed *v* relative to the barrel in the direction of flat car's motion. The speed of the flat car so that it may stop after the firing is:

(A) $\dfrac{mv}{M+m}$ **(B)** $\left(\dfrac{Mv}{M+m}\right)\cos a$ **(C)** $\left(\dfrac{m}{M}\right)v\cos\alpha$ **(D)** $(M+m)v\cos\alpha$

29. When a rubber band is stretched by a distance *x*, it exerts a restoring force of magnitude $F = ax + bx^2$, where *a* and *b* are constants. The work done in stretching the unstretched rubber band by *L* is :

(A) $aL^2 + bL^3$ **(B)** $\dfrac{1}{2}\left(aL^2 + bL^3\right)$ **(C)** $\dfrac{aL^2}{2} + \dfrac{bL^3}{3}$ **(D)** $\dfrac{1}{2}\left(\dfrac{aL^2}{2} + \dfrac{bL^3}{3}\right)$

30. Three balls *A*, *B*, and *C* ($m_A = m_C = 4m_B$) are placed on a smooth horizontal surface. Ball B collides with ball C with an initial velocity *v* as shown. Total number of collisions between the balls will be: (all collisions are elastic).

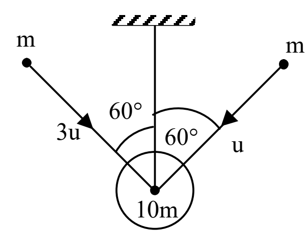


(A) One **(B)** Two **(C)** Three **(D)** Four

***31.** A cannon shell is fired to hit a target at a horizontal distance *R*. However, it breaks into two equal parts at its highest point. One part (*A*) return to the cannon. The other part.

(A) Will fall at a distance of *R* beyond the target **(B)** Will fall at a distance of 3R beyond the target
(C) Will hit the target **(D)** Have nine times the kinetic energy of *A*

32. A bob of mass 10*m* is suspended through an inextensible string of length *l*. When the bob is at rest in equilibrium position, two particles each of mass *m* strike it as shown. The particles stick after collision. Choose the correct statement from the following:



(A) Impulse in the string due to tension is $2mu$

(B) Velocity of the system just after collision is $v = \dfrac{u\sqrt{3}}{14}$

(C) Loss of energy is $\dfrac{137}{28}mu_2$ **(D)** Loss of energy is $\dfrac{137}{56}mu_2$

***33** In an elastic collision between spheres *A* and *B* of equal mass but unequal radii, *A* moves along the *x*-axis and *B* is stationary before impact. Which of the following is possible after impact?

(A) *A* comes to rest
(B) The velocity of *B* relative to *A* remains the same in magnitude but reverses in direction
(C) *A* and *B* move with equal speeds, making an angle of $45°$ each with the *x*-axis
(D) *A* and *B* move with unequal speeds, making angles of $30°$ and $60°$ with the *x*-axis respectively

34. This question has statement I and Statement II. Of the four choices given after the statements, choose the one that best describes the two statements.

Statement I A point particle of mass *m* moving with speed *v* collides with stationary point particle of mass *M*. If the maximum energy loss possible is given as $f\left(\dfrac{1}{2}mv^2\right)$, then $f = \left(\dfrac{m}{M+m}\right)$.

Statement II Maximum energy loss occurs when the particles get stuck together as a result of the collision.

(A) Statement I is true, Statement II is true; Statement II is the correct explanation of Statement I
(B) Statement I is true, Statement II is true; Statement II is not the correct explanation of Statement I
(C) Statement I is true, Statement II is false
(D) Statement I is false, Statement II is true

35. After a totally inelastic collision, two objects of the same mass and same initial speeds are found to move together at half of their initial speeds. The angle between the initial-velocities of the objects is :

(A) 120° **(B)** 60° **(C)** 150° **(D)** 45°

36. A uniform chain of length ℓ and mass m is placed on a smooth table with one-fourth of its length hanging over the edge. The work that has to be done to pull the whole chain back onto the table is:

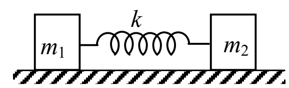
(A) $\frac{1}{4}mg\ell$ **(B)** $\frac{1}{8}mg\ell$ **(C)** $\frac{1}{16}mg\ell$ **(D)** $\frac{1}{32}mg\ell$

37. Consider two observers moving with respect to each other at a speed v along a straight line. They observe a block of mass m moving a distance l on a rough surface. The following quantities will be same as observed by the two observers.

(A) Kinetic energy of the block at time t **(B)** Work done by friction

(C) Total work done on the block **(D)** Acceleration of the block

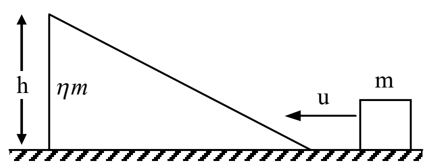
38. Two blocks m_1 and m_2 are pulled on a smooth horizontal surface, and are joined together with a spring of stiffness k as shown. Suddenly, block m_2 receives a horizontal velocity v, then the maximum extension x_m in the spring is :



(A) $v_0\sqrt{\frac{m_1m_2}{m_1+m_2}}$ **(B)** $v_0\sqrt{\frac{2m_1m_2}{(m_1+m_2)k}}$

(C) $v_0\sqrt{\frac{m_1m_2}{2(m_1+m_2)k}}$ **(D)** $v_0\sqrt{\frac{m_1m_2}{(m_1+m_2)k}}$

39. A block of mass m is pushed towards a movable wedge of mass ηm and height h, with a velocity u. All surfaces are smooth. The minimum value of u for which the block will reach the top of the wedge is :



(A) $\sqrt{2gh}$ **(B)** $\eta\sqrt{2gh}$

(C) $\sqrt{2gh(1+1/\eta)}$ **(D)** $\sqrt{2gh(1-1/\eta)}$

40. A small block of mass m is kept on a rough inclined surface of inclination θ fixed in an elevator. The elevator goes up with a uniform velocity v and the block does not slide on the wedge. The work done by the force of friction on the block in time t will be:

(A) Zero **(B)** $mgvt\ cos^2\theta$ **(C)** $mgvt\ sin^2\theta$ **(D)** $mgvt\ sin2\theta$

41. Two particles A and B, move with constant velocities $\vec{v}_1$ and $\vec{v}_2$. At the initial moment their position vectors are $\vec{r}_1$ and $\vec{r}_2$ respectively. The condition for particles A and B for their collision is :

(A) $\vec{r}_1\times\vec{v}_1=\vec{r}_2\times\vec{v}_2$ **(B)** $\vec{r}_1-\vec{r}_2=\vec{v}_1-\vec{v}_2$

(C) $\frac{\vec{r}_1-\vec{r}_2}{|\vec{r}_1-\vec{r}_2|}=\frac{\vec{v}_1-\vec{v}_2}{|\vec{v}_1-\vec{v}_2|}$ **(D)** $\vec{r}_1\cdot\vec{v}_1=\vec{r}_2\cdot\vec{v}_2$

42. The heart of a man pumps 5 litres of blood though the arteries per minute at a pressure of $150\,mm$ of mercury be $13.6\times10^3\,kg/m^3$ and $g=10\,m/s^2$ then the power :

(A) 3.0 **(B)** 1.50 **(C)** 1.70 **(D)** 2.35

43. On a frictionless surface, a block of mass M moving at speed at v collides elastically with another block of same mass M is initially at rest After collision the first block moves at an angle θ to its initial direction and has a speed $v/3$. The second blocks speed after the collision is :

(A) $\frac{3}{\sqrt{2}}v$ **(B)** $\frac{\sqrt{3}}{2}v$ **(C)** $\frac{2\sqrt{2}}{3}v$ **(D)** $\frac{3}{4}v$

44. A particle of mass is driven by a machine that delivers a constant power k watts. If the particle starts from rest the force on the particle at time t is :

(A) $\sqrt{2mk}\,t^{-1/2}$ **(B)** $\frac{1}{2}\sqrt{mk}\,t^{-1/2}$ **(C)** $\sqrt{\frac{mk}{2}}\,t^{-1/2}$ **(D)** $\sqrt{mk}\,t^{-1/2}$

45. A body of mass $(4m)$ is lying in x-y plane at rest. It has suddenly explodes into three pieces. Two pieces, each of mass (m) move perpendicular to each other with equal speeds (v). The total kinetic energy generated due to explosion is :

(A) mv^2 **(B)** $\frac{3}{2}mv^2$ **(C)** $2mv^2$ **(D)** $4mv^2$

46. A uniform force of $(3\hat{i}+\hat{j})$ newtons acts on a particle of mass $2\,kg$. Hence the particle is displaced from position $(2\hat{i}+\hat{k})$ meter to position $(4\hat{i}+3\hat{j}-\hat{k})$ meter. The work done by the force on the particle is :

(A) $13\,J$ **(B)** $15\,J$ **(C)** $9\,J$ **(D)** $6\,J$

47. The potential energy of a particle in a force field is $u=\frac{A}{r^2}-\frac{B}{r}$ where A and B are positive constants and r is the distance of particle from the centre of the field. For stable equilibrium, the distance of the particle is :

(A) $\frac{B}{2A}$ **(B)** $\frac{2A}{B}$ **(C)** $\frac{A}{B}$ **(D)** $\frac{B}{A}$

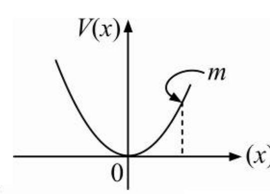
48. A car of mass m starts from rest and accelerates so that the instantaneous power delivered to the car has a constant magnitude P_0. The instantaneous velocity of this car is proportional to :

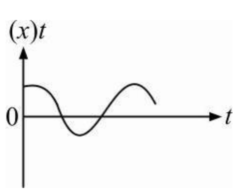
(A) t^2P_0 **(B)** $t^{1/2}$ **(C)** $t^{-1/2}$ **(D)** $\frac{t}{\sqrt{m}}$

49. A body projected vertically from the earth reaches a height equal to earth's radius before returning to the earth. The power exerted by the gravitational force is greatest :

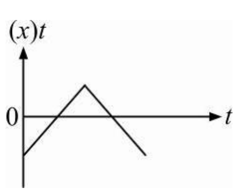
(A) at the highest position of the body **(B)** at the instant just before the body hits the earth
(C) it remains constant all through **(D)** at the instant just after the body is projected

50. A particle of mass m is released from rest and follows a parabolic path as shown. Assuming that the displacement of the mass from the origin is small, which graph correctly depicts the position of the particle as a function of time ?

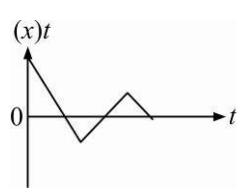




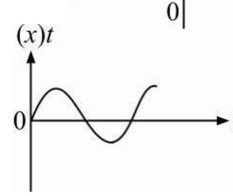
(A)



(B)



(C)



(D)

51. An engine pumps water though the pipe. Water passes through the pipe and leaves it with a velocity of $2\,m/s$. The mass per unit length of water in the pipe is $100\,kg/m$. What is the power of the engine ?

(A) 400 W **(B)** 200 W **(C)** 100 W **(D)** 800 W

52. An engine pumps water continuously through a hose. Water leaves the hose with a velocity v and m is the mass per unit length of the water jet. What is the rate at which kinetic energy is imparted to water ?

(A) mv^3 **(B)** $\frac{1}{2}mv^2$ **(C)** $\frac{1}{2}m^2v^2$ **(D)** $\frac{1}{2}mv^3$

53. Water falls from a height of 60 m at the rate of $15\,kg/s$ to operate a turbine. The losses due to frictional forces are 10% of energy. How much power is generated by the turbine? $(g = 10\,m/s^2)$

(A) 12.3 kW **(B)** 7.0 kW **(C)** 8.1 kW **(D)** 10.2 kW

54. A vertical spring with force constant k is fixed on a table. A ball of mass m at a height h above the free upper end of the spring falls vertically on the spring so that the spring is compressed by a distance d. The net work done in the process is :

(A) $mg(h+d)-\frac{1}{2}kd^2$ **(B)** $mg(h-d)-\frac{1}{2}kd^2$

(C) $mg(h-d)+\frac{1}{2}kd^2$ **(D)** $mg(h+d)+\frac{1}{2}kd^2$

55. A body of mass $3\,kg$ is under a constant force which causes a displacements s in metres in it, given by the relation $s = \frac{1}{3}t^2$, where t is in seconds. Work done by the force in 2 seconds is :

(A) $\frac{19}{5}J$ **(B)** $\frac{5}{19}J$ **(C)** $\frac{3}{8}J$ **(D)** $\frac{8}{3}J$

56. If $\vec{F} = \left(60\hat{i}+15\hat{j}-3\hat{k}\right)N$ and $\vec{v} = \left(2\hat{i}-4\hat{j}+5\hat{k}\right)m/s$, the instantaneous power is :

(A) 195 watt **(B)** 45 watt **(C)** 75 watt **(D)** 100 watt

57. A mass of 1 kg is thrown up with a velocity of $100\,m/s$. After 5 seconds, it explodes into two parts. One part of mass 400 g comes down with a velocity $25\,m/s$. The velocity of the other part is : (Take $g = 10\,ms^{-2}$)

(A) $40\,m/s\uparrow$ **(B)** $40\,m/s\downarrow$ **(C)** $100\,m/s\uparrow$ **(D)** $60\,m/s\uparrow$

58. A force acts on a $3\,g$ particle in such a way that the position of the particle as a function of time is given by $x = 3t - 4t^2 + t^3$, where x is in meters and t is in seconds. The work done during the first 4 second is :

(A) 490 mJ **(B)** 450 mJ **(C)** 576 mJ **(D)** 528 mJ

59. A metal ball of mass $2\,kg$ moving with speed of 36 km/h has a head on collision with a stationary ball of mass 3 kg. If after collision, both the balls move as a single mass, then the loss in K.E. due to collision is :

(A) $100\,J$ **(B)** $140\,J$ **(C)** $40\,J$ **(D)** $60\,J$

60. How much water a pump of 2 kW can raise in one minute to a height of $10\,m$? (Take $g = 10\,m/s^2$)

(A) 1000 litres **(B)** 1200 litres **(C)** 100 litres **(D)** 2000 litres

61. Two identical particles move towards each other with velocity 2v and v, respectively. The velocity of the centre of mass is :

(A) v **(B)** $v/3$ **(C)** $v/2$ **(D)** zero

62. A bomb of mass 9 *kg* at rest explodes into 2 pieces of mass 3 *kg* and 6 *kg*. The velocity of mass 3 *kg* is 1.6 *m*/*s*, the *K.E.* of mass 6 *kg* is :

(A) 3.84 *J* **(B)** 9.6 *J* **(C)** 1.92 *J* **(D)** 2.92 *J*

63. Consider the following two statements.

I. The linear momentum of a system of particles is zero.

II. The kinetic energy of a system of particles is zero.

Then :

(A) I implies II and II implies I. **(B)** I does not imply II and II does not imply I

(C) I implies II but II does not imply I **(D)** I does not imply II but II implies I

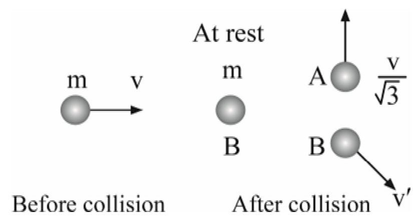
64. A ^{238}U nucleus decays by emitting an alpha particle of speed $v\, ms^{-1}$. The recoil speed of the residual nucleus is: $(in\ ms^{-1})$

(A) $-4v/234$ **(B)** $v/4$ **(C)** $-4v/238$ **(D)** $4v/238$

65. A uniform chain of length 2 *m* is kept on a table such that a length of 60 *cm* hangs freely from the edge of the table. The total mass of the chain in 4 *kg*. What is the work done in pulling the entire chain on the table?

(A) 7.2 *J* **(B)** 3.6 *J* **(C)** 120 *J* **(D)** 1200 *J*

66. A mass *m* moves with a velocity *v* and collides inclastically with another identical mass. After collision, the first mass moves with velocity $v/\sqrt{3}$ in a direction perpendicular to the initial direction of motion. Find the speed of the second mass after collision.



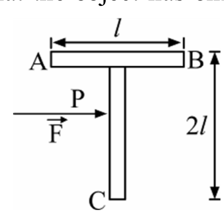
(A) $\frac{v}{\sqrt{3}}$ **(B)** $2\frac{v}{\sqrt{3}}$ **(C)** $\sqrt{3}\,v$ **(D)** v

67. A body *A* of mass *M*, while falling vertically downward under gravity, breaks into two parts – a body *B* of mass *M*/3 and a body *C* of mass 2*M*/3. The centre of mass of *B* and *C* taken together shifts compared to that of body *A* towards.

(A) body *B* **(B)** body *C*

(C) does not shift **(D)** depends on the height of breaking

68. A *T*-shaped object made from two rods of same material and area with length shown in the figure is lying on a smooth floor. A force $\vec{F}$ is applied at the point *P* parallel to *AB* such that the object has only translational motion without rotation. Find the location of *P* with respect to *C*.



(A) l **(B)** $\left(\frac{4}{3}\right)l$

(C) $\left(\frac{3}{2}\right)l$ **(D)** $\left(\frac{2}{3}\right)l$

69. A bomb of mass 16 *kg* at rest explodes into two pieces of masses 4 *kg* and 12 *kg*. The velocity of the 12 *kg* mass is 4 *m*/*s*. The kinetic energy of the other mass is :

(A) 192 *J* **(B)** 96 *J* **(C)** 144 *J* **(D)** 288 *J*

70. Consider a two-particle system with particles having masses m_1 and m_2. If the first particle is pushed towards the centre of mass through a distance d, by what distance should the second particle be moved, so as to keep the centre of mass at the same position?

(A) $\frac{m_1}{m_2}d$ **(B)** d **(C)** $\frac{m_2}{m_1}d$ **(D)** $\frac{m_1}{m_1+m_2}d$

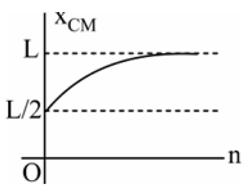
71. A circular disc of radius R is removed from a bigger circular disc of radius $2R$ such that the circumferences of the discs coincide. The centre of mass of the new disc is αR from the centre of the bigger disc. The value of α is:

(A) $\frac{1}{2}$ **(B)** $\frac{1}{6}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{1}{3}$

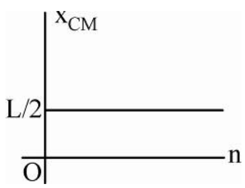
72. A block of mass 0.50 kg is moving with a speed of 2.00 m/s on a smooth surface. It strikes another mass of 1.00 kg and then they move together as a single body. The energy loss during the collision is :

(A) 0.16 J **(B)** 1.00 J **(C)** 0.67 J **(D)** 0.34 J

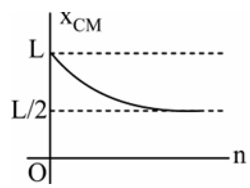
73. A thin rod of length L is lying along the x-axis with its ends at $x = 0$ and $x = L$. Its linear density (mass/length) varies with x as $k\,(x/L)^n$, where n can be zero or any positive number. If the position x_{CM} of the centre of mass of the rod is plotted against n, which of the following graphs best approximates the dependence of x_{CM} on n?



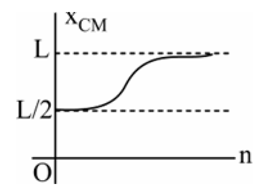
(A)



(B)



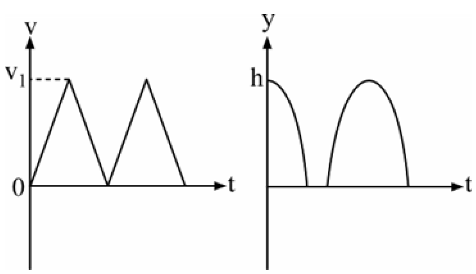
(C)



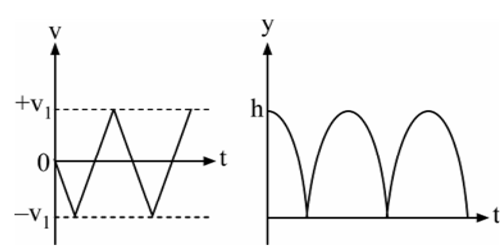
(D)

74. Consider a rubber ball freely falling from a height $h = 4.9\ m$ on a horizontal elastic plate. Assume that the duration of collision is negligible and the collision with the plate is totally elastic. Then the velocity as a function of time and the height as a function of time will be :

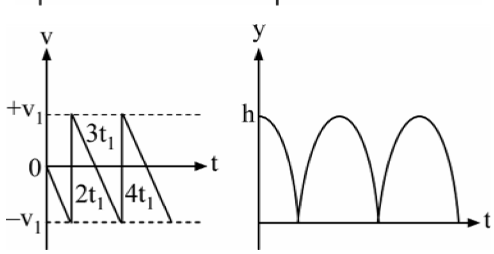
(A)



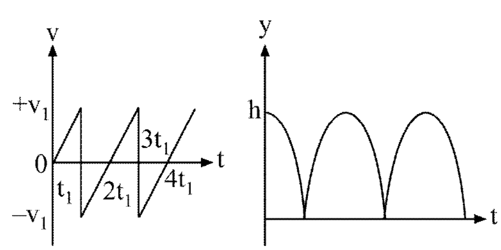
(B)



(C)



(D)

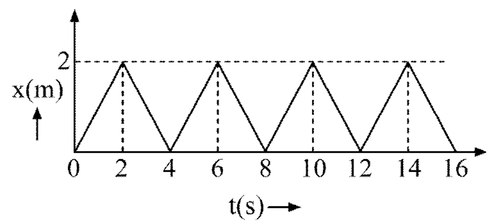


75. **Statement 1 :** Two particles moving in the same direction do not lose all their energy in a completely inelastic collision.

Statement 2 : The principle of conservation of momentum holds true for all kinds of collisions.

(A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

76. The figure shows the position – time ($x - t$) graph of one – dimensional motion of a body of mass 0.4 *kg*. The magnitude of each impulse is :



(A) 0.4 *Ns* **(B)** 0.8 *Ns*
(C) 1.6 *Ns* **(D)** 0.2 *Ns*

77. This questions has Statement I and Statement II. Of the four choices given after the Statements, choose the one that best describes the two statements

Statement 1 : A point particle of mass m moving with speed v collides with stationary point particle of mass M. If the maximum energy loss possible is given as

$$F = \left(\frac{1}{2}mfv^2\right) \text{ then } f = \left(\frac{m}{M+m}\right)$$

Statement 2 : Maximum energy loss occurs when the particles get stuck together as a result of the collision.

(A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

78. A particle of mass m moving in the x direction with speed $2v$ is hit by particle of mass $2m$ moving in the y direction with speed v. If the collision is perfectly inelastic, the percentage loss in the energy during the collision is close to :

(A) 44 % **(B)** 50 % **(C)** 56 % **(D)** 62 %

79. Distance of the centre of mass of a solid uniform cone from its vertex is z_0. If the radius of its base is R and its height is h then z_0 is equal to :

(A) $\frac{h^2}{4R}$ **(B)** $\frac{3h}{4}$ **(C)** $\frac{5h}{8}$ **(D)** $\frac{3h^2}{8R}$

80. From a building, two balls A and B are thrown such that A is thrown upward and B downward (both vertically) with same speed. If v_A and v_B are their respective velocities on reaching the ground, then :

(A) $v_B > v_A$ **(B)** $v_A = v_B$
(C) $v_A > v_B$ **(D)** their velocities depend on their masses

81. The speeds of two identical cars are u and $4u$ at a specific instant. The ratio of the respective distance in which the two cars are stopped from that instant is :

(A) 1 : 1 **(B)** 1 : 4 **(C)** 1 : 8 **(D)** 1 : 16

82. A spring of force constant 800 *N/m* has an extension of 5 *cm*. The work done in extending it from 5 *cm* to 15 *cm* is :

(A) 16 *J* **(B)** 8 *J* **(C)** 32 *J* **(D)** 24 *J*

83. Two masses of 1 *kg* and 16 *kg* are moving with equal kinetic energy. The ratio of magnitude of the linear momentum is :

(A) 1 : 2 **(B)** 1 : 4 **(C)** $1:\sqrt{2}$ **(D)** $\sqrt{2}:1$

84. A car moving with a speed of 50 *km/h* can be stopped by brakes after at least 6 *m*. If the same car is moving at a speed of 100 *km/h*, the minimum stopping distance is :

(A) 6 *m* **(B)** 2 *m* **(C)** 18 *m* **(D)** 24 *m*

85. A body is moved along a straight line by a machine delivering a constant power. The distance moved by the body in time *t* is proportional to :

(A) $t^{1/2}$ **(B)** $t^{3/4}$ **(C)** $t^{3/2}$ **(D)** $t^{1/4}$

86. A spring of constant 5×10^3 *N/m* is stretched initially by 5 *cm* from the unstreched position. Then the work required to stretch it further by another 5 *cm* is :

(A) 6.25 *N-m* **(B)** 12.50 *N-m* **(C)** 18.75 *N-m* **(D)** 25.00 *N-m*

87. The potential energy function for the force between two atoms in a diatomic molecule is approximately given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$, where *a* and *b* are constant and *x* is the distance between the atoms. If the dissociation energy of the molecules is $D = [U_{(x=\infty)} - U_{at\ equilibrium}]$, *D* is :

(A) $\frac{b^2}{2a}$ **(B)** $\frac{b^2}{12a}$ **(C)** $\frac{b^2}{4a}$ **(D)** $\frac{b^2}{6a}$

88. A particle moves in a straight line with retardation proportional to its displacement. Its loss of kinetic energy for any displacement *x* is proportional to :

(A) x **(B)** e^x **(C)** x^2 **(D)** $\log_e x$

89. A force $\vec{F}(5\vec{i} + 3\vec{j} + 2\vec{k})\ N$ is applied over a particle which displaces it from its origin to the point $\vec{r} = (2\vec{i} - \vec{j})m$ The work done on the particle (in *J*) is :

(A) + 10 **(B)** + 7 **(C)** –7 **(D)** + 13

90. A body of mass *m* accelerates uniformly from rest to v_1 in time t_1. The instantaneous power delivered to the body as a function of time t is :

(A) $\frac{mv_1t^2}{t_1}$ **(B)** $\frac{mv_1^2t}{t_1^2}$ **(C)** $\frac{mv_1t}{t_1}$ **(D)** $\frac{mv_1^2t}{t_1}$

91. A particle is acted upon by a force of constant magnitude which is always perpendicular to the velocity of the particle; the motion of the particle takes place in a plane. It follow that :

(A) its kinetic energy is constant **(B)** its acceleration is constant

(C) its velocity is constant **(D)** it moves in straight line

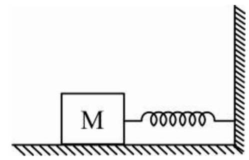
92. This question has Statement-1 and Statement-II. Of the four choices given after the statements, choose the one that best describes the two statements.

If two springs S_1 and S_2 of force constant k_1 and k_2, respectively, are stretched by the same force, it is found that more work is done on spring S_1 than on spring S_2.

Statement-I: If stretched by the same amount, work done on S_1, will be more that on S_2.
Statement-II: $k_1 < k_2$

(A) Statement-I is false, Statement-II is true
(B) Statement-I is true, Statement-I false
(C) Statement-I is true, Statement-II is true, Statement-II is the correct explanation for Statement-I.
(D) Statement-I is true, Statement-II is true, Statement-II is not the correct explanation of Statement-I.

93. The block of mass M moving on a frictionless horizontal surface collides with a spring of spring constant K and compresses it by length L. The maximum momentum of the block after collision is :



(A) $\frac{ML^2}{K}$ **(B)** Zero
(C) $\frac{KL^2}{2M}$ **(D)** $\sqrt{MK}\,L$

94. A particle of mass 0.3 *kg* in a straight line motion along x-axis is subjected to a force $F = -kx$ with $k = 15$ *N/m*. What will be its initial acceleration if it is released from a point 20 *cm* away from the origin ?

(A) 10 m/s^2 **(B)** 5 m/s^2 **(C)** 15 m/s^2 **(D)** 3 m/s^2

95. A spherical ball of mass 20 *kg* is stationary at the top of a hill of height 100 *m*. It rolls down a smooth surface to the ground, then climbs up another hill of height 30 *m* and finally rolls down to a horizontal base at a height of 20 *m* above the ground. The velocity attained by the ball is :

(A) $10\sqrt{30}\ m/s$ **(B)** 10 *m/s* **(C)** 20 *m/s* **(D)** 40 *m/s*

96. When a rubber-band is stretched by a distance x, it exerts a restoring force of magnitude $F = ax + bx^2$, where a and b are constants. The work done in stretching the unstretched rubber band by L is :

(A) $\frac{aL^2}{2} + \frac{bL^3}{3}$ **(B)** $\frac{1}{2}\left(\frac{aL^2}{2} + \frac{bL^3}{3}\right)$
(C) $aL^2 + bL^3$ **(D)** $\frac{1}{2}(aL^2 + bL^3)$

97. A particle of mass 100 *g is* thrown vertically upward with a speed of 5 *m/s*. The work done by the force of gravity during the time the particle goes up is :

(A) 1.25 *J* **(B)** 0.5 *J* **(C)** −0.5 *J* **(D)** −1.25 *J*

98. The potential energy of a 1 *kg* particle free to move along the x-axis is given by $V(x) = [(x^4/4) - (x^2/2)]\,J$. The total mechanical energy of the particle is 2 *J*. Then the maximum speed (in *m/s*) is :

(A) $\frac{1}{\sqrt{2}}$ **(B)** 2 **(C)** $\frac{3}{\sqrt{2}}$ **(D)** $\sqrt{2}$

99. A mass of *M kg* is suspended by a weightless string. The horizontal force required to displace it such that string makes an angle of 45° with the initial vertical direction is :

(A) $\frac{Mg}{\sqrt{2}}$ **(B)** $Mg\left(\sqrt{2}-1\right)$ **(C)** $Mg\left(\sqrt{2}+1\right)$ **(D)** $Mg\sqrt{2}$

100. A ball of mass 0.2 *kg* is thrown vertically upwards by applying a force by hand. If the hand moves 0.2 *m* while applying the force and the ball goes upto 2 *m* height further, find the magnitude of the force. Consider $g = 10\ m/s^2$.

(A) 20 *N* **(B)** 22 *N* **(C)** 4 *N* **(D)** 16 *N*

101. A 2 *kg* block slides on a horizontal floor with a speed of 4 *m/s*. It strikes an uncompressed spring, and compresses it till the block is motionless. The kinetic friction force is 15 *N* and spring constant is 10000 *N/m*. The spring compress by (in *cm*)

(A) 2.5 **(B)** 11.0 **(C)** 8.5 **(D)** 5.5

102. A particle is projected at 60° to the horizontal with a kinetic energy *K*. The kinetic energy at the highest point is :

(A) zero **(B)** $\frac{K}{4}$ **(C)** $\frac{K}{2}$ **(D)** K

103. An athlete in the Olympic games covers a distance of 100 *m* in 10 *s*. His kinetic energy can be estimated to be in the range.

(A) 200-500 J **(B)** $2\times10^5-3\times10^5\ J$

(C) $20000-50000\,J$ **(D)** $2000-5000\,J$

INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

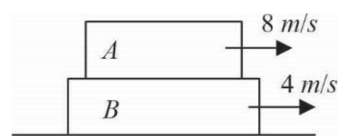
104. A stone of mass $10\ kg$ is laying at the bed of a lake $5m$ deep. The relative density of the stone is 2. Find the amount of work done to bring the stone to the top of the lake.

105. A rifle bullet looses ${}^{1}\!/_{20}th$ of its velocity in passing through a plank. Find the least number of planks required to just stop the bullet.

106. An electric pump is used to fill an overhead tank of capacity $9m^3$ kept at a height of $10m$ above the ground. If the pump takes 5 minutes to fill the tank by consuming $10kW$ power, find the efficiency of the pump (in percentage).

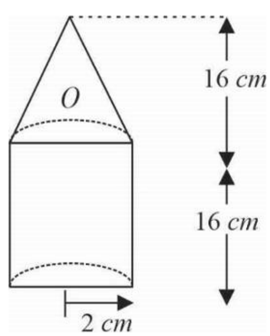
107. A ball is moving with velocity $2\ m/s$ towards a heavy wall. The wall is moving towards the ball with speed $1\ m/s$. They undergo an elastic head on collision. Find the velocity of the ball immediately after the collision.

108. Block A of mass $1\ kg$ is placed on the rough surface of block B of mass $3\ kg$. Block B is placed on a smooth horizontal surface. Blocks are given velocities as shown in the figure. Find the net work done by the frictional force (in $(-)ve\ J$).

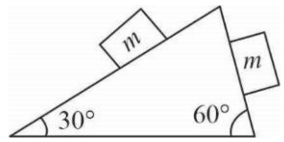


109. Sand drops vertically at the rate of $5kg/\sec$ onto a conveyor belt moving horizontally with a velocity of $0.8\ m/s$. Find the extra force required to keep the belt moving (in N).

110. Find the distance of the center of mass of a composite solid cone and solid cylinder made of same material from point O (in cm).



111. Two blocks of equal masses m are released from the top of a smooth fixed wedge as shown is the figure. Find the magnitude of the acceleration of the center of mass of the two blocks.

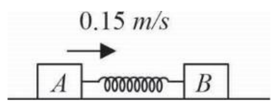


112. Two balls of equal masses are projected upward simultaneously; one from the ground with speed $50\ ^{m}/_{s}$ and other from a 40 m high tower with initial speed $30\ ^{m}/_{s}$. Find the maximum height attained by their center of mass. (Take $g = 10 m/s^2$)

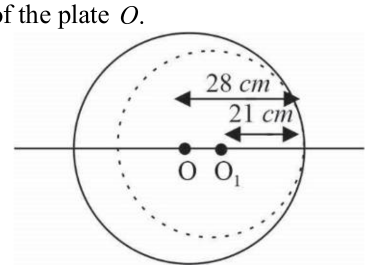
113. A bullet of mass 50gm is fired from a gun of mass 4kg with a muzzle speed of $560\ ^{m}/_{s}$. Find the recoil velocity of the gun (rounded off to closest integer).

114. An explosion blows a rock into three parts. Two parts go off at right angles to each other. These two are 1 kg first part moving with a velocity of $12\ ^{m}/_{s}$ and 2 kg second part moving with a velocity of $8\ ^{m}/_{s}$. Find the mass of the third part flies off with a velocity of $4\ ^{m}/_{s}$.

115. Two blocks A and B of masses 2 kg and 3 kg respectively are connected by a spring of spring constant $10.8\ ^{N}/_{m}$ and are placed on a frictionless horizontal surface. The block A is given an initial velocity of $0.15\ ^{m}/_{s}$ as shown in figure. Find the maximum compression in the spring (in cm).

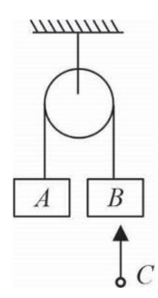


116. A circular plate of uniform thickness has a diameter of $56\,cm$. A circular portion of diameter $42\,cm$ is removed from one edge as shown in the figure. Find the distance of the center of mass of the remaining portion (in cm) from the center of the plate O.



117. Two bodies of masses $6\,kg$ and $1\,kg$ are tied to the ends of a string which passes over a light frictionless pulley. The masses are initially at rest and released. Find the acceleration of the center of mass.

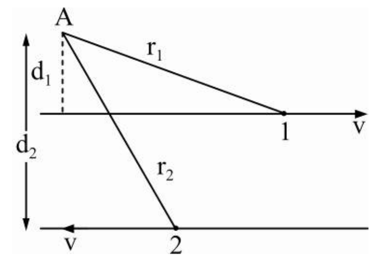
118. Two identical blocks A and B each of mass $2\,kg$ are hanging stationary by a light inextensible string, passing over a light and frictionless pulley as shown in the figure. A shell C of mass $1\,kg$ moving vertically upwards with velocity $9\,m/s$ collides with block B and gets stuck to it. Let t_0 be the time after which the string becomes taut again. Find $100t_0$.



Rotational Motion

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

***1.** Figure shows two identical particles 1 and 2, each of mass m, moving in opposite directions with same speed v along parallel lines. At a particular instant, r_1 and r_2 are their respective position vectors drawn from point A which is in the plane of the parallel lines. Choose the correct options :



(A) Angular momentum l_1 of particle 1 about A is $l = mvd_1 \odot$

(B) Angular momentum l_1 of particle 2 about A is $l_2 = mvr_2 \odot$

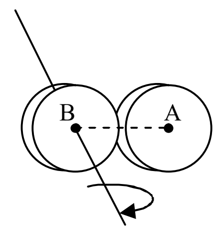
(C) Total angular momentum of the system about A is $l = mv(r_1 + r_2) \odot$

(D) Total angular momentum of the system about A is $l = mv(d_2 - d_1) \otimes$

$\odot$ represents a unit vector coming out of the page.

$\otimes$ represents a unit vector going into the page.

2. Two thin discs, each of mass M and radius r, are attached as shown in the figure, to form a rigid body. The rotational inertia of this body about an axis perpendicular to the plane of disc B and passing through its centre is :



(A) $2Mr^2$ **(B)** $3Mr^2$

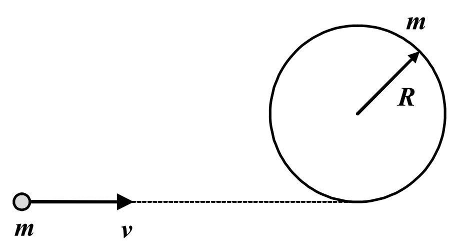
(C) $4Mr^2$ **(D)** $5Mr^2$

3. A circular hoop of mass m and radius R rests flat on a horizontal frictionless surface. A bullet, also of mass m, and moving with a velocity v, strikes the hoop and gets embedded in it. The thickness of the hoop is much smaller than R. The angular velocity with which the system rotates after the bullet strikes the hoop is :

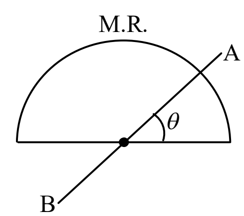
(A) $\frac{V}{4R}$

(B) $\frac{V}{3R}$

(C) $\frac{2V}{3R}$ **(D)** $\frac{3V}{4R}$



4. Moment of inertia of the semicircular ring of mass M and radius R about an axis AB as shown in the figure.



(A) dependent of angle θ **(B)** independent of angle θ

(C) $\frac{MR}{2}$, if $\theta = 45°$ **(D)** Mr^2, if $\theta = \frac{\pi}{2}$

***5.** The net external torque on a system of particles about an axis is zero. Which of the following are compatible with it ?

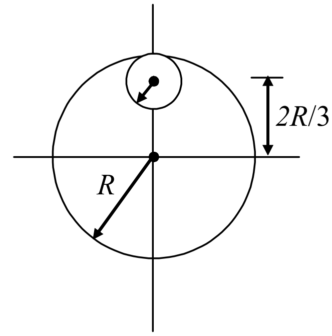
(A) The forces may be acting radically from a point on the axis

(B) The force may be acting on the axis of rotation

(C) The force may be acting parallel to the axis of rotation

(D) The torque caused by some forces may be equal and opposite to that caused by other forces

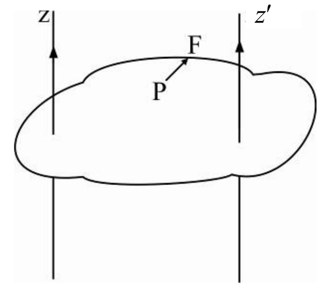
6. From a circular disc of radius R and mass $9M$, a small disc of radius $R/3$ is removed from the disc. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through O is :



(A) $4MR^2$ **(B)** $\frac{40}{9}MR^2$

(C) $10MR^2$ **(D)** $\frac{37}{9}MR^2$

***7.** Figure shows a lamina in x-y plane. Two axes z and z' pass perpendicular to its plane. A force F acts in the plane of lamina at point P as shown. Which of the following are true ? (The point P is closer to z'-axis than the z-axis.)



(A) Torque τ caused by F about z axis is along $-\hat{k}$

(B) Torque τ caused by F about z' axis is along $-\hat{k}$

(C) Torque τ caused by F about z axis is greater in magnitude than that about z axis

(D) Total torque is given be $\tau = \tau + \tau'$

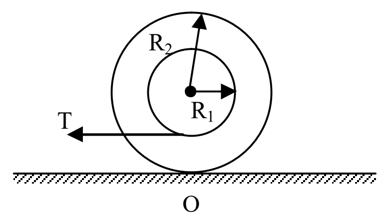
8. Four identical rods, each of mass m and length l are joined to form a rigid square frame. The frame lies in the xy plane, with its centre at the origin and the sides parallel to the x and y axes. Its moment of inertia about

(A) The x-axis is $\frac{2}{3}ml^2$ **(B)** The z-axis is $\frac{4}{3}ml^2$

(C) An axis parallel to the z-axis and passing through a corner is $\frac{10}{3}ml^2$

(D) One side is $\frac{5}{3}ml^2$

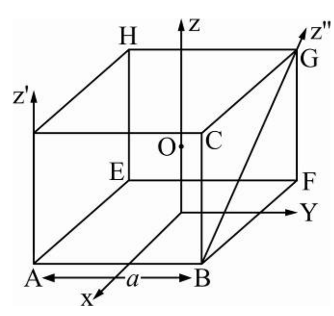
***9.** A stepped cylinder (shown in the figure) is having a mass of 50 kg and a radius of gyration k of 0.30 m. The radii R_1 and R_2 are respectively 0.30 m and 0.60 m. A pull T equal 200 N is exerted on the rope attached to the inner cylinder. The coefficients of static and dynamic friction between cylinder and ground are respectively 0.10 and 0.08. Which of the following statements are correct? ($g = 10 m/s^2$)



(A) The angular acceleration 2.67 rad/s^2 **(B)** The force kinetic friction is 40N

(C) The acceleration is $-3.2\ m/s^2$ **(D)** None of the above

***10.** With reference to fig. of a cube of edge a and mass m, state whether the following are true or false. (O is the centre of the cube.)



(A) The moment of inertia of cube about z-axis is $I_z = I_x + I_y$

(B) The moment of inertia of cube about z' is $I_z = I_z + \frac{ma^2}{2}$

(C) The moment of inertia of cube about z" is $= I_z + \frac{ma^2}{2}$

(D) $I_x = I_y$

11. Particle of mass m is projected with a velocity v_0 making an angle of $45°$ with horizontal. The magnitude of angular momentum of the projectile about the point of projection at its maximum height is :

(A) Zero **(B)** $mv^3/\sqrt{2g}$ **(C)** $mv_0^2/4\sqrt{2}g$ **(D)** $m\sqrt{2gh^3}$

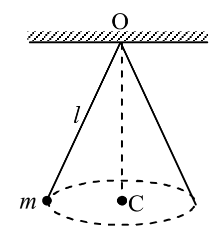
12. A solid sphere of mass m and radius R is gently placed on a conveyer belt moving with constant velocity v_0. If coefficient of friction between belt and sphere is 2/7, the distance traveled by the centre of the sphere before it starts pure rolling is :

(A) $\dfrac{v_0^2}{7g}$ **(B)** $\dfrac{2v_0^2}{49g}$ **(C)** $\dfrac{2v_0^2}{5g}$ **(D)** $\dfrac{2v_0^2}{7g}$

13. A uniform rod of mass m and length l rotates in a horizontal plane with an angular velocity ω about a vertical axis passing through one end. The tension in the rod at a distance x from the axis is :

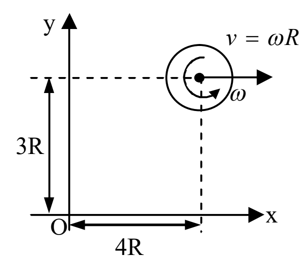
(A) $\frac{1}{2}m\omega^2 x$ **(B)** $\frac{1}{2}m\omega^2\frac{x^2}{l}$ **(C)** $\frac{1}{2}m\omega^2 l\left(1-\frac{x}{l}\right)$ **(D)** $\frac{1}{2}\frac{m\omega^2}{l}\left[l^2-x^2\right]$

***14.** A small ball of mass m suspended from the ceiling at a point O by a thread of length l moves along a horizontal circle with a constant angular velocity ω.



(A) Angular momentum is constant about O
(B) Angular momentum is constant about C
(C) Vertical component of angular momentum about O is constant
(D) Magnitude of angular momentum about O is constant

15. A disc of mass m and radius R moves in the $x-y$ plane as shown the figure. The angular momentum of the disc about the origin O at the instant shown is :



(A) $\frac{5}{2}mR^2\omega$ **(B)** $\frac{7}{3}mR^2\omega$
(C) $\frac{9}{2}mR^2\omega$ **(D)** $\frac{3}{2}mR^2\omega$

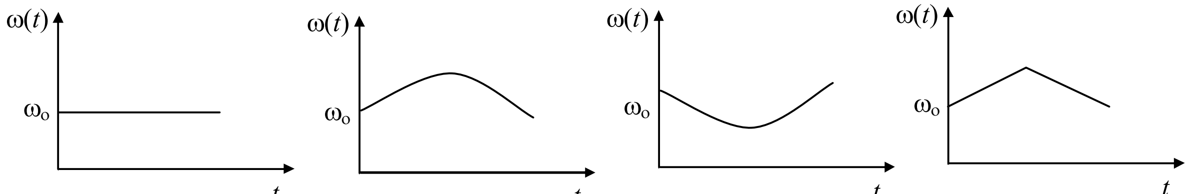
16. A particle is moving on a circular path in the horizontal plane with constant angular speed. The angular momentum will be constant about a point

(A) On the circumference **(B)** Outside the circle
(C) Inside the circle **(D)** On the center

***17.** A ring rolls without slipping on the ground. Its centre C moves with a constant speed u. P is any point on the ring. The speed of P with respect to the ground is v.

(A) $0 \le v \le 2u$
(B) $v = u$, if CP is horizontal.
(C) $v = u$, if CP makes angle of 60° with the horizontal and P is below the horizontal level of C.
(D) $v = \sqrt{2}\,u$, if CP is horizontal

18. Circular platform is free to rotate in a horizontal plane about a vertical axis passing through its center. A tortoise is sitting at the edge of the platform. Now, the platform is given an angular velocity ω_o. When the tortoise moves along a chord of the platform with a constant velocity (with respect to the platform), the angular velocity of the platform $\omega(t)$ will vary with time t as :

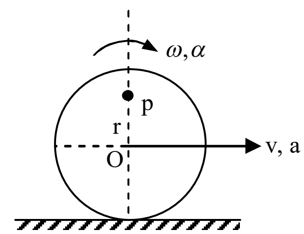


(A) **(B)** **(C)** **(D)**

19. A thin horizontal circular disc is rotating about a vertical axis passing through its centre. An insect is at rest at a point near the rim of the disc. The insect now moves along a diameter of the disc to reach its other end. During the journey of the insect, the angular speed of the disc

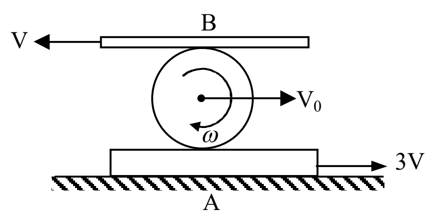
(A)	Continuously decreases	**(B)**	Continuously increase
(C)	First in increases and then decreases	**(D)**	Remains unchanged

20. A disc of radius R rolls on a horizontal ground with linear acceleration a and angular acceleration α and angular acceleration α as shown in the figure. The magnitude of acceleration of point P as shown in figure at an instant when its linear velocity is v and angular velocity is ω will be :



(A)	$\sqrt{(a + r\alpha)^2 + (r\omega^2)^2}$	**(B)**	$\frac{ar}{R}$
(C)	$\sqrt{r^2\alpha^2 + r^2\omega^4}$	**(D)**	$r\alpha$

***21.** The disc of radius r is confined to roll without slipping at A and B. If the plates have the velocities shown then :



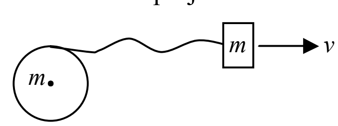
(A) Angular velocity of the disc of $2V/r$

(B) Linear velocity, $V_0 = V$

(C) Angular velocity of the disc is $3V/2r$

(D) None of these

22. A block of mass m is attached to a pulley disc of equal mass m and radius r by means of a slack string as shown. The pulley is hinged about its centre on a horizontal table and the block is projected with an initial velocity of 5 m/s. Its velocity when the string becomes taut will be :



(A)	3 m/s	**(B)**	2.5 m/s
(C)	5/3 m/s	**(D)**	10/3 m/s

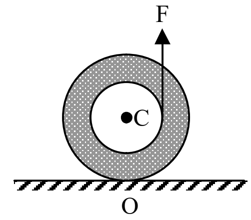
23. A particle rests on the top of a hemisphere of radius R. Find the smallest horizontal velocity that must be imparted to the particle if it is to leave the hemisphere without sliding down it.

(A) $\sqrt{gR}$ **(B)** $\sqrt{2gR}$ **(C)** $\sqrt{3gR}$ **(D)** $\sqrt{5gR}$

24. A hoop of radius r and mass m rotating with an angular velocity ω_0 is placed on a rough horizontal surface. The initial velocity of the centre of the hoop is zero. What will be the velocity of the centre of the hoop when it ceases to slip?

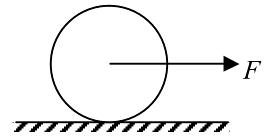
(A) $\frac{r\omega_0}{4}$ **(B)** $\frac{r\omega_0}{3}$ **(C)** $\frac{r\omega_0}{2}$ **(D)** $r\omega_0$

25. A yo-yo placed on a rough horizontal surface and a constant force F, which is less than its weight, pulls it vertically. Due to this



(A) friction force acts toward left, so it will move towards left
(B) friction force acts towards right so it will move towards right
(C) it will move towards left so friction force acts towards left
(D) it will move towards right so friction force acts towards right

26. A solid sphere of mass m is lying at rest on a rough horizontal surface. The coefficient of friction between the ground and sphere is μ. The maximum value of F, so that the sphere will not slip, is equal to :



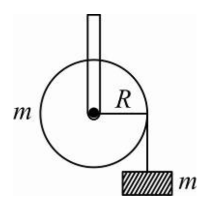
(A) $\frac{7}{5}\mu mg$ **(B)** $\frac{4}{7}\mu mg$ **(C)** $\frac{5}{7}\mu mg$ **(D)** $\frac{7}{2}\mu mg$

27. A bob of mass m attached to an inextensible string of length l is suspended from a point and moves as a conical pendulum with constant angular speed ω. About the point of suspension.
(A) Angular momentum is conserved
(B) Angular changes in magnitude but not in direction
(C) Angular momentum changes in direction but not in magnitude
(D) Angular momentum changes both in direction and magnitude

***28.** A horizontal disc rotates freely about a vertical axis through its centre. *A* ring, having the same mass and radius as the disc, is now gently placed on the disc. After some time, the two rotates with a common angular velocity, then:
(A) Finally some friction is acting between the disc and the ring
(B) The angular momentum of the “disc plus ring” is conserved
(C) The final common angular velocity is 2/3rd of the initial angular velocity of the disc
(D) (2/3)rd of the initial kinetic energy changes to heat

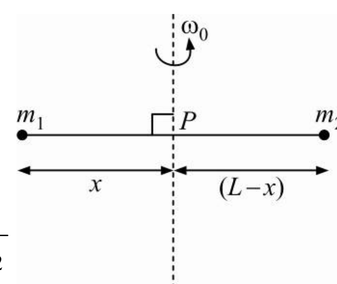
29. A pulley of radius 2 m is rotated about its axis by a force $F = (20t - 5t^2)\,N$ (where, t is measured in seconds) applied tangentially. If the moment of inertial of the pulley about its axis of rotation is 10 $kg\text{-}m^2$, then the number of rotations made by the pulley before its direction of motion is reserved, is :
(A) More than 3 but less than 6 **(B)** More than 6 but less than 9
(C) More than 9 **(D)** Less than 3

30. A mass m supported by a massless string wound around a uniform hollow cylinder of mass m and radius R. If the string does not slip on the cylinder, with what acceleration will the mass fall on release?



(A) $\frac{2g}{3}$ **(B)** $\frac{g}{2}$ **(C)** $\frac{5g}{6}$ **(D)** g

31. Point masses m_1 and m_2 are placed at the opposite ends of a rigid rod of length L, and negligible mass. The rod is to be set rotation about an axis perpendicular to it. The position of point P on this rod through which the axis should pass so that the work required to set the rod rotation with angular velocity ω_0 is minimum, is given by :



(A) $x = \frac{m_2}{m_1}L$ **(B)** $x = \frac{m_2 L}{m_1 + m_2}$
(C) $x = \frac{m_1 L}{m_1 + m_2}$ **(D)** $x = \frac{m_1}{m_2}L$

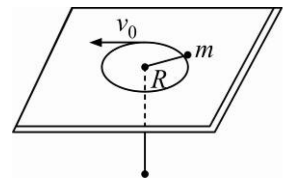
32. An automobile moves on a road with a speed of $54\,km\,h^{-1}$. The radius of its wheels is $0.45\,m$ and the moment of inertia of the wheel about its axis of rotation is $3\,kg\,m^2$. If the vehicle is brought to rest in $15\,s$, the magnitude of average torque transmitted by its breaks to the wheel is :

(A) $10.86\,kg\,m^2\,s^{-2}$ **(B)** $2.86\,kg\,m^2\,s^{-2}$ **(C)** $6.66\,kg\,m^2\,s^{-2}$ **(D)** $8.58\,kg\,m^2\,s^{-2}$

33. A force $\vec{F} = \alpha i + 3\hat{j} + 6\hat{k}$ is acting at a point $\vec{r} = 2\hat{i} - 6\hat{j} - 12\hat{k}$. The value of α for which angular momentum about origin is conserved is :

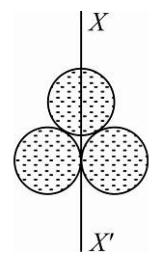
(A) zero **(B)** 1 **(C)** -1 **(D)** 2

34. A mass m moves in a circle on a smooth horizontal plane with velocity v_0 at a radius R_0. The mass is attached to a string which passes through a smooth hole in the plane as shown. Find the KE of mass if the radius is halved by pulling the string through hole.



(A) $2mv_0^2$ **(B)** $\frac{1}{2}mv_0^2$ **(C)** mv_0^2 **(D)** $\frac{1}{4}mv_0^2$

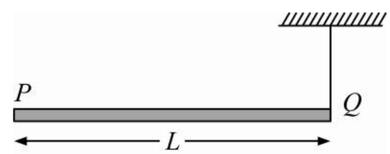
35. Three identical spherical shells, each of mass m and radius r are placed as shown in figure. Consider an axis XX' which is touching to two shells and passing through diameter of third shell. Moment of inertia of the system consisting of these three spherical shells about XX' axis is :



(A) $\frac{16}{5}mr^2$ **(B)** $4mr^2$

(C) $\frac{11}{5}mr^2$ **(D)** $3mr^2$

36. A rod PQ of mass M and length L is hinged at end P. The rod is kept horizontal by a massless string tied at point Q as shown in figure. When string is cut, the initial angular acceleration of the rod is :



(A) $\frac{2g}{L}$ **(B)** $\frac{2g}{2L}$

(C) $\frac{3g}{2L}$ **(D)** $\frac{g}{L}$

37. A small object of uniform density rolls up a curved surface with an initial velocity 'v'. It reaches upto a maximum height of $\frac{3v^2}{4g}$ with respect to the initial position. The object is :

(A) hollow sphere **(B)** disc **(C)** ring **(D)** solid sphere

38. The ratio of radii of gyration of a circular ring and a circular disc, of the mass and radius, about an axis passing through their centres and perpendicular to their planes are :

(A) $1:\sqrt{2}$ **(B)** $3:2$ **(C)** $2:1$ **(D)** $\sqrt{2}:1$

39. Two persons of masses $55\,kg$ and $65\,kg$ respectively, are at the opposite ends of a boat. The length of the boat is $3.0\,m$ and weight $100\,kg$. The $55\,kg$ man walks up to the $65\,kg$ man and sits with him. If the boat is in still water the centre of mass of the system shifts by :

(A) $3.0\,m$ **(B)** $2.3\,m$ **(C)** zero **(D)** $0.75\,m$

40. A circular platform is mounted on a frictionless vertical axle. Its radius $R = 2m$ and its moment of inertia about the axle is $200\,kg\ m^2$. It is initially at rest. A $50\,kg$ man stands on the edge of the platform and begins to walk along the edge at the speed of $1\,ms^{-1}$ relative to the ground. Time taken by the man to complete one revolution is :

(A) $\pi\,s$ **(B)** $\frac{3\pi}{2}s$ **(C)** $2\pi\,s$ **(D)** $\frac{\pi}{2}s$

41. The instantaneous angular position of a point on a rotating wheel is given by the equation $\theta(t) = 2t^3 - 6t^2$ The torque on the wheel becomes zero at :

(A) $t = 1s$ **(B)** $t = 0.5\,s$ **(C)** $t = 0.25\,s$ **(D)** $t = 2\,s$

42. A circular disk of moment of inertia I_t is rotating in a horizontal plane, about its symmetry axis, with a constant angular speed ω_i. Another disk of moment of inertia I_b is dropped coaxially onto the rotating disk. Initially the second disk has zero angular speed. Eventually both the both the disks rotate with a constant angular speed ω_f. The energy lost by the initially rotating disc to friction is :

(A) $\frac{1}{2}\frac{I_b^2}{(I_t+I_b)}\omega_i^2$ **(B)** $\frac{1}{2}\frac{I_t^2}{(I_t+I_b)}\omega_i^2$ **(C)** $\frac{I_b-I_t}{(I_t+I_b)}\omega_i^2$ **(D)** $\frac{1}{2}\frac{I_bI_t}{(I_t+I_b)}\omega_i^2$

43. A gramophone record is revolving with an angular velocity ω. A coin is placed at a distance r from the centre of the record. The static coefficient of friction is μ. The coin will revolved with the record if :

(A) $r = \mu g\omega^2$ **(B)** $r < \frac{\omega^2}{\mu g}$ **(C)** $r \le \frac{\mu g}{\omega^2}$ **(D)** $r \ge \frac{\mu g}{\omega^2}$

44. A thin circular ring of mass M and radius r is rotating about its axis with constant angular velocity ω. Two objects each of mass m are attached gently to the opposite ends of a diameter of the ring. The ring now rotates with angular velocity given by :

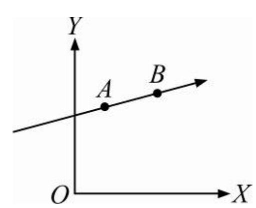
(A) $\frac{(M+2m)\omega}{2m}$ **(B)** $\frac{2M\omega}{M+2m}$ **(C)** $\frac{(M+2m)\omega}{M}$ **(D)** $\frac{M\omega}{M+2m}$

45. If $\vec{F}$ is the force acting on a particle having position vector $\vec{r}$ and $\vec{\tau}$ be the torque of this force about the origin, then :

(A) $\vec{r}\cdot\vec{\tau} > 0$ and $\vec{F}\cdot\vec{\tau} < 0$ **(B)** $\vec{r}\cdot\vec{\tau} = 0$ and $\vec{F}\cdot\vec{\tau} = 0$

(C) $\vec{r}\cdot\vec{\tau} = 0$ and $\vec{F}\cdot\vec{\tau} \ne 0$ **(D)** $\vec{r}\cdot\vec{\tau} \ne 0$ and $\vec{F}\cdot\vec{\tau} = 0$

46. A particle of mass m moves in the XY plane with a velocity v along the straight line AB. If the angular momentum of the particle with respect to origin O is L_A when it is at A and L_B when it is at B, then :



(A) $L_A = L_B$

(B) the relationship between L_A and L_B depends upon the slope of the line AB

(C) $L_A < L_B$ **(D)** $L_A > L_B$

47. A drum of radius R and mass M, rolls down without slipping along an inclined plane of angle θ. Then frictional force :

(A) does not act on the drum **(B)** does positive work on the drum

(C) does negative work on the drum **(D)** does no work on the drum

48. A particle of mass $m = 5$ is moving with a uniform speed $v = 3\sqrt{2}$ in the XOY plane along the line $Y = X + 4$. The magnitude of the angular momentum of the particle about the origin is :

(A) 60 units **(B)** $40\sqrt{2}$ units **(C)** zero **(D)** 7.5 units

49. A ball rolls without slipping. The radius of gyration of the ball about an axis passing through its centre of mass is *K*. If radius of ball be *R*, then the fraction of total energy associated with its rotational energy will be :

(A) $\frac{K^2+R^2}{R^2}$ **(B)** $\frac{K^2}{R^2}$ **(C)** $\frac{K^2}{K^2+R^2}$ **(D)** $\frac{R^2}{K^2+R^2}$

50. A point *P* consider at contact point of a wheel on group which rolls on ground without slipping then value of displacement of point *P* when wheel completes half of rotation (If radius of wheel is $1m$):

(A) $2m$ **(B)** $\sqrt{\pi^2+4}\,m$ **(C)** πm **(D)** $\sqrt{\pi^2+2}\,m$

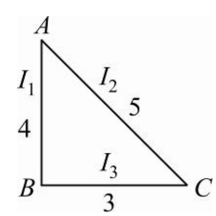
51. A circular disc is to be made by using iron and aluminium so that it acquired maximum moment of inertia about geometrical axis. It is possible with :

(A) aluminium at interior and iron surround to it
(B) iron at interior and aluminium surround to it
(C) using iron and aluminium layers in alternate
(D) sheet of iron is used at both external surface and aluminium sheet as internal layers

52. A disc is rolling, the velocity of its centre of mass is v_{cm}. Which one will be correct ?

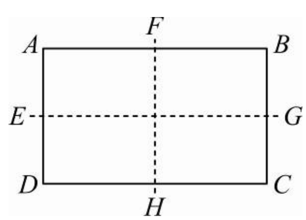
(A) the velocity of highest point is $2v_{cm}$ and point of contact is zero
(B) the velocity of highest point is v_{cm} and point of contact is v_{cm}
(C) the velocity of highest point is $2v_{cm}$ and point of contact is v_{cm}
(D) the velocity of highest point is $2v_{cm}$ and point of contact is $2v_{cm}$

53. For the adjoining diagram, the correct relation between I_1, I_2, I_3 is, (I - moment of inertia) :



(A) $I_1 > I_2$ **(B)** $I_2 > I_1$
(C) $I_3 > I_1$ **(D)** $I_3 > I_2$

54. In a rectangle $ABCD(BC = 2AB)$. The moment of inertia is minimum along axis through :



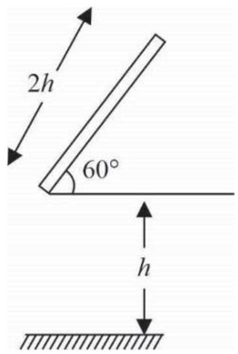
(A) BC **(B)** BD **(C)** HF **(D)** EG

INTEGER ANSWER TYPE QUESTIONS

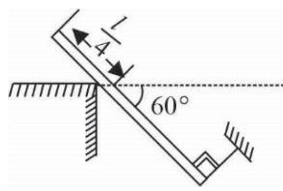
The Answers to the following questions are positive integers of 1/2/3 digits or zero.

55. Three particles A, B and C, each of mass m are connected to one another by three massless rigid rods to form a rigid, equilateral triangular body of side l. This body is placed on a horizontal frictionless table ($XY-$plane) and is hinged to it at the point A so that it can move without friction about the vertical axis through A with a constant angular velocity ω. The magnitude of horizontal force exerted by the hinge on the body is $\sqrt{P}ml\omega^2$ find the value of P

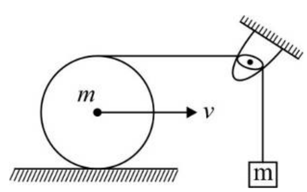
56. A uniform rod of mass m and length $l(=2h)$ strikes the rough ground after falling through a distance h. If the rod does not bounce. The angular velocity of the rod just after the impact is $\frac{3}{x}\sqrt{\frac{yg}{h}}$. Find $\frac{x}{y}$



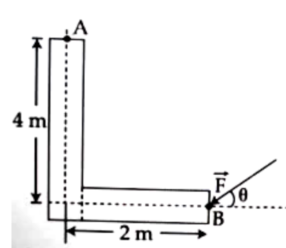
57. A uniform rod of mass m and length l is in equilibrium under the action of constraint forces, gravity and tension in the string. The friction force acting on the rod is $\frac{\sqrt{3}}{p}mg$. Find the value of P.



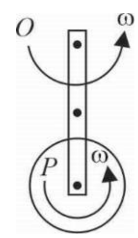
58. A sphere of mass m rolls without sliding by the thread which hangs a body of mass m. Ignoring the friction at the pulley, the speed of COM of sphere is $\sqrt{\frac{10}{9x}gy}$, when block moves vertically down by a distance y. Find x.



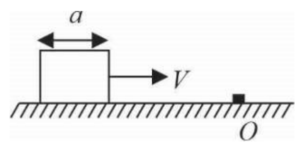
59. A force of 40 N acts on a point B at the end of an L-shaped object, as shown in the figure. The maximum torque of force about A is at angle $\theta = \tan^{-1}\left(\frac{x}{y}\right)$. Find $\frac{y}{x}$.



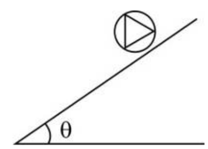
60. A rod of mass m and length $2R$ can rotate about an axis passing through O in vertical plane. A disc of mass m and radius $R/2$ is hinged to the other end P of the rod and can freely rotate about P. when disc is at lowest point both rod and disc has angular velocity ω. If rod rotates by maximum angle $\theta = 60°$ with downward vertical, then ω in terms of R and g (all hinges are smooth) is given by $\left(\frac{9g}{4xR}\right)^{1/2}$. Find x



61. A cubical block of side a is moving with velocity V on a horizontal smooth plane as shown in the figure. It hits a ridge at point O. The angular speed of the block after it hits O is $\frac{3V}{xa}$ Find the value of x.

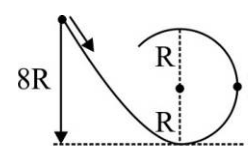


62. In the figure shown, a light ring with three rods, each of mass m is welded on this ring. The rods form an equilateral triangle. The rigid assembly is released on a rough fixed inclined plane. The minimum value of the coefficient of static friction, that will allow pure rolling of the assembly is given as $\frac{x}{y}\tan\theta$. Find the value of $x+2y$.

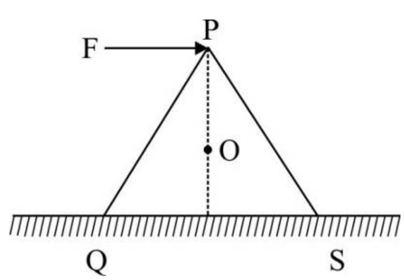


63. A hollow sphere of outer radius R is allowed to roll down on an incline without slipping and it reaches a speed v_0 at the bottom of the incline. The incline is then made smooth by waxing and the sphere is allowed to slide without rolling and now the speed attained is $\frac{5}{4}v_0$. The radius of gyration of the sphere about an axis passing through its centre is given by as $\frac{xR}{y}$. Find $x+y$

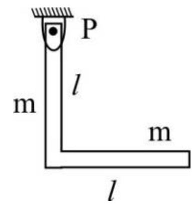
64. A small solid ball (mass = 0.1kg) rolls without slipping along the track shown in the figure. The radius of the circular track is R. if the ball starts from rest at a height 8R above the bottom, the horizontal force acting on it at point P is 5x newton. Find the value of x. (Given, $g = 10ms^{-2}$)



65. A wedge in the form of equilateral triangle is placed on a rough horizontal surface as shown in the figure. The minimum value of coefficient of friction, for which the wedge can topple without slipping is $\left(\frac{1}{x}\right)^{\frac{1}{2}}$. Find x.

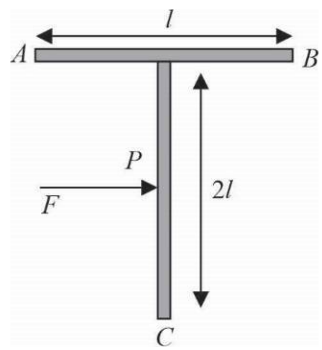


66. The L-shaped uniform rod having identical limbs each of mass m and length l is pivoted smoothly at P as shown in the figure. When the rod is released from rest from the given position, it swings in the vertical plane. The initial angular acceleration of the rod is $\frac{xg}{10l}$. Find the value of x.

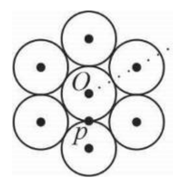


67. A cylinder is released from rest from the top of an incline plane of inclination 60° where friction coefficient varies with distance x as $\mu = \frac{2-3x}{\sqrt{3}}$. The distance travelled by the cylinder on incline before it starts slipping is $\frac{P}{3}$. Find P

68. A T shaped object with dimensions shown in the figure, is lying on a smooth floor. A force F is applied at the point P parallel to AB, such that the object has only the translational motion without rotation. The distance of P from C is given as $\frac{x}{y}l$. Find $x+y$.



69. Seven identical circular planar disks, each of mass M and radius R are welded symmetrically as shown. The moment of inertia of the arrangement about the axis normal to the plane and passing through the point P is $\frac{(12x+9)}{2}MR^2$. Point P is the contact point of central disc & any outer disc.



Gravitation

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

1. The energy required to move a satellite of mass m from an orbit of radius $2R$ to $3R$ is (where M is the mass of the earth and R is the radius of the earth)

(A) $\frac{GMm}{12R}$ **(B)** $\frac{GMm}{8R}$ **(C)** $\frac{GMm}{3R}$ **(D)** $\frac{GMm}{6R}$

2. If suddenly the gravitational force of attraction between the earth and a satellite revolving around it becomes zero, then the satellite will :

(A) continue to move in its orbit with the same velocity
(B) move tangentially to the original orbit with the same velocity
(C) become stationary in its orbit
(D) move towards the earth

3. The escape velocity of a body depends upon mass as :

(A) m^0 **(B)** m^1 **(C)** m^2 **(D)** m^3

4. The kinetic energy needed to project a body of mass m from the earth's surface (radius R) to infinity is :

(A) $\frac{mgR}{2}$ **(B)** $2mgR$ **(C)** mgR **(D)** $\frac{mgR}{4}$

5. The time period of a satellite of the earth is $5h$. If the separation between the earth and the satellite is increased to four times the pervious value, the new time period will become.

(A) $10\ h$ **(B)** $80\ h$ **(C)** $40\ h$ **(D)** $20\ h$

6. Two spherical bodies of masses m and $5M$ and radii R and $2R$, respectively, are released in free space with initial separation between their centres equal to $12R$. If they attract each other by gravitational force only, then the distance covered by the smaller body just before collision is :

(A) $2.5\ R$ **(B)** $4.5\ R$ **(C)** $7.5\ R$ **(D)** $1.5\ R$

7. The escape velocity for a body projected vertically upward from the surface of the earth is $11\ km/s$. If the body is projected at an angle of 45° with vertical, the escape velocity will be :

(A) $11\sqrt{2}\ km/s$ **(B)** $22\ km/s$ **(C)** $11\ km/s$ **(D)** $\frac{11}{\sqrt{2}}\ km/s$

8. A satellite of mass m revolves around the earth of radius R at a height x from its surface. If g is the acceleration due to gravity on the surface of the earth, the orbital speed of the satellite is :

(A) gx **(B)** $\left(\frac{gR^2}{R+x}\right)^{1/2}$ **(C)** $\frac{gR^2}{R+x}$ **(D)** $\frac{gR}{R-x}$

9. The time period of an earth satellite in circular orbit is independent of

(A) the mass of the satellite

(B) radius of its orbit

(C) both the mass and radius of the orbit

(D) neither the mass of the satellite nor the radius of its orbit

10. If g is the acceleration due to gravity on the earth's surface the gain in the potential energy of an object of mass m raised from the earth's surface to a height equal to the radius R of the earth is :

(A) $2mgR$ **(B)** $\frac{1}{2}mgR$ **(C)** $\frac{1}{4}mgR$ **(D)** mgR

11. Suppose the gravitational force varies inversely as the nth power of distance. Then the time period of a planet in a circular orbit of radius R around the sun will be proportional to :

(A) $R^{(n+1)/2}$ **(B)** $R^{(n-1)/2}$ **(C)** R^n **(D)** $R^{(n-2)/2}$

12. The average density of the earth

(A) does not depend on g. **(B)** is a complex function of g.

(C) is directly proportional to g **(D)** is inversely proportional to g.

13. The change in the value of g at a height h above the surface of the earth is the same as at a depth d below the earth. When both d and h are much smaller than the radius of the earth, which one of the following is correct?

(A) $d = \frac{h}{2}$ **(B)** $d = \frac{3h}{2}$ **(C)** $d = 2h$ **(D)** $d = h$

14. A particle of mass 10*g* is kept on the surface of a uniform sphere of mass 100 *kg* and radius 10*cm*. Find the work done against the gravitational force between them to take the particle far away from the sphere.

(A) $13.34 \times 10^{-10} J$ **(B)** $3.33 \times 10^{-10} J$ **(C)** $6.67 \times 10^{-9} J$ **(D)** $6.67 \times 10^{-10} J$

15. If g_E and g_M are the acceleration due to gravity on the moon, respectively, and if Millikan oil drop experiment could be performed on the two surfaces, one will find the ratio $\frac{\text{electronic charge on moon}}{\text{electronic charge on earth}}$ to be :

(A) 0 **(B)** $\frac{g_E}{g_M}$ **(C)** $\frac{g_M}{g_E}$ **(D)** 1

16. A planet in a distant solar system is 10 times more massive then the earth and its radius is 10 times smaller. Given that the velocity from the earth is 11 *km/s* the escape velocity from the surface of the planet would be :

(A) 1.1 *km/s* **(B)** 11 *km/s* **(C)** 110 *km/s* **(D)** 0.11 *km/s*

17. The height at which the acceleration due to gravity become *g/*9 (where *g* is the acceleration due to gravity on the surface of the earth) in terms of *R*, the radius of the earthy is:

(A) $2R$ **(B)** $\frac{R}{\sqrt{2}}$ **(C)** $\frac{R}{2}$ **(D)** $\sqrt{2}\,R$

18. **Statement : I** For *A* mass *M* kept at a centre of a cube of side *a*, the flux of gravitational field passing through its side is $4\pi GM$.

Statement : II If the direction of a field due to a point source is radial and its dependence on the distance *r* from the source is given is $1/r^2$, its flux through a closed surface depends only on the strength of the source enclosed by the surface and not on the size or shape of the surface.

(A) Statement-I is True, Statement-II is True and Statement-II is a correct explanation for Statement-I

(B) Statement-I is True, Statement-II is True and Statement-II is NOT a correct explanation for Statement-I

(C) Statement-I is True, Statement-II is False

(D) Statement-I is False, Statement-II is True

19. Two bodies of masses *m* and 4*m* are placed at a distance *r*. The gravitational potential at a point on the line joining them where the gravitational field is zero is :

(A) zero **(B)** $\frac{4Gm}{r}$ **(C)** $\frac{6Gm}{r}$ **(D)** $\frac{9Gm}{r}$

20. The mass of a spaceship is 1000 *kg*. It is to be launched from the earth's surface out into free space. The value of '*g*' and '*R*' (radius of earth) are 10 m/s^2 and 6400 *km*, respectively. The required energy for this work will be :

(A) $6.4 \times 10^{11} J$ **(B)** $6.4 \times 10^{8} J$ **(C)** $6.4 \times 10^{9} J$ **(D)** $6.4 \times 10^{10} J$

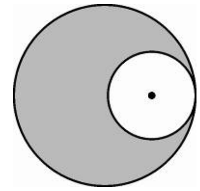
21. What is the minimum energy required to launch a satellite of mass *m* from the surface of a planet of mass *M* and radius *R* in a circular orbit at an altitude of 2*R*?

(A) $\frac{2GmM}{3R}$ **(B)** $\frac{GmM}{2R}$ **(C)** $\frac{GmM}{3R}$ **(D)** $\frac{5GmM}{6R}$

22. Four particles, each of mass *M* and equidistant from each other, move along a circle of radius *R* under the action of their mutual gravitational attraction. The speed of each particle is :

(A) $\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$ **(B)** $\frac{1}{2}\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$ **(C)** $\sqrt{\frac{GM}{R}}$ **(D)** $\sqrt{2\sqrt{2}\frac{GM}{R}}$

23. From a solid sphere of mass M and radius R, a spherical portion of radius $R/2$ is removed, as shown in the figure. Taking gravitational potential $V = 0$ at $r = \infty$, the potential at the centre of the cavity thus formed is (G = gravitational constant)



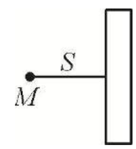
(A) $\dfrac{-GM}{2R}$ **(B)** $\dfrac{-GM}{R}$

(C) $\dfrac{-2GM}{3R}$ **(D)** $\dfrac{-2GM}{R}$

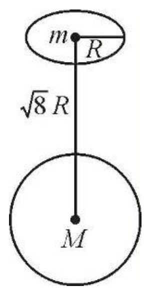
INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

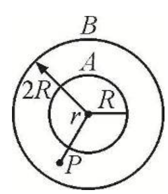
24. A point mass M is at a distance S from an infinitely long and thin rod of linear density D. If G is the gravitational constant then gravitational force between the point mass and the rod is $n\dfrac{MGD}{S}$ then n is.



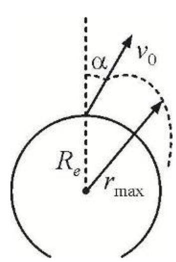
25. The centres of a ring of mass m and a sphere of mass $M = 54\sqrt{2}\,m$ of equal radius R, are at a distance $\sqrt{8}R$ apart as shown in figure. The force of attraction the ring and the sphere $= k\dfrac{Gm^2}{R^2}$ then $k =$



26. Two concentric spherical shells A and B of radii R and $2R$ and masses $4M$ and M respectively are as shown in figure. The gravitational potential at point 'P' at distance 'r' ($R < r < 2R$) from centre of shell is ($r = 1.5\ R$). Given by $-\dfrac{P}{q}\dfrac{GMm}{R}$ then $|P - 2q|$ is:

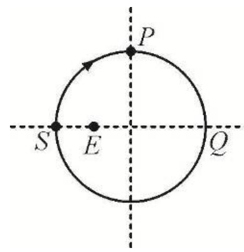


27. A projectile of mass m is fired from the surface of the earth at an angle $\alpha = 60°$ from the vertical. The initial speed v_0 is equal to $\sqrt{\dfrac{GM_e}{R_e}}$. Neglect air resistance and the earth's rotation, if its maximum height is $\dfrac{R_e}{x}$ then x is:

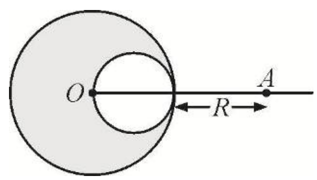


28. A binary star consists of two stars A(mass = $2.2\,M_s$) and B(mass = $11\,M_s$), where M_s is mass of the sun. They are separated by distance d and are rotating about their centre of mass which is stationary. The ratio of the total angular momentum of the binary star to the angular momentum of star B about the centre of the mass is:

29. The satellite is moving in an elliptical or about the earth as shown in figure. The minimum and maximum distance of satellite from earth are 3 units and 5 units, respectively. The distance of satellite from earth when it is at P:



30. A solid sphere of uniform density and radius R applies a gravitational force of attraction equal to F_1 on a particle placed at a distance $2R$ from the centre of the sphere. A spherical cavity of radius $(R/2)$ is now made in the sphere as shown in figure. The sphere with the cavity now applies a gravitational force F_2 on the same particle. The ratio $(F_2/F_1) = (P/q)$ then $|P - q|$ is.

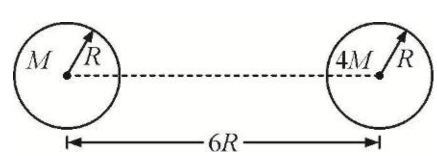


31. A bullet is fired vertically upwards with velocity v from the surface of a spherical planet. When it reaches its maximum height, its acceleration due to the planet's gravity is $1/4^{\text{th}}$ of its value at the surface of the planet. If the escape velocity from the planet is $v_{\text{esc}} = v\sqrt{N}$, then the value of N is ………. (Ignore energy loss due to atmosphere)

32. A large spherical mass M is fixed at one position and two identical point masses m are kept on a line passing through the centre of M (see figure). The point masses are connected by a rigid massless rod of length l and this assembly is free to move along the line connecting them. All three masses interact only through their mutual gravitational interaction. When the point mass nearer to M is a distance $r = 3l$ from M, the tension in the rod is zero for $m = k\left(\dfrac{M}{288}\right)$. The value of k is.

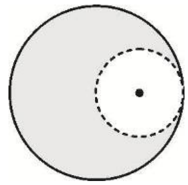
33. Three particles, each of mass M, are moving in a circle under their mutual gravitational forces such that they always form an equilateral triangle of side l while rotating. Speed of each particle is $N^{1/8}\sqrt{\dfrac{Fr}{M}}$ then N is:

34. Two uniform solid spheres of equal radii R, but mass M and $4M$ have a centre to centre separation $6R$, as shown in figure. A projectile of mass m is projected from the surface of the sphere of mass M directly towards the centre of the second sphere. The minimum kinetic energy of the projectile so that it reaches the surface of the second sphere is $\dfrac{x}{20}\dfrac{GMm}{R}$ then $x =$

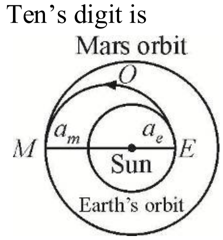


35. A particle of mass M is situated at the centre of a spherical shell of same mass M and radius R. The gravitational potential at a point situated at $R/2$ distance from the centre is given as $-x\frac{GM}{R}$ then value of
$2x$ is:

36. From a solid sphere of mass M and radius R, a spherical portion of radius $R/2$ is removed, as shown in figure. Taking gravitational potential $V = 0$ at $r = \infty$, the potential at the centre of the cavity thus formed is $-x\frac{GM}{R}$ then value of $2x$ is : (G = gravitational constant)



37. India's Mangalyan was send to the Mars by launching it into a transfer orbit EOM around the sun. It leaves the earth at E and meets Mars at M. If the semi-major axis of Earth's orbit is $a_e = 1.5 \times 10^{11} m$, that of Mars orbit $a_m = 2.28 \times 10^{11} m$, using Kepler's laws give the estimate of time for Mangalyan to reach Mars from Earth be N day's then in 'N' Ten's digit is

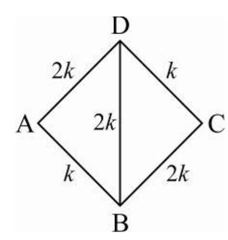


38. A very long (length L) cylindrical galaxy is made of uniformly distributed mass and has radius $R(R << L)$. A star outside the galaxy is orbiting the galaxy in a plane perpendicular to the galaxy and passing through its centre. If the time period of star is T and its distance from the galaxy's axis is r, then $T \propto r^x$ then $|2x - 3|$ is.

Properties of Matter

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

PARAGRAPH FOR QUESTIONS 1 - 3



A system of rods is assembled such that each rod has a length ℓ and cross-sectional areas S. The mode of heat transfer is conduction and the system is in steady state. The temperature of junction B is T and that of D is $2T$. Now answer the following questions.

1. Temperature of junction A is :

(A) $\frac{5}{3}T$ **(B)** $\frac{2}{3}T$ **(C)** $1.8\ T$ **(D)** $1.5\ T$

2. Temperature of junction C is :

(A) $\frac{5}{3}T$ **(B)** $1.6\ T$ **(C)** $\frac{4}{3}T$ **(D)** $1.5\ T$

3. The rate of heat flow along DB is :

(A) Zero **(B)** $\frac{kTS}{\ell}$ **(C)** $\frac{1}{2}\frac{kTS}{\ell}$ **(D)** $2\frac{kTS}{\ell}$

4. Emissivity e is a property of surface. Suppose, for a surface emissivity e varies with Kelvin with temp. T as $e = CT$ (C is constant). If energy emission rate at temp. 600 K from the surface is 160 W, what will be the energy emission rate (in watt) at 300 K.

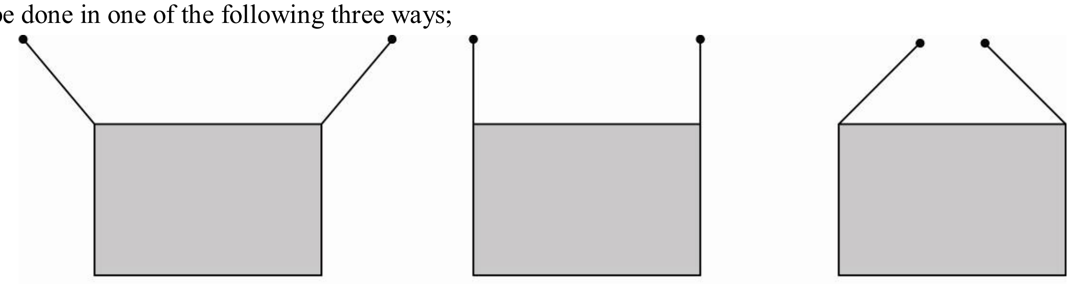
(A) 320 **(B)** 5 **(C)** 32 **(D)** 16

5. Two identical blocks of metal are at $20^\circ C$ and $80^\circ C$ respectively. The specific heat of the material of the two blocks increases with temperature. Which of the following is true about the final temperature T_f when the two blocks are brought into contact (Assuming that no heat is lost to the surrounding)

(A) T_f will be $50^\circ C$ **(B)** T_f wll be more than $50^\circ C$ **(C)** T_f will be less than $50^\circ C$

(D) T_f can be more than or less than $50^\circ C$ depending on the precise variation of the specific heat with temperature

6. A rectangular frame is to be suspended symmetrically by two strings of equal length on two supports figure). It can be done in one of the following three ways;



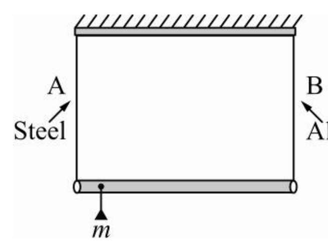
(i) (ii) (iii)

The tension in the strings will be :

(A) the same in all cases **(B)** least in (i) **(C)** least in (ii) **(D)** least in (iii)

7. A rod of length l and negligible mass is suspended at its two ends by two wires of steel (wire A) and aluminium (wire B) of equal length. The cross-sectional areas of wires A and B are $1.0\,mm^2$ and $2.0\,mm^2$, respectively.

$\left(Y_{Al} = 70 \times 10^9\,Nm^{-2} \text{ and } Y_{steel} = 200 \times 10^9\,Nm^{-2}\right)$



(A) Mass m should be suspended close to wire A to have equal stresses in both the wires

(B) Mass m should be suspended close to wire B to have equal stresses in both the wires

(C) Mass m should be suspended at the middle of the wires to have equal stresses in both the wires

(D) Mass m should be suspended close to wire A to have greater stress in wire B

8. When a stone is suspended by a wire, the extension of the wire is ℓ. When the wire is rotated by holding the top end, it traces a conical pendulum of semi vertical angle θ. The extension of the wire is : (Neglect the mass of wire)

(A) $\frac{\ell}{cos\,\theta}$ **(B)** $\ell\, cos\,\theta$ **(C)** $\ell\, sin\,\theta$ **(D)** $\frac{\ell}{sin\,\theta}$

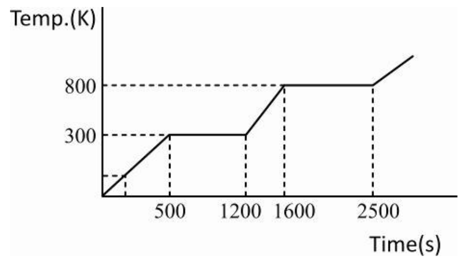
9. Three spheres A, B and C having radii R, $2R$ and $3R$ respectively are coated with carbon black on their outer surface. The wavelengths corresponding to maximum intensity are $3000\,Å, 4000\,Å, 5000\,Å$ respectively. The power radiated by them are Q_A, Q_B and Q_C.

(A) Q_A is maximum **(B)** Q_B is maximum **(C)** Q_C is minimum **(D)** $Q_A = Q_B = Q_C$

10. The pressure that has to be applied to the ends of a steel wire of length 10 cm to keep its length constant when its temperature is raised by $100°C$ is (For steel Young's modulus is $2 \times 10^{11}\ Nm^{-2}$ and coefficient of thermal expansion is $1.1 \times 10^{-5} K^{-1}$) (If the deformation is small, then the stress in a body is directly proportional to the corresponding strain.)

(A) $2.2 \times 10^8\ Pa$ **(B)** $2.2 \times 10^9\ Pa$ **(C)** $2.2 \times 10^7\ Pa$ **(D)** $2.2 \times 10^6\ Pa$

11. A heating curve has been plotted for a solid object as shown in the figure. If the mass of the object is 200 g, then latent heat of vaporization for the material of the object, is : [Power supplied to the object is constant and equal to 1 kW]



(A) $4.5 \times 10^6\ J/kg$ **(B)** $4.5 \times 10^6\ cal/kg$

(C) $4.5 \times 10^4\ J/kg$ **(D)** $4.5 \times 10^4\ cal/kg$

***12.** For an ideal liquid :

(A) the bulk modulus is infinite **(B)** the bulk modulus is zero

(C) the shear modulus is infinite **(D)** the shear modulus is zero

***13.** A copper and a steel wire of the same diameter are connected end to end. A deforming force F is applied to this composite wire which causes a total elongation of 1 cm. The two wires will have. $(Y_{Cu} \neq Y_{St})$.

(A) the same stress **(B)** different stress **(C)** the same strain. **(D)** different strain

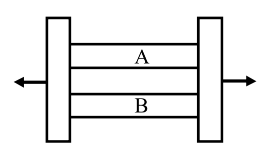
14. Three rods of copper, brass and steel are welded together to form a Y-shaped structure. Area of cross-section of each rod = 4 cm^2. End of copper rod is maintained at $100°C$ whereas ends of brass and steel are kept at $0°C$. Lengths of the copper, brass and steel rods are 46, 13 and 12 cm, respectively. Thermal conductivities of copper, brass and steel are 0.92, 0.26 and 0.12 CGS units, respectively. Rate of heat flow through copper rod is :

(A) 1.2 cal/s **(B)** 2.4 cal/s **(C)** 4.8 cal/s **(D)** 6.0 cal/s

REASONING TYPE FOR QUESTION 15 - 16

(A) Statement-1 is True, Statement-2 is True and Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True and Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

15. **Statement 1 :** Two rods A and B of different material but of the same length and equal areas of cross-section are held fixed at both ends as shown. If a force is now applied, the energy densities are the same.



Statement 2 : The strains in the two materials are the same.

16. **Statement 1 :** Water is filled up in a hollow cylinder container of conducting bases and adiabatic curved surface and kept vertical in an isolated system. If temperature of cylinder is decreased slowly from the bottom. The ice formation will start from the bottom.

Statement 2 : The temperature of liquid which is at the top will be lowest first.

PARAGRAPH FOR QUESTIONS 17 - 19

A rod of length ℓ, cross section area *A*, mass *m* and Young's modulus *Y* is suspended vertically. It has some elongation. The rod is now pivoted at one end so that it is horizontal. Then it is rotated uniformly in the horizontal plane about the vertical axis through the pivot. It has the same elongation.

17. When suspended vertically, the elongation in the top half is :

(A) $\frac{mg\ell}{8AY}$ **(B)** $\frac{mg\ell}{4AY}$ **(C)** $\frac{3mg\ell}{8AY}$ **(D)** $\frac{mg\ell}{2AY}$

18. If the rod is rotated with angular velocity ω, the tension at the pivot is :

(A) $\frac{m}{2}\omega^2\ell$ **(B)** $\frac{m}{2}\omega^2\ell g$ **(C)** $\frac{m}{3}\omega^2\ell$ **(D)** $m\omega^2\ell$

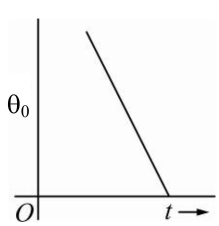
19. In the above case value of ω to produce the same extension as in the first case in which the rod is suspended vertically is :

(A) $\sqrt{\frac{g}{\ell}}$ **(B)** $\sqrt{\frac{g}{2\ell}}$ **(C)** $\sqrt{\frac{3g}{\ell}}$ **(D)** $\sqrt{\frac{3g}{2\ell}}$

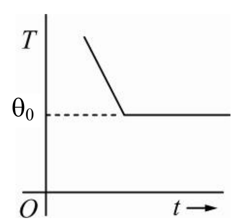
20. A uniform cylinder of length *L* and mass *M* having cross-section area *A* is suspended with its length vertical from a fixed point by a massless spring such that it is half submerged in a liquid of density σ at equilibrium position. The extension x_0 of the spring when it is in equilibrium, is :

(A) $\frac{Mg}{k}$ **(B)** $\frac{Mg}{k}\left(1-\frac{LA\sigma}{M}\right)$ **(C)** $\frac{Mg}{k}\left(1-\frac{LA\sigma}{2M}\right)$ **(D)** $\frac{Mg}{k}\left(1+\frac{LA\sigma}{M}\right)$

21. If a piece of metal is heated to temperatures and then allowed to cool in a room which is at temperature θ_0, the graph between the temperature *T* of the metal and time *t* will be closed to :



(A)



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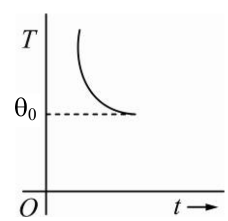
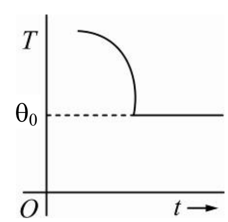


(C)



(D)

22. A bob of mass 10 *kg* is attached to a wire 0.3 *m* long. Its breaking stress is $4.8 \times 10^7\ N/m^2$. The area of cross-section of the wire is $10^{-6}\ m^2$. What is the maximum angular velocity with which it can be rotated in a horizontal circle ?

(A) 8 *rad/s* **(B)** 4 *rad/s* **(C)** 2 *rad/s* **(D)** 1 *rad/s*

23. The specific heat capacity of a metal at low temperature (*T*) is given as $C_p\left(kj\ K^{-1}\ kg^{-1}\right)=32\left(\frac{T}{400}\right)^3$.

A 100 *g* vessel of this metal is to be cooled from 20 *K* to 4 *K* by a special refrigerator operating at room temperature $(27°C)$. This amount of work required to cool the vessel is :

(A) Equal to 0.002 *kJ* **(B)** Greater than 0.148 *kJ*

(C) Between 0.148 *kJ* and 0.028 *kJ* **(D)** Less than 0.028 *kJ*

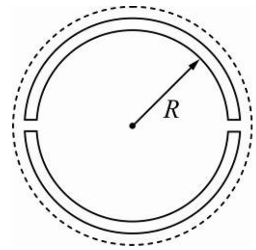
24. If a spring of stiffness k is cut into two parts A and B of length $l_A : l_B = 2:3,$ then the stiffness of spring A is given by :

(A) $\frac{5}{2}k$ **(B)** $\frac{3k}{2}$ **(C)** $\frac{2k}{5}$ **(D)** k

25. A sonometer wire of length $1.5\,m$ is made of steel. The tension in it produces an elastic strain of 1%. What is the fundamental frequency of steel, if density and elasticity of steel are $7.7 \times 10^3\, kg/m^3$ and $2.2 \times 10^{11}\, N/m^2$, respectively ?

(A) $188.5\,Hz$ **(B)** $178.2\,Hz$ **(C)** $200.5\,Hz$ **(D)** $770\,Hz$

26. A wooden wheel of radius R is made of two semi-circular parts (see figure). The two parts are held together by a ring made of a metal strip of cross-sectional area S and length L. L is slightly less than $2\pi R$. To fit the ring on the wheel, it is heated so that its temperature rises by ΔT and it just steps over the wheel. As it cools down to surrounding temperature, it presses the semi-circular parts together. If the coefficient of linear expansion of the metal is α and its Young's modulus is Y, the force that one part of the wheel applies on the other part is :



(A) $2\pi\, SY\alpha\, \Delta T$ **(B)** $SY\alpha\, \Delta T$ **(C)** $\pi SY\alpha\Delta T$ **(D)** $2SY\alpha\, \Delta T$

27. Two wires are made of the same material and have the same volume. However, wire 1 has cross-section area A and wire 2 has cross-section area $3A$. If the length of wire 1 increases by Δx on applying force F, how much force is needed to stretch wire 2 by the same amount ?

(A) F **(B)** $4F$ **(C)** $6F$ **(D)** $9F$

28. A spherical solid ball of volume V is made of a material of density ρ_1. It is falling though a liquid of density ρ_2 $(\rho_2 < \rho_1)$. [Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed v, *i.e.*, $F_{viscous} = -kv^2\,(k > 0)$]. The terminal speed of the ball is :

(A) $\sqrt{\frac{Vg(\rho_1 - \rho_2)}{k}}$ **(B)** $\frac{Vg\rho_1}{k}$ **(C)** $\sqrt{\frac{Vg\rho_1}{k}}$ **(D)** $\frac{Vg(\rho_1 - \rho_2)}{k}$

29. The value of coefficient of volume expansion of glycerin is $5 \times 10^{-4} K^{-1}$. The fractional change in the density of glycerin for a rise of 40°C in its temperature, is :

(A) 0.025 **(B)** 0.010 **(C)** 0.015 **(D)** 0.020

30. The Young's modulus of steel is twice that of brass. Two wires of same length and same area of cross section, one of steel and another of brass are suspended from the same roof. If we want the lower ends of the wires to be at the same level, then the weights added to the steel and brass wires must be in the ratio of :

(A) $4:1$ **(B)** $1:1$ **(C)** $1:2$ **(D)** $2:1$

31. The two ends of a metal rod are maintained at temperatures $100°C$ and $110°C$. The rate of heat flow in the rod is found to be $4.0\,J/s$. If the ends are maintained at temperatures $200°C$ and $210°C$, the rate of heat flow will be :

(A) 8 J/s **(B)** 4 J/s **(C)** 2 J/s **(D)** 6 J/s

32. On observing light from three different starts P, Q and R, it was found that intensity of violet colour is maximum in the spectrum of P, the intensity of green colour is maximum in the spectrum of R and the intensity of red colour is maximum in the spectrum of Q. If T_P, T_Q and T_R are the respective absolute temperatures of P, Q and R then it can be concluded from the observations that :

(A) $T_P < T_R < T_Q$ **(B)** $T_P < T_Q < T_R$ **(C)** $T_P > T_Q > T_R$ **(D)** $T_P > T_R > T_Q$

33. The approximate depth of an ocean is $2700\,m$. The compressibility of water is $45.4\times10^{-11}\,Pa^{-1}$ and density of water is $10^3\,kg/m^3$. What fractional compression of water will be obtained at the bottom of the ocean ?

(A) 1.2×10^{-2} **(B)** 1.4×10^{-2} **(C)** 0.8×10^{-2} **(D)** 1.0×10^{-2}

34. Steam at 100°C is passed into 20g of water at 10°C. When water acquires a temperature of 80°C, the mass of water present will be : [Take specific heat of water $= 1\,cal\,g^{-1}\,°C^{-1}$ and latent heat of steam $= 540\,cal\,g^{-1}$]

(A) $24\,g$ **(B)** $31.5\,g$ **(C)** $42.5\,g$ **(D)** $22.5\,g$

35. Certain quantity of water cools from 70°C to 60°C in the first 5 minutes and 60° to 54°C in next 5 minutes. The temperature of the surrounding is :

(A) 45°C **(B)** 20°C **(C)** 42°C **(D)** 10°C

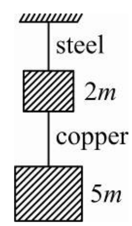
36. A piece of iron is heated in a flame. It first becomes dull red then becomes reddish yellow and finally turns to white hot. The correct explanation for the above observation is possible by using :

(A) Kirchhoff's Law **(B)** Newton's Law of cooling
(C) Stefan's Law **(D)** Wien's displacement Law

37. The following four wires are made of the same material. Which of these will have the largest extension when the same tension is applied ?

(A) length $= 200\,cm$, diameter $= 2\,mm$ **(B)** length $= 300\,cm$, diameter $= 3\,mm$
(C) length $= 50\,cm$, diameter $= 0.5\,mm$ **(D)** length $= 100\,cm$, diameter $= 1\,mm$

38. If the ratio of diameters, lengths and Young's modulus of steel and copper wires shown in the figure are p, q and s respectively, then the corresponding ratio of increase in their lengths would be :



(A) $\dfrac{5q}{(7sp^2)}$ **(B)** $\dfrac{7q}{(5sp^2)}$
(C) $\dfrac{2q}{(5sp)}$ **(D)** $\dfrac{7q}{(5sp)}$

39. If the radius of a star is R and it acts as a black body, what would be the temperature of the star, in which the rate of energy production is Q ?

(A) $\dfrac{Q}{4\pi R^2\sigma}$ **(B)** $\left(\dfrac{Q}{4\pi R^2\sigma}\right)^{-1/2}$ **(C)** $\left(\dfrac{4\pi R^2 Q}{\sigma}\right)^{1/4}$ **(D)** $\left(\dfrac{Q}{4\pi R^2\sigma}\right)^{1/4}$

40. A slab of same area $0.36\,m^2$ and thickness $0.1\,m$ is exposed on the lower surface to steam 100°C. A block of ice at 0°C rests on the upper surface of the slab. In one hour $4.8\,kg$ of ice is melted. The thermal conductivity of slab is : (Given latent heat of fusion of ice $= 3.36\times10^5\,J\,kg^{-1}$)

(A) $1.24\,J/m/s/°C$ **(B)** $1.29\,J/m/s/°C$ **(C)** $2.05\,J/m/s/°C$ **(D)** $1.02\,J/m/s/°C$

41. A cylindrical metallic rod in thermal contact with two reservoirs of heat at its two ends conducts an amount of heat Q in time t. The metallic rod is melted and the material is formed into a rod of half the radius of the origin rod. What is the amount of heat conduced by the new rod, when placed in thermal contact with the two reservoirs in time t ?

(A) $\dfrac{Q}{4}$ **(B)** $\dfrac{Q}{16}$ **(C)** $2Q$ **(D)** $\dfrac{Q}{2}$

42. Assuming the sun to have a spherical outer surface of radius r, radiating like a black body at temperature $t°C$, the power received by a unit surface, (normal to the incident rays) at a distance R from the centre of the sun is :

(A) $\dfrac{r^2\sigma(t-273)^4}{4\pi R^2}$ **(B)** $\dfrac{16\pi^2 r^2\sigma t^4}{R^2}$ **(C)** $\dfrac{r^2\sigma(t+273)^4}{R^2}$ **(D)** $\dfrac{4\pi r^2\sigma t^4}{R^2}$

43. A black body at $1227°C$ emits radiations with maximum intensity at a wavelength of $5000\,Å$. If the temperature of the body is increased by $1000\,Å$, the maximum intensity will be observed at :

(A) $3000\,Å$ **(B)** $4000\,Å$ **(C)** $5000\,Å$ **(D)** $6000\,Å$

44. For a black body at temperature $727°C$, its radiating power is 60 watt and temperature of surrounding is $227°C$. If temperature of black body is changed to $1227°C$ then its radiating power will be :

(A) 304 W **(B)** 320 W **(C)** 240 W **(D)** 120 W

45. A cylindrical rod having temperature T_1 and T_2 at its end. The rate of flow of heat Q_1 cal/sec. If all the linear dimension are doubled keeping temperature constant, then rate of flow of heats Q_2 will be :

(A) $4Q_1$ **(B)** $2Q_1$ **(C)** $\dfrac{Q_1}{4}$ **(D)** $\dfrac{Q_1}{2}$

46. A black body has maximum wavelength λm at $2000\,K$. Its corresponding maximum wavelength at $3000\,K$ will be :

(A) $\dfrac{3}{2}\lambda m$ **(B)** $\dfrac{2}{3}\lambda m$ **(C)** $\dfrac{16}{81}\lambda m$ **(D)** $\dfrac{81}{16}\lambda m$

47. A beaker full of hot water is kept in a room. If it cools from $80°C$ to $75°C$ in t_1 minutes, $75°C$ to $70°C$ in t_2 minutes and $70°C$ to $65°C$ in t_3 minutes, then :

(A) $t_1 < t_2 < t_3$ **(B)** $t_1 > t_2 > t_3$ **(C)** $t_1 = t_2 = t_3$ **(D)** $t_1 > t_2 = t_3$

INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

48. A steel wire of diameter 0.8 *mm* and length 1 *m* is clamped firmly at two points A and B which are 1 *m* apart and in the same plane. A body is hung from the middle point of the wire such that the middle point sags 1 *cm* lower from the original position. Calculate the mass of the body in gram. Given that Young's modulus of the material of wire is $2\times10^{11}\,N/m^2$.

49. A smooth uniform string of natural length l, cross-sectional area A and Young's modulus Y is pulled along its length by a force F on a horizontal surface. Find the elastic potential energy U stored in the string. If $U = \dfrac{F^2 l}{k\,AY}$, find value of K?

50. A copper sphere is suspended in an evacuated chamber maintained at 300 K. The sphere is maintained at a constant temperature of 500 K by heating it electrically. A total of 300 W of electric power is needed to do it. When half of the surface of the copper sphere is completely blackened, 600 W is needed to maintain the same temperature of the sphere. Calculate the emissivity of copper. If $e = \dfrac{1}{K}$, find K?

51. In a 20 *m* deep lake, the bottom is at a constant temperature of 4°C. The air temperature is constant at –10°C. The thermal conductivity of ice is 4 times that water. Neglecting the expansion of water on freezing, the maximum thickness of ice will be equal to $\frac{2\times10^K}{11}m,$ find *K*?

52. A horizontally oriented copper rod of length l = 1.0 *m* is rotated about a vertical axis passing through its middle. What is the number of *rps* (revolutions per second) at which this rod ruptures?

Breaking strength of copper $\sigma_b = 3.0\times10^7\,N/m^2,$ density of copper $= 8.9\times10^3\,kg/m^3.$

53. A ring of radius *r* = 25 *cm* made of lead wire is rotated about a stationary vertical axis passing through its centre and perpendicular to the plane of the ring. What is the number of *rps* at which the ring ruptures?

Breaking strength of lead $= 1.5\times10^7\,N/m^2,$ and density of lead $\rho = 1.13\times10^4\,kg/m^3.$

54. A thin uniform metallic rod of length 0.5 *m* and radius 0.1 *m* rotates with an angular velocity 400 *rad/s* in a horizontal plane about a vertical axis passing through one of its ends the elongation of the rod. The density of material of the rod is $10^4\,kg/m^3$ and the Young's modulus is $2\times10^{11}\,N/m^2.$ If $\Delta L = \frac{1}{3}\times10^{-k}\,m$ then find *k*.

55. An iron wire *AB* of length 3*m* at 0°*C* is stretched between the opposite walls of a brass casing at 0°*C*. The diameter of the wire is 0.6 *mm*. If extra tension will be set up in the wire when the temperature of the system is raised to 40°*C* is 1.420×10^k then find value of *k*.

Given $\alpha_{\text{brass}} = 18\times10^{-6}/K$

$\alpha_{\text{iron}} = 12\times10^{-6}/K$

$Y_{\text{iron}} = 21\times10^{10}\,N/m^2$

56. A calorimeter contains 400 *g* of water at a temperature of 5°*C*. Then, 200 *g* of water at a temperature of +10°*C* and 400 *g* of ice at a temperature of –60°C are added. What is the final temperature of the contents of calorimeter?

Specific heat capacity of water = 1000 *cal/kg/K*

Specific latent heat of fusion of ice = 80 × 1000 *cal/kg*

Relative specific heat of ice = 0.5

57. A body cools down from 60°*C* to 55°*C* in 30 *s*. Using Newton's law of cooling, calculate the time taken by same body to cool down from 55°*C* to 50°*C*. Assume that the temperature of surrounding is 45°*C*. Mark answer as GIF of time found.

58. Some water at 0°*C* is placed in a large insulated enclosure (vessel). The water vapour formed is pumped out continuously. What fraction of the water will ultimately freeze, if the latent heat of vapourization is seven times the latent heat of fusion? If the fraction is 7/*k* find *k*?

59. When 2 *kg* block of copper at 100°*C* is put in an ice container with 0.75 *kg* of ice at 0°*C*, find the equilibrium temperature and final composition of the mixture. Given that specific heat of copper is 378 *J/kg K* and that of water is 4200 *J/kg K* and the latent heat of fusion of ice is $3.36\times10^5\,J/kg.$

60. A metal block of density $5000\,kg/m^3$ and mass 2 *kg* is suspended by a spring of force constant 200 *N/m*. The spring block system is submerged in a water vessel. Total mass of water in it is 300 *g* and in equilibrium the block is at a height 40 *cm* above the bottom of vessel. If the support is broken. If the rise in temperature of water is $4.9\times10^{-k}.$ Find value of *k*? Specific heat of the material of block is 250 *J/kg K* and that of water is 4200 *J/kg K*. Neglect the heat capacities of the vessel and the spring.

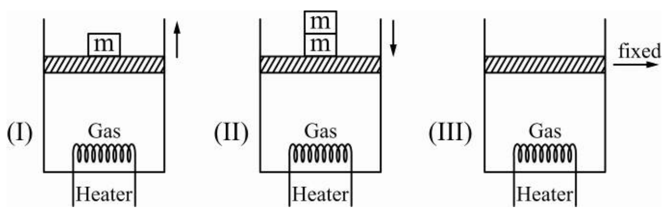
61. A metal container of mass 500 *gm* contains 200 *gm* of water at 20°*C*. A block of iron also of mass 200 *gm* at 100°*C* is dropped into water. Find the equilibrium temperature of the water. Given that specific heats of metal of container and iron and that of water are 910 *J/kg K*. 470 *J/kg K* and 4200 *J/kg K* respectively.

62. Three rods of the same length are arranged to form an equilateral triangle. Two rods made of the same material of coefficient of linear expansion α_1 and the third rod which forms the base of the triangle has coefficient of expansion α_2. The altitude of the triangle will remain the same at all temperatures if α_1/α_1 is nearly $= \frac{1}{k}$ find *k*. Mark your answer as GIF of equilibrium temperature.

Gaseous State & Thermodynamics

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

1. A sample of gas is heated by three different methods from same initial state as shown. In each methods heat supplied is the same. In (*I*) piston moves up by some amount. In (*II*) piston moves down and in (*III*) piston does not move. Specific heat of the gas calculated in each of the methods to be C_I, C_{II} *and* C_{III}.



(A) $C_I > C_{II} > C_{III}$ **(B)** $C_{II} > C_I > C_{III}$ **(C)** $C_{III} > C_{II} > C_I$ **(D)** $C_I > C_{III} > C_{II}$

2. A cubic vessel (with faces horizontal + vertical) contains an ideal gas at NTP. The vessel is being carried by a rocket which is moving at a speed of $500\,m\,s^{-1}$ in vertical direction. The pressure of the gas inside the vessel as observed by us on the ground :

(A) remains the same because $500 m s^{-1}$ is very much smaller than υ_{rms} of the gas.

(B) remains the same because motion of the vessel as a whole does not affect the relative motion of the gas and the walls.

(C) will increase by a factor equal to $\left(\upsilon_{rms}^2 + (500)^2\right)/\upsilon_{rms}^2$ where υ_{rms} was the original mean square velocity of the gas.

(D) will be different on the top wall and bottom wall of the vessel.

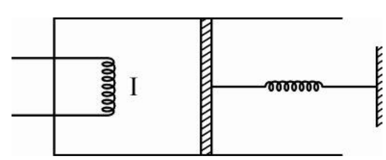
3. An ideal gas is enclosed in a non-conducting cylinder as shown in the figure. The piston is connected with an ideal spring whose one end is fixed. When heater supplies heat to the gas, gas does work

(A) Against atmospheric pressure plus spring

(B) Against spring only

(C) Equal to heat supplies by the heater

(D) None of these



4. At ordinary temperatures, the molecules of an ideal gas have only translational and rotational kinetic energies. At higher temperatures, they may also have vibrational energy. As a result, at higher temperatures :

(A) $C_p = 3\frac{R}{2}$ for monatomic gas **(B)** $C_v > 3\frac{R}{2}$ for monatomic gas

(C) $C_v < 5\frac{R}{2}$ for diatomic gas **(D)** $C_v > 5\frac{R}{2}$ for diatomic gas

***5.** When an ideal monatomic gas is heated at constant pressure, which of the following may be true :

(A) $\frac{dU}{dQ} = \frac{3}{5}$ **(B)** $\frac{dW}{dQ} = \frac{2}{5}$ **(C)** $\frac{dU}{dQ} = \frac{4}{5}$ **(D)** $dW + dU + d\theta = 0$

6. Which of the following parameters is the same for molecules of all gases at a given temperature ?

(A) mass **(B)** speed **(C)** momentum **(D)** kinetic energy

7. A gas behaves more closely as an ideal gas at :

(A) low pressure and low temperature **(B)** low pressure and high temperature

(C) high pressure and low temperature **(D)** high pressure and high temperature

8. The energy of a given sample of an ideal depends only on its :

(A) volume **(B)** pressure **(C)** density **(D)** temperature

9. The first law of thermodynamics is a statement of :

(A) conservation of heat **(B)** conservation of work

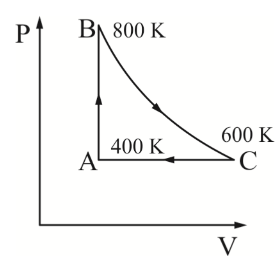
(C) conservation of momentum **(D)** conservation of energy

10. If heat is supplied to an ideal gas in an isothermal process,

(A) the internal energy of the gas will increase **(B)** the gas will do positive work

(C) the gas will do negative work **(D)** the said process is not possible

11. One mole of a diatomic ideal gas undergoes a cyclic process *ABC* as shown in the figure. The process *BC* is adiabatic. The temperatures at *A*, *B*, and *C* are 400*K*, 800*K* and 600*K* respectively. Choose the correct statement.



(A) The change in internal energy in whole cyclic process is 250 R

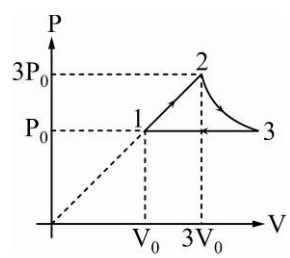
(B) The change in internal energy in the process *CA* is 700 R

(C) The change in internal energy in the process *AB* is $-350R$

(D) The change in internal energy in the process *BC* is $-500R$

PARAGRAPH FOR QUESTION 12 - 13

One mole of an ideal monoatomic gas undergoes a cyclic process as shown in figure. Temperature at point $1 = 300\,K$ and process 2-3 is isothermal.



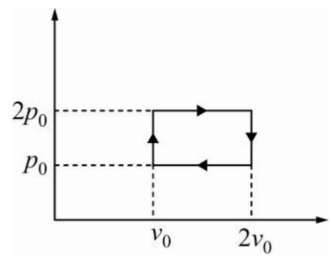
12. Net work done by gas in process $1 \to 2$ is :

(A) $4P_0V_0$ **(B)** $3P_0V_0$ **(C)** $2P_0V_0$ **(D)** $6P_0V_0$

13. Heat capacity of process $2 \to 3$ is :

(A) $\frac{R}{2}$ **(B)** $\frac{3R}{2}$ **(C)** $\frac{5R}{2}$ **(D)** infinite

14. The given *p*-*V* diagram represents the thermodynamic cycle of an engine, operating with an ideal monatomic gas. The amount of heat, extracted from the source in a single cycle is :



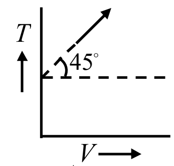
(A) p_0V_0 **(B)** $\left(\frac{13}{2}\right)p_0V_0$

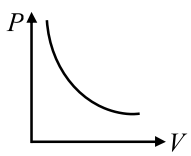
(C) $\left(\frac{11}{2}\right)p_0V_0$ **(D)** $4p_0V_0$

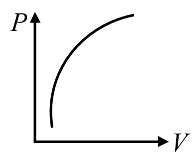
15. One *kg* of a diatomic gas is at a pressure of $8\times10^4\ Nm^{-2}$. The density of the gas is $4\ kgm^{-3}$. What is the energy of the gas due to its thermal motion :

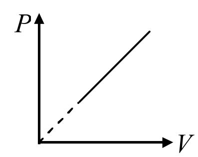
(A) $3\times10^4\ J$ **(B)** $5\times10^4\ J$ **(C)** $6\times10^4\ J$ **(D)** $7\times10^4\ J$

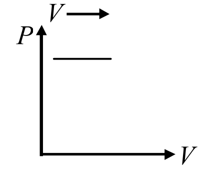
16. The given curve represents the variation of temperature as a function of volume for one mole of an ideal gas. Which of the following curves best represents the variation of pressure as a function of volume ?



(A) 

(B) 

(C) 

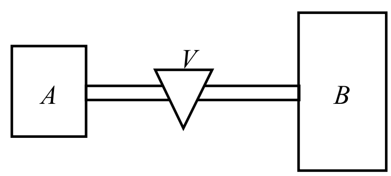
(D) 

17. An ideal gas enclosed in a vertical cylindrical container supports a freely moving piston of mass *M*. The piston and the cylinder have equal cross-sectional area *A*. When the piston is in equilibrium, the volume of the gas is V_0 and its pressure is p_0. The piston is slightly displaced from the equilibrium position and released. Assuming that the system is completely isolated from its surrounding the piston executes a simple harmonic motion with frequency.

(A) $\frac{1}{2\pi}\frac{A\gamma p_0}{V_0M}$ **(B)** $\frac{1}{2\pi}\frac{V_0Mp_0}{A^2\gamma}$ **(C)** $\frac{1}{2\pi}\sqrt{\frac{A^2\gamma p_0}{M\ V_0}}$ **(D)** $\frac{1}{2\pi}\sqrt{\frac{MV_0}{A\gamma p_0}}$

PARAGRAPH FOR QUESTIONS 18 - 20

Container *A* holds an ideal gas at a pressure $1\times10^5\ Pa$ and at 300 *K*. Container *B* whose volume is 4 times the volume of *A* has the same ideal gas at 400 *K* and at a pressure of 5×10^5 *Pa*.



18. The value *V* is adjusted to allow the pressure to equalize, but the temperature of each container is kept constant at the initial value, then the pressure in the two containers is :

(A) $4\times10^5\ Pa$ **(B)** $3\times10^5\ Pa$ **(C)** $2\times10^5\ Pa$ **(D)** $1\times10^5\ Pa$

19. Before the value *V* is opened, v_A and v_B are *rms* velocities of the molecules in container *A* and *B* then $\frac{v_A}{v_B}=$

(A) $\frac{2}{\sqrt{3}}$ **(B)** $\frac{\sqrt{3}}{2}$ **(C)** $\frac{3}{4}$ **(D)** $\frac{4}{3}$

20. Before the value *V* is opened, the ratio of number of moles of gas in *A* and *B* is :

(A) 15/1 **(B)** 1/30 **(C)** 1/15 **(D)** 30/1

21. An open glass tube is immersed in mercury in such a way that a length of 8 *cm* extends above the mercury level. The open end of the tube is then closed and sealed and the tube is raised vertically up by additional 46 *cm*. What will be length of the air column above mercury in the tube now? (Atmospheric pressure 76 *cm* of *Hg*)

(A) 16 *cm* **(B)** 22 *cm* **(C)** 38 *cm* **(D)** 6 *cm*

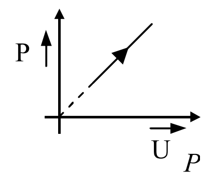
22. Pressure exerted by an ideal gas on the walls of the vessel containing it is due to the

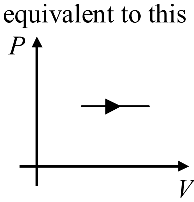
(A) change in kinetic energy of the gas molecules as the strike the walls
(B) collisions between the gas molecules
(C) repulsive force between the gas molecules
(D) change in momentum of the gas molecules as they strike the walls

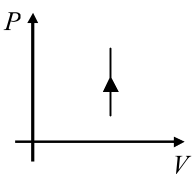
23. One mole of an ideal gas (mono-atomic) at temperature T_0 expands slowly according to law $P^2 = cT$ (c is constant). If final temperature is $2T_0$, heat supplied to gas is:

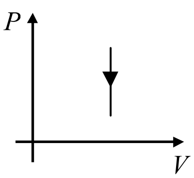
(A) $2RT_0$ **(B)** $\frac{3}{2}RT_0$ **(C)** RT_0 **(D)** $\frac{RT_0}{2}$

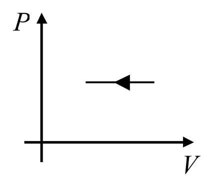
24. The given *P*-*U* graph shows the variation of internal energy of an ideal gas with increase in pressure. Which of the following pressure-volume graph is equivalent to this graph?



(A) 

(B) 

(C) 

(D) 

25. An ideal gas is compressed to half its initial volume by means of several processes. Which of the process results in the maximum work done on the gas ?

(A) Isochoric **(B)** Isothermal **(C)** Adiabatic **(D)** Isobaric

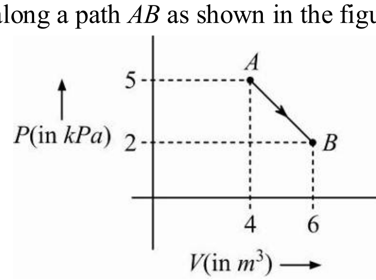
26. Two vessels separately contains two ideal gases *A* and *B* at the same temperature, the pressure of *A* being twice that of *B*. Under such conditions, the density of *A* is found to be 1.5 times the density of *B*. The ratio molecular weight of *A* and *B* is :

(A) 2 **(B)** $\frac{1}{2}$ **(C)** $\frac{2}{3}$ **(D)** $\frac{3}{4}$

27. A Carnot engine, having an efficiency of $\frac{1}{10}$ as heat engine, is used as a refrigerator. If the work done on the system is $10\,J$, the amount of energy absorbed from the reservoir at lower temperature is :

(A) $90\,J$ **(B)** $1\,J$ **(C)** $100\,J$ **(D)** $99\,J$

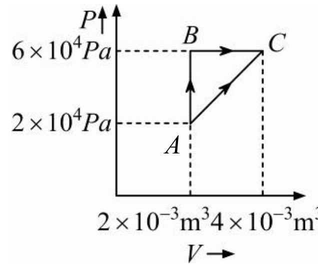
28. One mole of an ideal diatomic gas undergoes a transition from *A* to *B* along a path *AB* as shown in the figure.



The change in internal energy of the gas during the transition is :

(A) $20\,J$ **(B)** $-12\,J$ **(C)** $20\,kJ$ **(D)** $-20\,kJ$

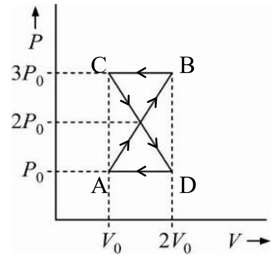
29. Figure below shows two paths that may be taken by a gas to go from a state A to a state C.



In process AB, $400\,J$ of heat is added to the system and in process BC, $100\,J$ of heat is added to the system. The heat absorbed by the system in the process AC will be :

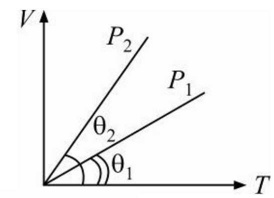
(A) $460\,J$ **(B)** $300\,J$
(C) $380\,J$ **(D)** $500\,J$

30. A thermodynamic system undergoes cyclic process $ABCDA$ as shown in figure. The work done by the system in the cycle is :



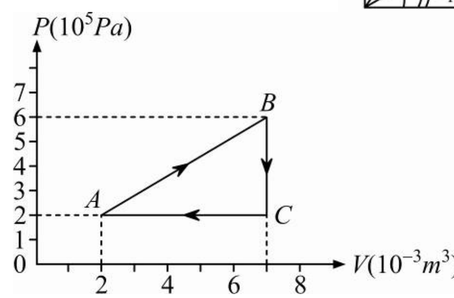
(A) P_0V_0 **(B)** $2P_0V_0$
(C) $\frac{P_0V_0}{2}$ **(D)** zero

31. In the given $(V-T)$ diagram, what is the relation between pressures P_1 and P_2 ?



(A) $P_2 < P_1$ **(B)** Cannot be predicted
(C) $P_2 = P_1$ **(D)** $P_2 > P_1$

32. A gas is taken through the cycle $A \to B \to C \to A$, as shown. What is the net work done by the gas ?

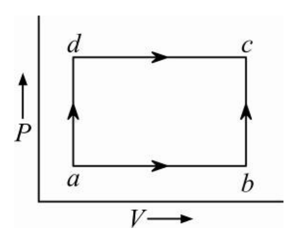


(A) Zero
(B) $-2000\,J$
(C) $2000\,J$
(D) $1000\,J$

33. The amount of heat energy required to raise the temperature of $1\,g$ of Helium at NTP, from T_1K to T_2K is ?

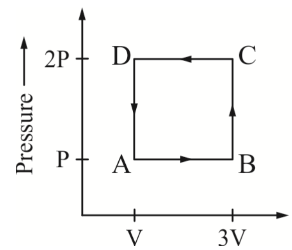
(A) $\frac{3}{4}N_ak_B\left(T_2-T_1\right)$ **(B)** $\frac{3}{4}N_ak_B\left(\frac{T_2}{T_1}\right)$
(C) $\frac{3}{8}N_ak_B\left(T_2-T_1\right)$ **(D)** $\frac{3}{2}N_ak_B\left(T_2-T_1\right)$

34. A system is taken from state a to state c by two paths adc and abc as shown in the figure. The internal energy at a is $U_a = 10\,J$. Along the path adc the amount of heat absorbed $dQ_1 = 50\,J$ and the work obtained $dW_1 = 20\,J$ whereas along the path abc the heat absorbed $dQ_2 = 36\,J$. The amount of work along the path abc is :



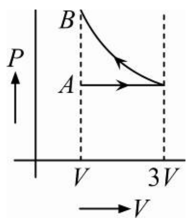
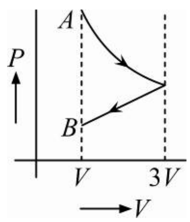
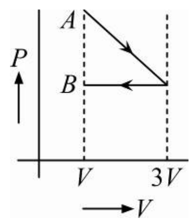
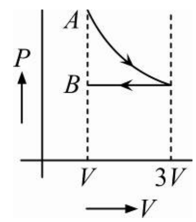
(A) $10\,J$ **(B)** $12\,J$
(C) $36\,J$ **(D)** $6\,J$

35. A thermodynamic system is taken through the cycle $ABCD$ as shown in figure. Heat rejected by the gas during the cycle is :

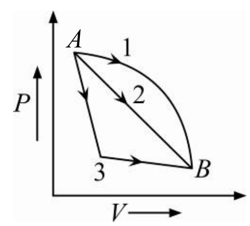


(A) $2PV$
(B) $4PV$
(C) $\frac{1}{2}PV$
(D) PV

36. One mole of an ideal gas goes from an initial state A to final state B via two processes : It first undergoes isothermal expansion from volume V to $3V$ and then its volume is reduced from $3V$ to V at constant pressure. The correct $P-V$ diagram representing the two processes is :

(A) 
(B) 
(C) 
(D) 

37. An ideal gas goes from state A to state B via three different processes as indicated in the $P-V$ diagram. If Q_1, Q_2, Q_3 indicate the heat absorbed by the gas along the three processes and $\Delta u_1, \Delta u_2, \Delta u_3$ indicate the change in internal energy along the three processes respectively, then :



(A) $Q_1 > Q_2 > Q_3$ and $\Delta u_1 = \Delta u_2 = \Delta u_3$
(B) $Q_3 > Q_2 > Q_1$ and $\Delta u_1 = \Delta u_2 = \Delta u_3$
(C) $Q_1 = Q_2 = Q_3$ and $\Delta u_1 > \Delta u_2 > \Delta u_3$
(D) $Q_3 > Q_2 > Q_1$ and $\Delta u_1 > \Delta u_2 > \Delta u_3$

38. During an isothermal expansion, a confined ideal gas does 150 J of work against its surroundings. This implies that:
(A) $150 J$ of heat been removed from the gas
(B) $300 J$ of heat has been added to the gas
(C) no heat is transferred because the process is isothermal
(D) $150 J$ of heat has been added to the gas

39. If Δu and ΔW represent the increase in internal energy and work done by the system respectively in a thermodynamics process, which of the following is true ?
(A) $\Delta u = -\Delta W$, in a adiabatic process
(B) $\Delta u = \Delta W$, in a isothermal process
(C) $\Delta u = \Delta W$, in a adiabatic process
(D) $\Delta u = -\Delta W$, in a isothermal process

40. If c_p and c_v denote the specific heats (per unit mass of and ideal gas of molecular weight M, then :
(A) $c_p - c_v = R/M^2$
(B) $c_p - c_v = R$
(C) $c_p - c_v = R/M$
(D) $c_p - c_v = MR$
Where R is the mole gas constant.

41. The internal energy change in a system that has absorbed 2 kcal of heat and done $500 J$ of work is :
(A) $6400 J$
(B) $5400 J$
(C) $7900 J$
(D) $8900 J$

42. An engine has an efficiency of $1/6$. When the temperature of sink is reduced by $62°C$, its efficiency is doubled. Temperatures of the source is :

(A) $37°C$ **(B)** $62°C$ **(C)** $99°C$ **(D)** $124°C$

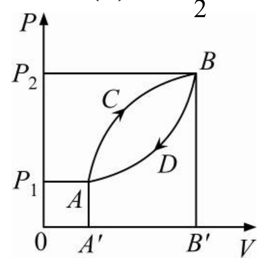
43. If a ratio of specific heat of a gas at constant pressure to that at constant volume is γ, the change in internal energy of a mass of gas, when the volume changes from V to $2V$ at constant pressure P, is :

(A) $\frac{PV}{(\gamma-1)}$ **(B)** PV **(C)** $\frac{R}{(\gamma-1)}$ **(D)** $\frac{\gamma PV}{(\gamma-1)}$

44. In an adiabatic change, the pressure and temperature of a monatomic gas are related as $P \propto T^C$, where C equals:

(A) $\frac{3}{5}$ **(B)** $\frac{5}{3}$ **(C)** $\frac{2}{5}$ **(D)** $\frac{5}{2}$

45. A thermodynamic system is taken from state A to B alone ACB and is bought back to A along BDA as shown in the PV diagram. The net work done during the complete cycle is given by the area :



(A) $P_1ACBP_2P_1$ **(B)** $ACBB'A'A$

(C) $ACBDA$ **(D)** $ADBB'A'A$

46. For hydrogen gas $C_P - C_V = a$ and for oxygen gas $C_P - C_V = b$, so the relation between a and b, so the relation between a and b is given by :

(A) $a = 16b$ **(B)** $16b = a$ **(C)** $a = 4b$ **(D)** $a = b$

47. Three containers of the same volume contain three different gases. The masses of the molecules are m_1, m_2 and m_3 and the number of molecules in their respective containers are N_1, N_2 and N_3. The gas pressure in the containers are P_1, P_2 and P_3 respectively. All the gases are now mixed and put in the one these containers. The pressure P of the mixture will be :

(A) $P < (P_1 + P_2 + P_3)$ **(B)** $P = \frac{P_1 + P_2 + P_3}{3}$ **(C)** $P = P_1 + P_2 + P_3$ **(D)** $P > (P_1 + P_2 + P_3)$

INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

48. The root mean square (rms) speed of oxygen molecules (O_2) at a certain absolute temperature is V. If the temperature is doubled and oxygen gas dissociates into atomic oxygen then rms speed was found to be nV. Value of n must be.

49. During an adiabatic process the pressure of the gas is found to be proportional to the cube of its absolute temperature. The ratio of $\frac{C_p}{C_v}$ is γ. Find 2γ.

50. A mixture of n_1 moles of monoatomic gas and n_2 moles of diatomic gas has $\frac{C_p}{C_v} = \gamma = 1.5$. The relation between n_1 and n_2 was found to be $n_1 = Kn_2$. Determine the value of K.

51. 2000 J of heat leaves the system and 2500J of work is done on the system. Then the change is internal energy of the system was found to be $\frac{x}{10}$. Find value of x.

52. A sample of ideal gas is expanded to twice its original volume of 1.00 m^3 is a qnasi-static process for which $P = \alpha V^2$, with $\alpha = 5.00\ atm/m^6$. Amount of work done by the gas was found to be 1.18×10^n J. Find the value of n.

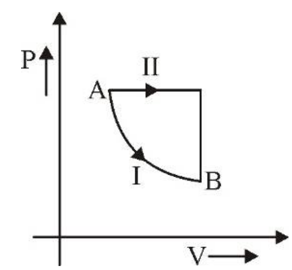
53. A gas has molar heat capacity C = 4.5R follows the process PT = constant. Then, number of degree of freedom will be ____.

54. A heat engine receives 50 Kcal of heat from the source per cycle, and operates with an efficiency of 20%. Heat rejected to the sink per cycle was found to be $y \times 10$ Kcal. Then y is _____.

55. If P-V diagram of a diatomic gas is plotted, it is a straight line passing through origin. The molar heat capacity of the gas in the process is nR where n is an integer. The value of n is.

56. A vessel contains helium, which expands at constant pressure when 15kJ of heat is supplied to it. The variation of the internal energy of the gas (in kJ) will be _____.

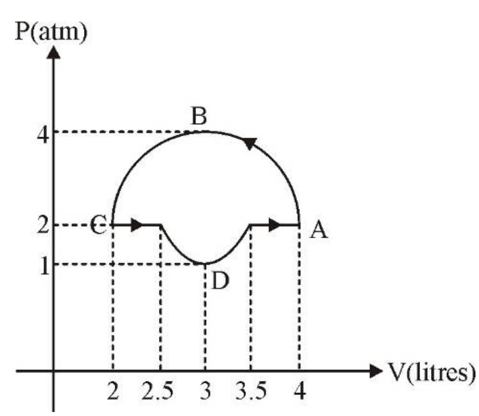
57. A certain mass of gas is taken from an initial thermodynamics state A to another state B by process I and II. In process I the gas does 5J of work and absorbs 4J of heat energy. In process II the gas absorbs 5J of heat. The work done by the gas in process II is :-



58. The value of $\gamma = \frac{C_p}{C_v}$ is $\frac{4}{3}$ for an adiabatic process of an ideal gas for which internal energy $U = K + nPV$. The value of n (k is a constant) is :

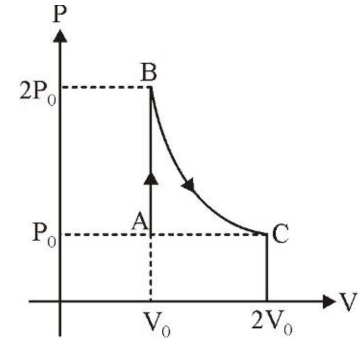
59. Heat $Q = \frac{3}{2}RT$ is supplied to 4 moles of an ideal diatomic gas at temperature T, which remains constant. Number of moles of gas dissociated into atoms is :

60. Work done by the gas in the process shown is figure was found to be $\frac{-n\pi}{4}$ atm L where n is an integer. Find n.



61. In an adiabatic process, $R = \frac{2}{3}C_V$. The pressure of the gas was found to be proportional to T^x. Then 4x will be:

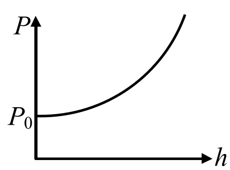
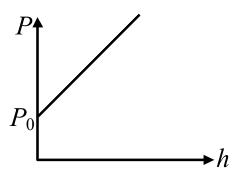
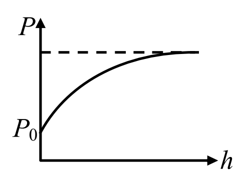
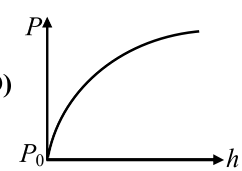
62. A diatomic ideal gas undergoes a thermodynamic change according to the P-V diagram shown in the given figure. The heat given to the gas is nearly nP_0V_0 where n is some real number. Then nearest integer to10n will be (process BC is isothermal)



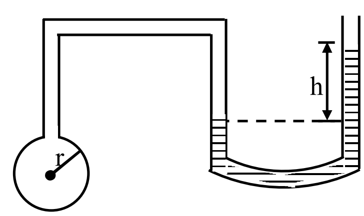
Liquids

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

1. At shallow depth, *h* the pressure in the ocean is simply given by $P = P_0 + \rho gh$, in which ρ is the density of water and P_0 is the air pressure, As we go deeper, the high pressure causes the water to compress and become denser. Which of the following sketches illustrates the correct dependence of the pressure on the depth *h*?

(A) 
(B) 
(C) 
(D) 

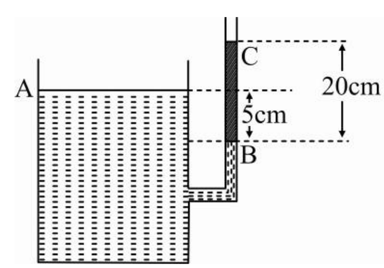
2. If the radius of the air-bubble on the side of tube is *r* and difference in height of liquid of density ρ in manometer is *h*, then surface tension of liquid used to make the bubble is:



(A) $T = 2r\,\rho hg$ **(B)** $T = \frac{rh\rho g}{4}$

(C) $T = \frac{2\pi rh\rho g}{2}$ **(D)** $T = \frac{rh\rho g}{2}$

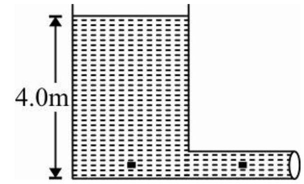
3. A container has a vertical tube, connected to it at its side. An unknown liquid reaches level *A* in the container and level *B* in the tube level *A* being 5.0 *cm* higher than level *B*. The liquid supports a 20.0 *cm* high column of oil, between levels *B* and *C*, whose density is 500 kg/m^3. In figure, density of unknown liquid is :



(A) 1800 kg/m^2 **(B)** 2000 kg/m^3

(C) 1400 kg/m^3 **(D)** 1600 kg/m^3

4. A vented tank of large cross-sectional area has a horizontal pipe 0.12 *m* in diameter at the bottom. The tank holds a liquid whose density is 1500 kg/m^3 to a height of 4.0 *m*. Assume the liquid is an ideal fluid in laminar flow. In figure, the velocity with which fluid flows out is :

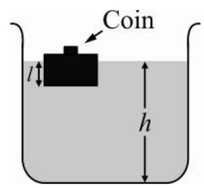


(A) $2\sqrt{5}\ m/s$ **(B)** $\sqrt{5}\ m/s$
(C) $4\sqrt{5}\ m/s$ **(D)** $\sqrt{10}\ m/s$

5. An ideal fluid flows through a pipe of circular cross-section made of two sections with diameters 2.5 *cm* and 3.75 *cm*. The ratio of the velocities in the pipes is :

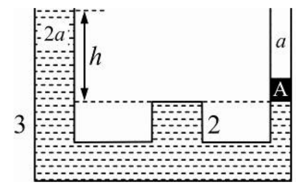
(A) 9 : 4 **(B)** 3 : 2 **(C)** $\sqrt{3} : \sqrt{2}$ **(D)** $\sqrt{2} : \sqrt{3}$

***6.** A wooden block with a coin placed on its top, floats in water shown in figure.
The distance *l* and *h* are shown in the figure. After some time the coin falls into the water. Then :



(A) *l* decreases **(B)** *h* decreases
(C) *l* increases **(D)** *h* increases

7. In the adjoining figure, the cross sectional area of the smaller tube is '*a*' and the larger tube is 2*a*. A block of mass *m* is kept in the smaller tube having the same base area *a*, as that of the tube. The difference in between water levels of the two tubes are :



(A) $\dfrac{P_0}{\rho g}+\dfrac{m}{a\rho}$ **(B)** $\dfrac{P_0}{\rho g}+\dfrac{m}{2a\rho}$ **(C)** $\dfrac{m}{a\rho}$ **(D)** $\dfrac{m}{2a\rho}$

8. Water is being poured into a vessel at a constant rate $\phi\ m^3/s$. There is a small aperture of cross sectional area a at the bottom of the vessel. The maximum level of water in the vessel is proportional is :

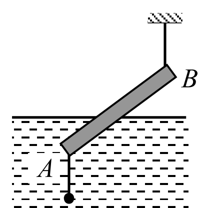
(A) ϕ **(B)** ϕ^2 **(C)** $\dfrac{1}{a}$ **(D)** $\dfrac{1}{a^2}$

9. A small spherical ball falling through a viscous medium of negligible density has terminal velocity *v*. Another ball of the same mass but of radius twice that of the earlier falling through the same viscous medium will have terminal velocity.

(A) *v* **(B)** *v*/4 **(C)** *v*/2 **(D)** 2*v*

10. *A* uniform rod *AB*, 12*m* long weighing 24 *kg*, is supported at end *B* by a flexible light string and a lead weight (of very small size) of 12 *kg* attached at end *A*. The rod floats in water with one-half of its length submerged. For this situation, mark out the correct statement. [Take $g = 10\ m/s^2$, density of water = 100 kg/m^3]

(A) The tension in the string is 36 *g*
(B) The tension in the string is 12 *g*
(C) The volume of the rod is $6.4 \times 10^{-2}\ m^3$
(D) The point of application of the buoyancy force is passing through *C* (centre of mass of rod)



11. In a cylinder piston arrangement, air is under a pressure P_1. A soap bubble of radius r lies inside the cylinder. Soap bubble has the surface tension T. The radius of soap bubble is to be reduced to half. Find the pressure P_2 to which air should be compressed isothermally :

(A) $P_1 + \frac{4T}{r}$ **(B)** $4P_1 + \frac{12T}{r}$ **(C)** $8P_1 + \frac{24T}{r}$ **(D)** $P_1 + \frac{2T}{r}$

12. A tank is filled with water of density $1 g cm^{-3}$ and oil of density $0.9 g\, cm^{-3}$. The height of water layer is 100 *cm* and of the oil layer is 400 *cm*. If g = 980 $cm\, s^{-2}$, then the velocity of efflux from an opening in the bottom of the tank is :

(A) $\sqrt{900 \times 980}\ cms^{-1}$ **(B)** $\sqrt{1000 \times 980}\ cms^{-1}$ **(C)** $\sqrt{920 \times 980}\ cms^{-1}$ **(D)** $\sqrt{950 \times 980}\, cms^{-1}$

13. A cylinder is filled with liquid of density d upto a height h. If the cylinder is at rest, then the mean pressure on the wall is :

(A) $hdg/4$ **(B)** $hdg/2$ **(C)** $2\ hdg$ **(D)** hdg

14. Two drops of equal radius coalesce to form a bigger drop. What is ratio of final and initial surface energy?

(A) $2^{1/2} : 1$ **(B)** $1 : 1$ **(C)** $1 : 2^{1/3}$ **(D)** None of these

15. The reading of a spring balance when a block is suspended from it in air is 60 *N*. The reading is changed to 40 *N* when the block is submerged in water. The specific gravity of the block must be therefore.

(A) 3/2 **(B)** 6 **(C)** 2 **(D)** 3

16. The terminal velocity of spherical ball of radius a falling through a viscous liquid is proportional to :

(A) a **(B)** a^2 **(C)** a^3 **(D)** a^{-1}

17. A small wooden ball of density D is immersed in water of density d to a depth h below the surface of water and then released. Upto what height will the ball jump out of water? ($D < d$)

(A) $\frac{d}{D}h$ **(B)** $\left(\frac{d}{D} - 1\right)h$ **(C)** h **(D)** Zero

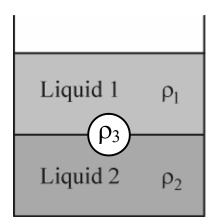
18. A piece of ice is floating in a beaker containing thick sugar solution of water. As the ice melts, the total level of the liquid.

(A) increases **(B)** decreases **(C)** remains unchanged **(D)** insufficient data

19. Two rain drops reach the earth with different terminal velocities having ratio 9 : 4. Then the ratio of their volumes is :

(A) 3 : 2 **(B)** 4 : 9 **(C)** 9 : 4 **(D)** 27 : 8

20. A jar is filled with two non-mixing liquids 1 and 2 having densities ρ_1 and ρ_2 respectively. A solid ball, made of a material of density ρ_3, is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1, ρ_2 and ρ_3 ?



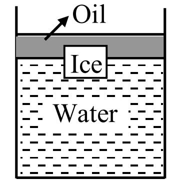
(A) $\rho_3 < \rho_1 < \rho_2$ **(B)** $\rho_1 > \rho_3 > \rho_2$

(C) $\rho_1 < \rho_2 < \rho_3$ **(D)** $\rho_1 < \rho_3 < \rho_2$

21. A hole is made at the bottom of the tank filled with water (density 1000 kg/m^3). If the total pressure at the bottom of the tank is 3 *atm* (1 *atm* = $10^5\ N/m^2$), then the velocity of efflux is :

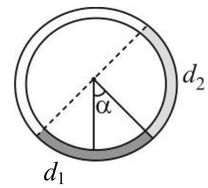
(A) $\sqrt{200} m/s$ **(B)** $\sqrt{400} m/s$ **(C)** $\sqrt{500} m/s$ **(D)** $\sqrt{800}\ m/s$

22. An ice cube is floating in water above which of layer of lighter oil is poured. As the ice melts completely, the level of interface and the upper most level of oil will respectively :



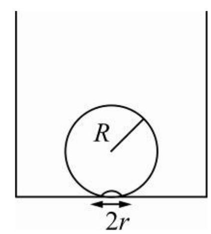
(A) rise and fall **(B)** fall and rise
(C) not change and no change **(D)** not change and fall

23. There is a circular tube in a vertical plane. Two liquids which do not mix and of densities d_1 and d_2 are filled in the tube. Each liquid subtends $90°$ angle at centre. Radius joining their interface makes an angle α with vertical. Ratio $\frac{d_1}{d_2}$ is :



(A) $\frac{1+\sin\alpha}{1-\sin\alpha}$ **(B)** $\frac{1+\cos\alpha}{1-\cos\alpha}$ **(C)** $\frac{1+\tan\alpha}{1-\tan\alpha}$ **(D)** $\frac{1+\sin\alpha}{1-\cos\alpha}$

24. On heating water bubbles being formed at the bottom of the vessel detatch and rise. Take the bubbles to be spheres of radius R and making a circular contact of radius r with the bottom of the vessel. If $r << R$ and the surface tension of water is T, value of r just before bubbles detatch is (density of water is ρ)



(A) $R^2\sqrt{\frac{2\rho_w g}{3T}}$ **(B)** $R^2\sqrt{\frac{\rho_w g}{6T}}$ **(C)** $R^2\sqrt{\frac{\rho_w g}{T}}$ **(D)** $R^2\sqrt{\frac{3\rho_w g}{T}}$

25. Assume that a drop of liquid evaporates by decrease in its surface energy, so that its temperature remains unchanged. What should be the minimum radius of the drop for this to be possible? The surface tension is T, density of liquid is ρ and L is its latent heat of vaporisation.

(A) $\rho L/T$ **(B)** $\sqrt{T/\rho L}$ **(C)** $T/\rho L$ **(D)** $2T/\rho L$

26. If a ball of steel (density, $\rho = 7.8\,g\,cm^{-3}$) attains a terminal velocity of $10\,cms^{-1}$ when falling in tank of water (coefficient of velocity $\eta_{water} = 8.5\times10^{-4}$ $Pa\text{-}s$), then its terminal velocity in glycerine ($\rho = 1.2\,g\,cm^{-3}$, $\eta = 13.2\,Pa\text{-}s$) would be nearly.

(A) $1.6\times10^{-5}\,cms^{-1}$ **(B)** $6.45\times10^{-4}\,cms^{-1}$
(C) $6.25\times10^{-4}\,cms^{-1}$ **(D)** $1.5\times10^{-5}\,cm\,s^{-4}$

27. Water is flowing continuously from a tap having an internal diameter $8\times10^{-3}m$. The water velocity as it leaves the tap is $0.4\,ms^{-1}$. The diameter of the water stream at a distance $2\times10^{-1}m$ below the tap is close to :

(A) $7.5\times10^{-3}\,m$ **(B)** $9.6\times10^{-3}\,m$ **(C)** $3.6\times10^{-3}\,m$ **(D)** $5.0\times10^{-3}\,m$

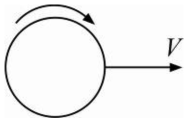
28. An incompressible, non-viscous fluid flows steadily through a cylindrical pipe, which has radius $2R$ at point A and radius R at point B farther along the flow direction. If the velocity of flow at point A is V, the velocity of flow at point B will be :

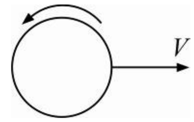
(A) $2V$ **(B)** V **(C)** $V/2$ **(D)** $4V$

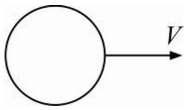
29. A barometer kept in an elevator accelerating upward reads $76\,cm$. The air pressure in the elevator is :

(A) $76\,cm$ **(B)** $< 76\,m$ **(C)** $> 76\,m$ **(D)** zero

30. To get maximum flight, a ball must be thrown as :

(A) 

(B) 

(C) 

(D) Any of (A), (B) and (C)

31. The weight of a body in water is one third of its weight in air. The density of the body is :

(A) $0.5\, gm/cm^3$ **(B)** $1.5\, gm/cm^3$ **(C)** $2.5\, gm/cm^3$ **(D)** $3.5\, gm/cm^3$

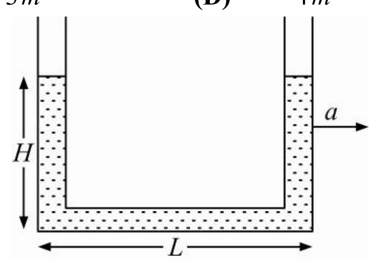
32. A solid uniform ball having volume V and density ρ floats at the interface of two immiscible liquids. The densities of the upper and the lower liquids are ρ_1 and ρ_2 respectively, such that $\rho_1 < \rho < \rho_2$. The fraction of the volume of the ball in the lower liquid is :

(A) $\frac{\rho - \rho_2}{\rho_1 - \rho_2}$ **(B)** $\frac{\rho_1}{\rho_1 - \rho_2}$ **(C)** $\frac{\rho_1 - \rho}{\rho_1 - \rho_2}$ **(D)** $\frac{\rho_1 - \rho_2}{\rho_2}$

33. The specific gravity of ice is 0.9. The area of the smallest slab of ice of height $0.5\, m$ floating in fresh water that will just support a $100\, kg$ man is :

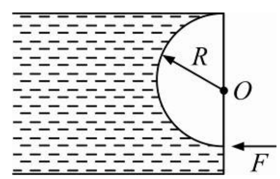
(A) $1.5\, m^2$ **(B)** $2\, m^2$ **(C)** $3\, m^2$ **(D)** $4\, m^2$

34. A liquids stands at the same level in the U-tube when at rest. If area of cross-section of both the limbs are equal, the difference in heights h of the liquid in the two limbs of U-tube, when the system is given an acceleration a in horizontal direction as shown, is :



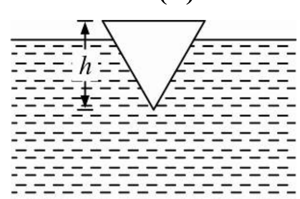
(A) $\frac{gL^2}{aH}$ **(B)** $\frac{La}{g}$ **(C)** $\frac{L^2 a}{Hg}$ **(D)** $\frac{Hg}{a}$

35. The figure shows a semi-cylindrical massless gate (of radius R) pivoted at the point O holding a stationary liquid of density ρ. A horizontal force F is applied at its lowest position to keep it stationary. The magnitude of the force is :



(A) $\frac{9}{2}\rho g R^3$ **(B)** $\frac{3}{2}\rho g R^3$ **(C)** $\rho g R^3$ **(D)** Zero

36. A conical block, floats in water with 90% height immersed in it. Height h of the of the block is equal to the diameter of the block i.e., 20 cm The mass to be kept on the block, so that the block just floats at the surface of water, is :



(A) $568\, g$ **(B)** $980\, g$ **(C)** $112\, g$ **(D)** $196\, g$

37. A gas having density ρ flows with a velocity v along a pipe of cross sectional area s and bent at an angle of 90° at a point A. The force exerted by the gas on the pipe at A is :

(A) $\frac{\sqrt{2}\, sv}{\rho}$ **(B)** $\sqrt{2}\, sv^2 \rho$ **(C)** $\frac{\sqrt{3}\, sv^2 \rho}{2}$ **(D)** $\sqrt{3}\, sv^2 \rho$

38. With increase in temperature the viscosity of :

(A) both gases and liquids increases **(B)** both gases and liquids decreases

(C) gases increases and liquids decreases **(D)** gases decreases and liquids increases

Paragraph for Questions 39 - 41

Water rises in a vertical capillary tube upto a height of $10\,cm$. The tube is now inclined at $45°$ with horizontal.

39. The length of water risen in the tube will be :

(A) $10\sqrt{2}\,cm$ **(B)** $10\,cm$ **(C)** $10/\sqrt{2}\,cm$ **(D)** None of these

40. If the angle of contact is $0°$ the shape of meniscus is :

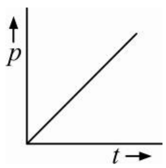
(A) Plane **(B)** parabolic **(C)** cylindrical **(D)** hemispherical

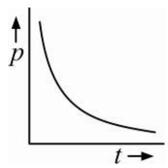
41. The pressure just below the meniscus is :

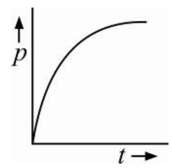
(A) is greater than just above it **(B)** is lesser than just above it

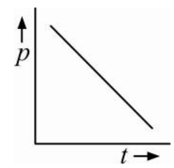
(C) is same as just above it **(D)** is always equal to atmosphere pressure

42. A soap bubble is blown slowly at the end of a tube by a pump supplying air at a constant rate. Which one of the following graphs represents the correct variation of the excess of pressure inside the bubble with time :

(A) 

(B) 

(C) 

(D) 

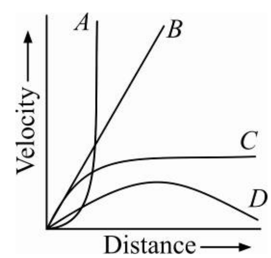
43. If a small sphere is let fall vertically in a liquid of density smaller than that of the material of the sphere :

(A) at first velocity increases, but soon approaches a constant value

(B) it falls with constant velocity all along from the very beginning

(C) at first it falls with a constant velocity which after some time goes on decreasing

(D) nothing can be said about its motion



44. A small spherical solid ball is dropped in a viscous liquid. Its journey in the liquid is best described in the figure drawn by :

(A) curve A **(B)** curve B

(C) curve C **(D)** curve D

45. Air is pushed into a soap bubble of radius r to double its radius. If the surface tension of the soap solution is S, the work done in the process is :

(A) $8\pi r^2 S$ **(B)** $12\pi r^2 S$ **(C)** $16\pi r^2 S$ **(D)** $24\pi r^2 S$

46. An air bubble of radius r in water is at a depth h below the water surface at some instant. If P is atmosphere pressure and d and T are the density and surface tension of water respectively, the pressure inside the bubble will be :

(A) $P + hdg - \frac{4T}{r}$ **(B)** $P + hdg + \frac{2T}{r}$ **(C)** $P + hdg - \frac{2T}{r}$ **(D)** $P + hdg + \frac{4T}{r}$

47. Water rises in a capillary tube to a certain height such that the upward force due to surface tension is balanced by $75 \times 10^{-4} N$, force due to the weight of the liquid. If the surface tension of water is $6 \times 10^{-2}\ N/m$, the inner circumference of the capillary must be :

(A) $1.25 \times 10^{-2} m$ **(B)** $0.50 \times 10^{-2} m$ **(C)** $6.5 \times 10^{-2} m$ **(D)** $12.5 \times 10^{-2} m$

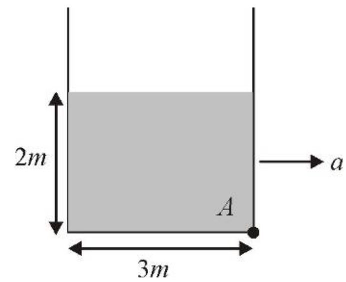
INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

48. A U-tube filled with water with density of 1000 kg/m^3. One column of tube is filled with glycerin with density 1300 kg/m^3. If height of glycerin is 5 cm. If height difference of both the column is h find $10h$(cm).

49. A cube of wood supporting 200 g mass just floats in water. When the mass is removed, the cube rises by 2cm. What is the side length of cube (in cm).

50. The minimum horizontal acceleration of the so that presence at point A of the contains becomes atmospheric is (tank is of sufficient height) $\left(\frac{a}{b}\right)g$ (g is acceleration due to gravity). Find $(a + b)$.

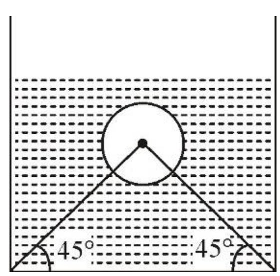


51. A lawn sprinkler has 20 holes, each of sectional area $2\times10^{-2}\,cm^2$ and is connected to a hose pipe of cross section area a 2.4 cm^2. If the speed of water in the hose pipe is 1.5 m/s, the speed of water as it emerges from the holes is (m/s).

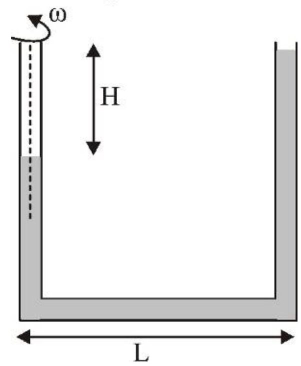
52. A beaker containing water is placed on platform of a spring balance. The balance reads 0.5kg. A stone of mass 0.5kg and density 10^4 kg/m^3 is immersed in water without touching the walls of the beaker. What will be the balance reading now (in kg).

53. A liquid is kept in a cylindrical vessel which is rotated along its axis. The liquid rises at its side. If the radius of vessel is 5cm and speed of rotation is 5 rev/s, then the difference in the height of the liquid at the centre of the vessel and its sides is (in cm).

54. A hollow sphere of mass M = 50 kg and radius $r=\left(\frac{3}{40\pi}\right)^{1/3} m$ is immersed in a tank of water $(\rho_w = 10^3\,kg/m^3)$. The sphere is tied to the bottom of a tank by two wires A and b as shown. Tension in wire A is T_0. Find $\frac{T}{\sqrt{2}}$.



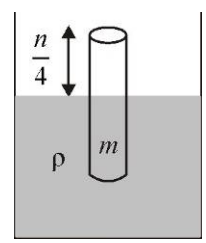
55. A U-shaped tube contains a liquid of density e and it is rotated about the left dotted line a shown in figure, with angular speed $w = \sqrt{\frac{4g}{L}}$. Find $2L$.



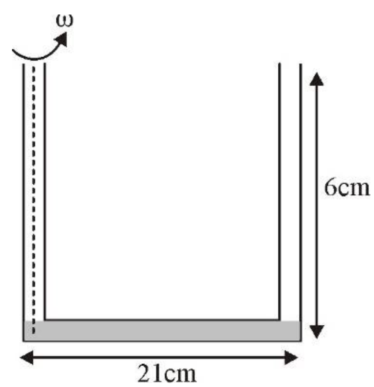
56. A sphere is just immersed in a liquid. The ratio of hydrostatic force acting on bottom and top half of the sphere will be.

57. A capillary tube of length l = 50cm and radius $r = \frac{1}{4} mm$ is immersed vertically into water. The capillary rise will be : (Angle of contact θ = 0°, surface tension = 72 dyne/cm, g = 1000 cm/s^{-2}).

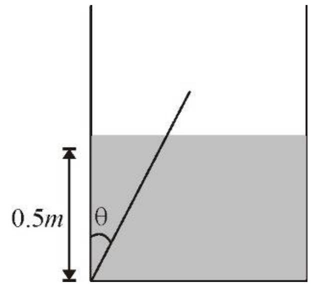
58. A solid cylinder of height h and mass m is floating in a liquid of density ρ as shown in the figure. Find the acceleration (round off to nearest integer) of the vessel (in m/s^2) containing liquid for which the relative acceleration of the completely immersed cylinder w.r.t vessel becomes equal to one-third of that of vessel take (g = 10 m/s^2)



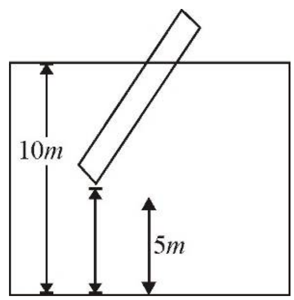
59. Length of horizontal arm of a uniform cross section U-tube is $l = 21$cm and ends of both vertical arms are open to surrounding of pressure $10 \times 500\,\mathrm{N/m^2}$. A liquid of density $(\rho = 101\,\mathrm{g/m^3})$ is poured into the tube such that liquid just files the horizontal part of the tube. None one of the open ends is sealed and the tube is then rotated about a vertical axis passing through the other vertical arm with angular velocity $w_0 = 10\,\mathrm{rad/sec}$. If length of each vertical arm is a = 6cm, calculate the length of air column in the sealed arm (in cm).



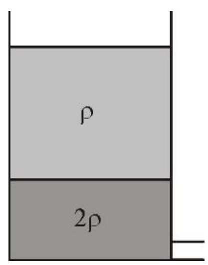
60. A wooden plank of length 1m and uniform cross section is hinged at one end of the bottom of the tank as shown in the figure. The tank is filled with water upto a height of 0.5m. The specific gravity of the plank is 0.5. If the angle θ by the inclination of that the plank makes with the vertical in the equilibrium position (enclude the case θ = 0) Find the value of $\frac{1}{\cos^2 \theta}$



61. A rod of length 6m has specific gravity $\rho = \left(\frac{35}{36}\right)$. One end of the rod is tied to a 5m long rope, which in turn is tied to a floor of a pool 10m deep, as shown. Find the length (in m) of the part of the rod which is out of water.



62. The velocity of the liquid out of a small hole of a vessel containing two different liquids of densities 2ρ and ρ as shown in the figure is :- $n\sqrt{gh}$ then the value of n is?



Simple Harmonic Motion

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

1. A body executing *SHM* on a straight line *ABC* has extreme position at points *A* and *C*, such that $AB = a$ and $BC = b$. The velocity of particle at mid point of line *ABC* is *u*. The time period of *SHM* is :

(A) $\frac{2\pi(b-a)}{u}$ **(B)** $\frac{\pi(b-a)}{u}$ **(C)** $\frac{\pi(a+b)}{u}$ **(D)** $\frac{2\pi(a+b)}{u}$

2. A spring has an equilibrium length of 2.0 meters and a spring constant of 10 N/m. Alice is pulling on one end of the spring with a force 3.0 N. Bob is pulling on the opposite end of the spring with a force of 3.0 N, in the opposite direction. What is the resulting length of the spring?

(A) 1.7 *m* **(B)** 2.0 *m* **(C)** 2.3 *m* **(D)** 2.6 *m*

3. An object of mass 0.2 *kg* executes simple harmonic oscillations along *X*-axis with a frequency of $(25/\pi)$ Hz. At the position $x = 0.04\ m$, the object has kinetic energy of 0.5 *J* and potential energy of 0.4 *J*. Find the amplitude of oscillations (in *cm*). [assume zero potential energy at equilibrium position]

(A) 2 *cm* **(B)** 4 *cm* **(C)** 5 *cm* **(D)** 6 *cm*

4. The displacement of a particle varies with time according to the relation : $y = a \sin \omega t + b \cos \omega t$

(A) The motion is oscillatory but not S.H.M. **(B)** The motion is S.H.M. with amplitude $a + b$

(C) The motion is S.H.M. with amplitude $a^2 + b^2$

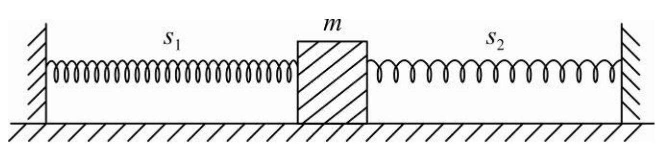
(D) The motion is S.H.M. with amplitude $\sqrt{a^2 + b^2}$

5. The equation of motion of a particle is $x = a \cos(\alpha t)^2$. The motion is :

(A) periodic but not oscillatory **(B)** periodic and oscillatory

(C) oscillatory but not periodic **(D)** neither periodic nor oscillatory

6. When a mass *m* is connected individually to two springs S_1 and S_2 the oscillation frequencies are ν_1 and ν_2. If the same mass is attached to the springs as shown in figure, the oscillation frequency would be :



(A) $\nu_1 + \nu_2$ **(B)** $\sqrt{\nu_1^2 + \nu_2^2}$ **(C)** $\left(\frac{1}{\nu_1} + \frac{1}{\nu_2}\right)^{-1}$ **(D)** $\sqrt{\nu_1^2 - \nu_2^2}$

7. A particle executing a simple harmonic motion has a period of 6 *s*. The time taken by the particle to move from the position of half the amplitude, starting from the mean position is :

(A) 1/4*s* **(B)** 3/4*s* **(C)** 1/2*s* **(D)** 3/2*s*

8. The speed of propagation of a wave in a medium is $300\,m/s^{-1}$. The equation of motion of point at $x = 0$ is given $y = 0.04 \sin 600\ \pi t$ (meter). The displacement of a point $x = 75\ cm$ at $t = 0.01s$ is:

(A) 0.02 *m* **(B)** 0.04 *m* **(C)** Zero **(D)** 0.028 *m*

9. Consider the following statements :
The total energy of a particle executing simple harmonic motion depends on its

I. amplitude **II.** period **III.** displacement

(A) I and II are correct **(B)** II and III are correct

(C) I and III are correct **(D)** I, II and III are correct

10. The equation of a simple harmonic progressive wave is given by $y = A\sin(100\pi t - 3x)$. Find the distance between 2 particles having a phase difference of $\pi/3$.

(A) $\pi/9\ m$ **(B)** $\pi/18\ m$ **(C)** $\pi/6\ m$ **(D)** $\pi/3\ m$

11. $y = 3\sin\pi\left(\frac{t}{2} - \frac{x}{4}\right)$ represents an equation of a progressive wave, where t is in second and x is in meter. The distance travelled by the wave in 5 s is :

(A) 8 m **(B)** 10 m **(C)** 5 m **(D)** 32 m

PARAGRAPH FOR QUESTIONS 12 - 14

Incident wave $y = A\sin\left(ax + bt + \frac{\pi}{2}\right)$ is reflected by an obstacle at $x = 0$ which reduces intensity of reflected wave by 36%. Due to superposition a resulting wave consist of standing wave and traveling $y = -1.6A\sin ax\ \sin bt + cA\cos(bt + ax)$ where A, a b and c are positive constants.

12. Amplitude of reflected wave is :

(A) $0.6A$ **(B)** $0.8A$ **(C)** $0.4A$ **(D)** $0.2A$

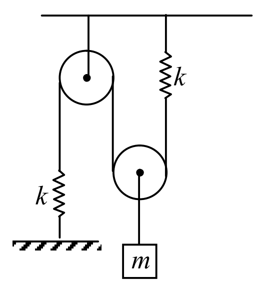
13. Value of c is :

(A) 0.2 **(B)** 0.4 **(C)** 0.6 **(D)** 0.3

14. Position of second antinode is :

(A) $x = \frac{\pi}{3a}$ **(B)** $x = \frac{3\pi}{a}$ **(C)** $x = \frac{3\pi}{2a}$ **(D)** $x = \frac{2\pi}{3a}$

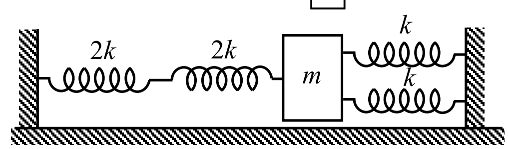
15. Find time period of oscillation for arrangement shown in figure.



(A) $2\pi\sqrt{\frac{m}{2k}}$ **(B)** $2\pi\sqrt{\frac{2m}{k}}$

(C) $\pi\sqrt{\frac{m}{k}}$ **(D)** $\pi\sqrt{\frac{m}{2k}}$

16. A mass is m is attached to four springs of spring constants $2k$, $2k$, k, k as shown in figure. The mass is capable of oscillating on a frictionless horizontal floor. If it is displaced slightly and released the frequency of resulting *SHM* would be :

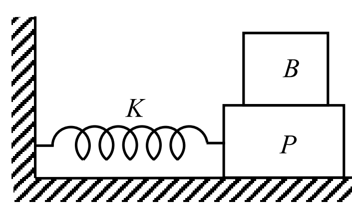


(A) $\frac{1}{2\pi}\sqrt{\left(\frac{11k}{2m}\right)}$ **(B)** $\frac{1}{2\pi}\sqrt{\left(\frac{2k}{3m}\right)}$ **(C)** $\frac{1}{2\pi}\sqrt{\left(\frac{3k}{m}\right)}$ **(D)** $\frac{1}{2\pi}\sqrt{\left(\frac{4k}{m}\right)}$

17. The graph between the time period and the length of a simple pendulum is :

(A) straight line **(B)** curve **(C)** ellipse **(D)** parabola

18. A flat plate P of mass M executes *S.H.M.* on a horizontal plane by sliding over a frictionless surface with a frequency υ. A block B of mass m rests on the plate as shown in figure. Coefficient of friction between the surfaces of B and P is μ_s. If the block B is not to slip on the plate, then the maximum amplitude of oscillation that the plate block system can have is:

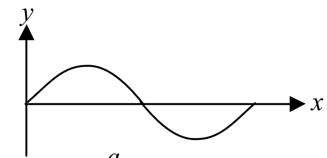


(A) $\frac{\mu g}{4\pi v^2}$ **(B)** $\frac{\mu^2 g}{\pi^2 v^2}$ **(C)** $\frac{\mu^2 g}{4\pi^2 v^2}$ **(D)** $\frac{\mu g}{4\pi^2 v^2}$

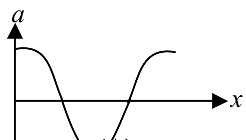
19. When a wave travels in a medium, the particle displacement is given by the equation $y = a\sin 2\pi(bt - cx)$ where a, b and c are constants. The maximum particle velocity will be twice the wave velocity if :

(A) $c = \frac{1}{\pi a}$ **(B)** $c = \pi a$ **(C)** $b = ac$ **(D)** $b = \frac{1}{ac}$

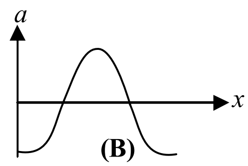
20. Corresponding to y-x graph of a transverse harmonic wave shown in figure



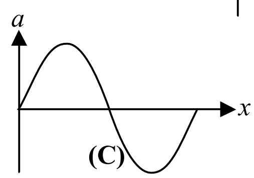
Choose the correct options at same time



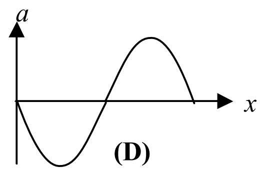
(A)



(B)



(C)



(D)

21. Ratio of kinetic energy at mean position to potential energy at $A/2$ of a particle performing SHM :

(A) 2 : 1 **(B)** 4 : 1 **(C)** 8 : 1 **(D)** 1 : 1

22. The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in $5s$. In another 10 s, it will decrease to α times its original magnitude, where α equals :

(A) 0.7 **(B)** 0.81 **(C)** 0.729 **(D)** 0.6

23. If a simple pendulum has significant amplitude (upto a factor of $1/e$ of original) only in the period between $t = 0$ to $t = \tau$ then τ may be called the average life of the pendulum. When the spherical bob of the pendulum suffers a retardation (due to viscous drag) proportional to its velocity with b as constant of proportionally, the average life time of the pendulum (assuming damping is small) in seconds is :

(A) $\frac{0.693}{b}$ **(B)** b **(C)** $\frac{1}{b}$ **(D)** $\frac{2}{b}$

24. A string is stretched between fixed points separated by $75.0\,cm$. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. The lowest resonant frequency for this string is:

(A) 10.5 Hz **(B)** 105 Hz **(C)** 155 Hz **(D)** 205 Hz

25. A particle executing a simple harmonic motion. Its maximum acceleration is α and maximum velocity is β. Then, its time period of vibration will be :

(A) $\frac{\beta^2}{\alpha}$ **(B)** $\frac{2\pi\beta}{\alpha}$ **(C)** $\frac{\beta^2}{\alpha^2}$ **(D)** $\frac{\alpha}{\beta}$

26. A particle is executing SHM along a straight line. Its velocities at distances x_1 and x_2 from the mean position are V_1 and V_2, respectively. Its time period is :

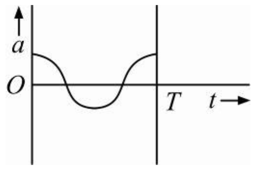
(A) $2\pi\sqrt{\frac{V_1^2 + V_2^2}{x_1^2 + x_2^2}}$ **(B)** $2\pi\sqrt{\frac{V_1^2 - V_2^2}{x_1^2 - x_2^2}}$ **(C)** $2\pi\sqrt{\frac{x_1^2 + x_2^2}{V_1^2 + V_2^2}}$ **(D)** $2\pi\sqrt{\frac{x_2^2 - x_1^2}{V_1^2 - V_2^2}}$

27. When two displacement represented by $y_1 = a\sin(\omega t)$ and $y_2 = b\cos(\omega t)$ are superimposed the motion is :

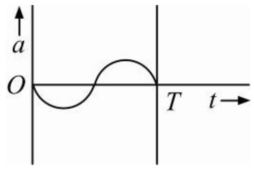
(A) simple harmonic with amplitude $\sqrt{a^2 + b^2}$ **(B)** simple harmonic with amplitude $\frac{(a^2 + b^2)}{2}$

(C) not a simple harmonic **(D)** simple harmonic with amplitude $\frac{a}{b}$

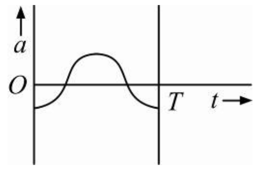
28. The oscillation of a body on a smooth horizontal surface is represented by the equation, $X = A\cos(\omega t)$; where $X =$ displacement at time t and $\omega =$ frequency of oscillation. Which one of the following graphs shows correctly the variation a with t ?



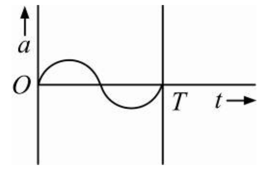
(A)



(B)



(C)



(D)

Here $a =$ acceleration at time t and $T =$ time period

29. Out of the following functions representing motion of a particle which represents SHM :

I. $y = \sin\omega t - \cos\omega t$ **II.** $y = \sin^3 \omega t$

III. $y = 5\cos\left(\frac{3\pi}{4} - 3\omega t\right)$ **IV.** $y = 1 + \omega t + \omega^2 t^2$

The correct choice is :

(A) Only I **(B)** Only IV does not represent SHM
(C) Only I and III **(D)** Only I and II

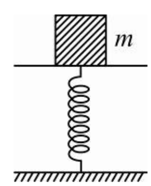
30. Two particles are oscillating along two close parallel straight lines side by side, with the same frequency and amplitudes. They pass each other, moving in opposite directions when their displacement is half of the amplitude. The mean positions of two particles lie on a straight line perpendicular to the paths of the two particles. The phase difference is :

(A) $\frac{\pi}{6}$ **(B)** 0 **(C)** $\frac{2\pi}{3}$ **(D)** π

31. The given displacement of a particle along the x-axis is given by $x = a\sin^2 \omega t$. The motion of the particle corresponds to:

(A) simple harmonic motion of frequency ω/π **(B)** simple harmonic motion of frequency $3\omega/2\pi$
(C) non simple harmonic motion **(D)** simple harmonic motion of frequency $\omega/2\pi$

32. A mass of $2.0\,kg$ is put on a flat pan attached to a vertical spring fixed on the ground as shown in the figure. The mass of the spring and the pan is negligible. When passed slightly and released the mass executes a simple harmonic motion. The spring constant is $200\,N/m$. What should be the minimum amplitude of the motion so that the mass gets detached from the pan ? (Take $g = 10\,m/s^2$).



(A) $10.0\,cm$ **(B)** any value less than $12.0\,cm$
(C) $4.0\,cm$ **(D)** $8.0\,cm$

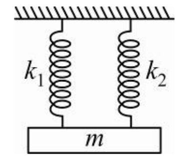
33. The particle executing simple harmonic motion has a kinetic energy $K_0 \cos^2 \omega t$. The maximum values of the potential energy and the total energy are respectively :

(A) $K_0/2$ and K_0 **(B)** K_0 and $2K_0$ **(C)** K_0 and K_0 **(D)** 0 and $2K_0$

34. The time period of mass suspended from a spring is T. If the spring is cut into four equal parts and the same mass is suspended from one of the parts, then the new time period will be :

(A) $T/4$ **(B)** T **(C)** $T/2$ **(D)** $2T$

35. A mass is suspended separately by two different springs in successive order then time periods is t_1 and t_2 respectively. If it is connected by both spring as shown in figure then time period is t_0 the correct relation is :



(A) $t_0^2 = t_1^2 + t_2^2$ **(B)** $t_0^{-2} = t_1^{-2} + t_2^{-2}$

(C) $t_0^{-1} = t_1^{-1} + t_2^{-1}$ **(D)** $t_0 = t_1 + t_2$

36. A particle, with restoring force proportional to displacement and resisting force proportional to velocity is subjected to a force $F \sin \omega_0 t$. If the amplitude of the particle is maximum for $\omega = \omega_1$ and the energy of the particle maximum for $\omega = \omega_2$, then :

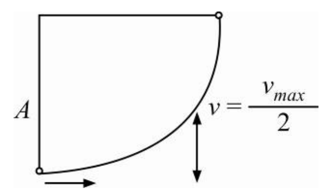
(A) $\omega_1 \neq \omega_0$ and $\omega_2 = \omega_0$ **(B)** $\omega_1 = \omega_0$ and $\omega_2 = \omega_0$

(C) $\omega_1 = \omega_0$ and $\omega_2 \neq \omega_0$ **(D)** $\omega_1 \neq \omega_0$ and $\omega_2 \neq \omega_0$

37. Two SHM's with same amplitude and time period, when acting together in perpendicular directions with a phase difference of $\pi/2$, given rise to :

(A) straight motion **(B)** elliptical motion **(C)** circular motion **(D)** None of these

38. A particle starts with S.H.M. from the mean position as shown in the figure. Its amplitude is A and its time period is T. At one time, its speed is half that of the maximum speed. What is this displacement ?



(A) $\frac{2A}{\sqrt{3}}$ **(B)** $\frac{3A}{\sqrt{2}}$

(C) $\frac{\sqrt{2}A}{3}$ **(D)** $\frac{\sqrt{3}A}{2}$

39. A second pendulum is mounted in a rocket. Its period of oscillation with decreases when rocket is :

(A) moving down with uniform acceleration **(B)** moving around the earth in geostationary orbit

(C) moving up with uniform velocity **(D)** moving up with uniform acceleration

40. A loaded vertical spring executes S.H.M. with a time period of 4 sec. The difference between the kinetic energy and potential energy of this system varies with a period of :

(A) 2 sec **(B)** 1 sec **(C)** 8 sec **(D)** 4 sec

41. A simple pendulum is suspended from the roof of trolley which moves in a horizontal direction with an acceleration a, then the period is given by $T = 2\pi\sqrt{(l/g')}$, where g' is equal to :

(A) g **(B)** $g - a$ **(C)** $g + a$ **(D)** $\sqrt{(g^2 + a^2)}$

42. A body is executing simple harmonic motion. When the displacements from the mean position is 4 cm and 5 cm, the corresponding velocities of the body is $10\,cm/\text{sec}$ and $8\,cm/\text{sec}$. Then the time period of the body is :

(A) 4πs **(B)** 3πs **(C)** πs **(D)** 2πs

43. The angular velocity and the amplitude of a simple pendulum is ω and a respectively. At a displacement x from the mean position if its kinetic energy T and potential energy is V, then the ratio of T to V is :

(A) $\frac{(a^2 - x^2\omega^2)}{x^2\omega^2}$ **(B)** $\frac{x^2\omega^2}{(a^2 - x^2\omega^2)}$ **(C)** $\frac{(a^2 - x^2)}{x^2}$ **(D)** $\frac{x^2}{a^2 - x^2}$

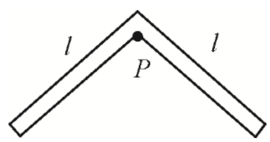
INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

44. A mass at the end of a spring executes harmonic motion about an equilibrium position with an amplitude 1 *m*. Its speed as it passes through the equilibrium position is $3 m/s$. If extended by 2 *m* and released, the speed of the mass passing through the equilibrium position will be.

45. A man of mass 60 kg standing on a plateform executing S.H.M. in the vertical plane. The displacement from the mean position varies as $y = 0.025 sin(kt)$. The minimum value of *k*, for which the man will feel weight less ness at the highest point is : (*y* is in metres and acceleration due to gravity is $10 m/s^2$)

46. A system of two identical rods (L-shaped) of mass $m = 1kg$ and length $L = 60\sqrt{2}$ are resting on a peg *P* as shown in the figure. If the system is displaced in its plane by a small angle q, the period of oscillation is $2k\pi$. Find *k*.



47. A block of mass 1 kg hangs without vibrating at the end of a spring with a force constant $10 N/m$ attached to the ceiling of an elevator. The elevator is rising with an upward acceleration of $g/4$. The acceleration of the elevator suddenly ceases. What is the amplitude of the resulting oscillations.(in cm)

48. If the potential energy of a harmonic oscillator of mass 2 kg on its equilibrium position is 5 joules and the total energy is 9 joules when the amplitude is one meter the period of the oscillator (in sec) is Kπ. Find K:

49. Two particles of mass $3M/4$ and *M*, are connected by a massless spring of free length *L* and force constant *k*. These masses are initially at rest *L* apart on a horizontal frictionless table. A particle of mass $M/4$ moving with speed *v* along the line joining the two connected masses, collides with and sticks to the particle of mass $3M/4$. Find the amplitude with which the spring between the two masses vibration. ($M = 64Kg, k = 0.5 N/m, v = 1 m/s$)

50. A particle is executing SHM on a straight line. *A* and *B* are two points at which its velocity is zero. It passes through a certain point $P\ (AP < PB)$ at successive intervals of 0.5 and 1.5 sec with a speed of $3 m/s$. Find maximum kinetic energy if mass is 1 kg.

51. A platform is executing simple harmonic motion in a vertical direction with an amplitude of 5 cm and a frequency of $10/\pi$ vibrations per seconds. A block is placed on the platform at the lowest point of its path. At what height above the lowest point will the block leave the plat form? (in mm)

52. A particular S.H.M. has an amplitude of *A* & period *T*. The square of the ratio of its maximum velocity to of its velocity $T/8$ seconds after the particle reaches the extreme position would be.

53. Two particles execute SHM of same amplitude of 20 cm with same period along the same line about the same equilibrium position. The maximum distance between the two is 20 cm. Their phase difference in degrees is

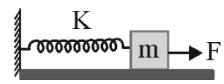
54. A particle executes SHM with time period *T* and amplitude *A*. The maximum possible average velocity in time $T/4$ is KA/T. Find K^2.

55. Two particles P and Q describe simple harmonic motions of same period, same amplitude, along the same line about the same equilibrium position O. When P and Q are on opposite sides of O at the same distance from O they have the same speed of $1.2\,m/s$ in the same direction, when their displacements are the same they have the same speed of $1.6\,m/s$ in opposite directions. The maximum velocity in m/s of either particle is

56. The acceleration of a particle moving along x-axis is $a = -100x + 50$. It is released from $x = 2$. Here 'a' and 'x' are in S.I units. The angular frequency of particle is.

57. A seconds pendulum A (time period 2 second) and another simple pendulum B of slightly less length than A are made to oscillate at $t = 0$ in same phase. If they are again in the same phase first time, after 18 seconds, then the time period of B is(round to nearest one digit).

58. A constant force produces maximum velocity V on the block connected to the spring of force constant K as shown in the fig. When the force constant of spring becomes $4K$, the maximum velocity of the block is a times V. Find 10 times a.



Wave Motion

CHOOSE THE CORRECT ALTERNATIVE. ONLY ONE CHOICE IS CORRECT. HOWEVER, QUESTIONS MARKED '*' MAY HAVE MORE THAN ONE CORRECT OPTION.

1. Equations of a stationary and a travelling waves are as follows : $y_1 = a\,sin\,kx\,cos\,\omega t$ and $y_2 = a\,sin(\omega t - kx)$ The phase difference between two points $x_1 = \frac{\pi}{3k}$ and $x_2 = \frac{3\pi}{2k}$ are ϕ_1 and ϕ_2 respectively for the two waves. The ratio is $\frac{\phi_1}{\phi_2}$:

(A) 1 **(B)** 5/6 **(C)** 3/4 **(D)** 6/7

***2.** Speed of sound wave in air :

(A) is independent of temperature.
(B) increases with pressure.
(C) increases with increase in humidity.
(D) decreases with increase in humidity.

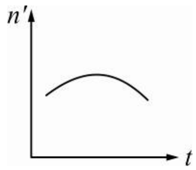
3. A sound wave is passing through air column in the form of compression and rarefaction. In consecutive compressions and rarefactions,

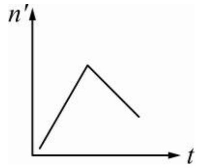
(A) density remains constant
(B) Boyle's law is obeyed
(C) bulk modulus of air oscillates
(D) there is no transfer of heat

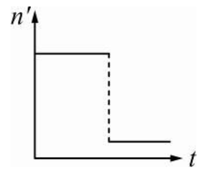
4. Equation of a plane progressive wave is given by $y = 0.6\,sin\,2\pi\left(t - \frac{x}{2}\right)$. On reflection from a denser medium its amplitude becomes $2/3$ of the amplitude of the incident wave. The equation of the reflected wave is :

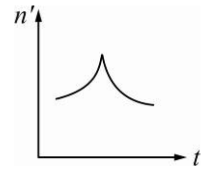
(A) $y = 0.6\,sin\,2\pi\left(t + \frac{x}{2}\right)$
(B) $y = -0.4\,sin\,2\pi\left(t + \frac{x}{2}\right)$
(C) $y = 0.4\,sin\,2\pi\left(t + \frac{x}{2}\right)$
(D) $y = -0.4\,sin\,2\pi\left(t - \frac{x}{2}\right)$

5. A train whistling at constant frequency is moving towards a station at a constant speed V. The train goes past a stationary observer on the station. The frequency n of the sound as heard by the observer is plotted as a function of time t. Identify the expected curve.

(A) 

(B) 

(C) 

(D) 

***6.** During propagation of a plane progressive mechanical wave :

(A) all the particles are vibrating in the same phase
(B) amplitude of all the particles is equal.
(C) particles of the medium executes S.H.M.
(D) wave velocity depends upon the nature of the medium.

***7.** The transverse displacement of a string (clamped at its both ends) is given by :

$$y(x, t) = 0.06\, sin(2\pi x/3)\, cos(120\pi t).$$

All the points on the string between two consecutive nodes vibrate with :

(A) same frequency **(B)** same phase **(C)** same energy **(D)** different amplitude

***8.** A train, standing in a station yard, blows a whistle of frequency 400 Hz in still air. The wind starts blowing in the direction from the yard to the station with a speed of $10\,m/s$. Given that the speed of sound in still air is $340\,m/s$,

(A) the frequency of sound as heard by an observer standing on the platform is $400\,Hz$
(B) the speed of sound for the observer standing on the platform is $350\,m/s$
(C) the frequency of sound as heard by the observer standing on the platform will increase
(D) the frequency of sound as heard by the observer standing on the platform will decrease

9. A string of length 1 m and linear mass density 0.01 kg/m is stretched to a tension of 100 N. When both ends of the string are fixed, the three lowest frequencies for standing wave are f_1, f_2 and f_3. When only one end of the string is fixed, the three lowest frequencies for standing wave are n_1, n_2 and n_3. Then:

(A) $n_3 = 5n_1 = f_3 = 125Hz$ **(B)** $f_3 = 5f_1 = n_2 = 125Hz$

(C) $f_3 = 2n_2 = 3f_1 = 150Hz$ **(D)** $n_2 = \dfrac{f_1 + f_3}{2} = 75Hz$

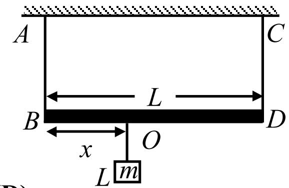
10. A closed organ pipe of cross sectional area $100\,cm^2$ resonates with a tuning fork of frequency $1000\,Hz$ in fundamental tone. The minimum volume of water to be drained out so that the pipe again resonates with the same tuning fork is (take velocity of wave $= 320\,m/s$)

(A) $800\,cm^3$ **(B)** $1200\,cm^3$ **(C)** $1600\,cm^3$ **(D)** $2000\,cm^3$

11. An organ pipe of 3.9 πm long, open at both ends is driven to third harmonic standing wave. If the amplitude of pressure oscillation is 1% of mean atmospheric pressure $\left[p_0 = 10^5\,N/m^2\right]$. The maximum displacement of particle from mean position will be: [Given velocity of sound = 200 m/s and density of air = 1.3 kg/m^3]

(A) 2.5 cm **(B)** 5 cm **(C)** 1 cm **(D)** 2 cm

12. A massless rod of length L is suspended by two identical strings AB and CD of equal length. A block of mass m is suspended from point O such that BO is equal to $'x'$. Further it is observed that the frequency of 1[st] harmonic in AB is equal to 2[nd] harmonic frequency in CD. $'x'$ is:



(A) $\frac{L}{5}$ **(B)** $\frac{4L}{5}$ **(C)** $\frac{3L}{4}$ **(D)** $\frac{L}{4}$

13. The ratio of the velocity of sound in hydrogen $\left(\gamma = \frac{7}{5}\right)$ to that in helium $\left(\gamma = \frac{5}{3}\right)$ at the same temperature is:

(A) $\sqrt{\frac{5}{42}}$ **(B)** $\sqrt{\frac{5}{21}}$ **(C)** $\frac{\sqrt{42}}{5}$ **(D)** $\sqrt{\frac{21}{5}}$

14. A wire under tension vibrates with a fundamental frequency of 600 Hz. If the length of the wire is doubled, the radius is halved and the wire is made to vibrate under one ninth the tension. Then the fundamental frequency will became.

(A) 400 Hz **(B)** 600 Hz **(C)** 300 Hz **(D)** 200 Hz

15. A string fixed at both ends oscillates in 5 segments, length 10 m and velocity of wave is $20 ms^{-1}$. What is the frequency?

(A) 5Hz **(B)** 15 Hz **(C)** 10 Hz **(D)** 2 Hz

16. A string vibrates according to the equation $y = 5 \sin\left(\frac{2\pi x}{3}\right) \cos 20\,\pi t$

Where x and y are in cm and t in second. The distance between two adjacent nodes is:

(A) 3 cm **(B)** 4.5 cm **(C)** 6 cm **(D)** 1.5 cm

17. When two progressive waves $y_1 = 4 \sin(2x - 6t)$ and $y_2 = 3 \sin\left(2x - 6t - \frac{\pi}{2}\right)$ are superimposed, the amplitude of the resultant wave is :

(A) 5 **(B)** 6 **(C)** 5/6 **(D)** 1/2

18. A transverse wave is described by the equation $y = y_0 \sin 2\pi\left(ft - \frac{x}{\lambda}\right)$. The maximum particle velocity is equal to four times the wave velocity, if :

(A) $\lambda = \frac{\pi y_0}{4}$ **(B)** $\lambda = \frac{\pi y_0}{2}$ **(C)** $\lambda = \pi y_0$ **(D)** $\lambda = 2\pi y_0$

19. A car sounding its horn at 480 Hz moves towards a high wall at a speed of $20 ms^{-1}$. If the speed of sound is $340 ms^{-1}$, the frequency of the reflected sound heard by the girl sitting in the car will be closest to :

(A) 540 Hz **(B)** 524 Hz **(C)** 568 Hz **(D)** 480 Hz

20. **Statement I** : Two longitudinal waves given by equations $y_1(x,t) = 2a \quad \sin(\omega t - kx)$ and $y_2(x,t) = a \quad \sin(2\omega t - 2kx)$ will have equal intensity.

Statement II : Intensity of waves of given frequency in same medium is proportional to the square of amplitude only.

(A) If Statement-I is True, Statement-II is True; Statement-II is a correct explanation for Statement-I

(B) If Statement-I is True, Statement-II is True; Statement-II is NOT a correct explanation for Statement-I

(C) If Statement-I is True, Statement-II is False

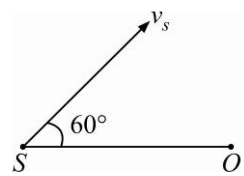
(D) If Statement-I is False, Statement-II is True

21. The transverse displacement y (x, t) of a wave on a string is given $b\ y(x,t)=e^{-\left(ax^2+bt^2+2\sqrt{ab}\ xt\right)}$. This represents a :

(A) Wave moving in $-x$ direction with speed $\sqrt{\frac{b}{a}}$ **(B)** Standing wave of frequency $\sqrt{b}$

(C) Standing wave of frequency $\sqrt{\frac{1}{b}}$ **(D)** Wave moving in $+x$ direction with speed $\sqrt{\frac{a}{b}}$

22. A source of sound S emitting waves of frequency 100 Hz and an observer O are located at some distance from each other. The source is moving with a speed of $19.4\,ms^{-1}$ at an angle of 60° with the source observer line as shown in the figure. The observer is at rest.



The apparent frequency observed by the observer (velocity of sound in air $330\,ms^{-1}$), is :

(A) 106 Hz **(B)** 97 Hz **(C)** 100 Hz **(D)** 103 Hz

23. The fundamental frequency of a closed organ pipe of length $20\,cm$ is equal to the second overtone of an organ pipe open at both the ends. The length of organ pipe open at both the ends is :

(A) $120\,cm$ **(B)** $140\,cm$ **(C)** $80\,cm$ **(D)** $100\,cm$

24. If n_1, n_2 and n_3 are the fundamental frequencies of three segments into which a string is divided, then the original fundamental frequency n of the string is given by :

(A) $\frac{1}{n}=\frac{1}{n_1}+\frac{1}{n_2}+\frac{1}{n_3}$ **(B)** $\frac{1}{\sqrt{n}}=\frac{1}{\sqrt{n_1}}+\frac{1}{\sqrt{n_2}}+\frac{1}{\sqrt{n_3}}$

(C) $\sqrt{n}=\sqrt{n_1}+\sqrt{n_2}+\sqrt{n_3}$ **(D)** $n=n_1+n_2+n_3$

25. The number of possible natural oscillation of air column in a pipe closed at one end of length $85\,cm$ whose frequencies lie below 1250 Hz are : (Velocity of sound $=340\,ms^{-1}$) :

(A) 4 **(B)** 5 **(C)** 7 **(D)** 6

26. A speeding motorcyclist sees traffic jam ahead him. He slows down to $36\,km$ hour^{-1}. He finds that traffic has eased and a car moving ahead of him at $18\,km$ hour^{-1} is honking at a frequency of 1392 Hz. If the speed of sound is $343\,ms^{-1}$, the frequency of the honk as heard by him will be :

(A) 1332 Hz **(B)** 1372 Hz **(C)** 1412 Hz **(D)** 1454 Hz

27. If we study the vibration of a pipe open at both ends, then the following statement is not true.

(A) All harmonics of the fundamental frequency will be generated.
(B) Pressure change will be maximum at both ends.
(C) Open end will be antinode.
(D) Odd harmonics of the fundamental frequency will be generated.

28. A wave travelling in the $+ve$ x-direction having displacement along y-direction as $1m$, wavelength $2\pi m$ and frequency of $\frac{1}{\pi}$ Hz is represented by :

(A) $y=\sin(10\pi x-20\pi t)$ **(B)** $y=\sin(2\pi x-2\pi t)$

(C) $y=\sin(x-2t)$ **(D)** $y=\sin(2\pi x-20\pi t)$

29. A source of unknown frequency gives 4 beats/s when sounded with a source of known frequency 250 Hz. The second harmonic of the source of unknown frequency gives five beats per second, when sounded with a source of frequency 513 Hz. The unknown frequency is :

(A) 240 Hz **(B)** 260 Hz **(C)** 254 Hz **(D)** 246 Hz

30. The length of the wire between two ends of a sonometer is $100\,cm$. What should be the positions of two bridges below the wire so that the three segments of the wire so that the three segments of the wire have their fundamental frequencies in the ratio 1 : 3 : 5.

(A) $\frac{1500}{23}cm, \frac{500}{23}cm$ **(B)** $\frac{1500}{23}cm, \frac{300}{23}cm$

(C) $\frac{300}{23}cm, \frac{1500}{23}cm$ **(D)** $\frac{1500}{23}cm, \frac{2000}{23}cm$

31. When a string is divided into three segments of length l_1, l_2 and l_3 the fundamental frequencies of these three segments are υ_1, υ_2 and υ_3 respectively. The original fundamental frequency (υ) of the string is :

(A) $\sqrt{\upsilon} = \sqrt{\upsilon_1} + \sqrt{\upsilon_2} + \sqrt{\upsilon_3}$ **(B)** $\upsilon = \upsilon_1 + \upsilon_2 + \upsilon_3$

(C) $\frac{1}{\upsilon} = \frac{1}{\upsilon_1} + \frac{1}{\upsilon_2} + \frac{1}{\upsilon_3}$ **(D)** $\frac{1}{\sqrt{\upsilon}} = \frac{1}{\sqrt{\upsilon_1}} + \frac{1}{\sqrt{\upsilon_2}} + \frac{1}{\sqrt{\upsilon_3}}$

32. Two sources of sound placed closed to each other, are emitting progressive wave given by

$y_1 = 4\sin 600\pi t$ and $y_2 = 5\sin 608\pi t$

An observer located near these two sources of sound will hear :

(A) 4 beats per second with intensity ratio $25:16$ between waxing and waning

(B) 8 beats per second with intensity ratio $25:16$ between waxing and waning

(C) 8 beats per second with intensity ratio $81:1$ between waxing and waning

(D) 4 beats per second with intensity ratio $81:1$ between waxing and waning

33. The equation of a simple harmonic wave is given by $y = 3\sin\frac{\pi}{2}(50t - x)$, where x and y are in metres and t is in seconds. The ratio of maximum particle velocity to the wave velocity is :

(A) 2π **(B)** $\frac{3}{2}\pi$ **(C)** 3π **(D)** $\frac{2}{3}\pi$

34. Two waves are represented by the equations $y_1 = a\sin(\omega t + kx + 0.57)$ and $y_2 = a\cos(\omega t + kx)m$, where x is in meter and t in sec. The phase difference between them is :

(A) 1.0 radian **(B)** 1.25 radian **(C)** 1.57 radian **(D)** 0.57 radian

35. Sound waves travel at 350 *m/s* through a warm air and at 3500 *m/s* through brass. The wavelength of a 700 Hz acoustic wave as it enters brass from warm air :

(A) decrease by a factor 10 **(B)** increase by a factor 20

(C) increase by a factor 10 **(D)** decrease by a factor 20

36. Two identical piano wires, kept under the same tension *T* have a fundamental frequency of 600 Hz. The fractional increase in the tension of one of the wires which will lead to occurrence of 6 beats/s when both wires oscillate together would be :

(A) 0.01 **(B)** 0.02 **(C)** 0.03 **(D)** 0.04

37. A tuning fork of frequency 512 Hz makes 4 beats per second with the vibrating string of a piano. The beat frequency decreases to 2 beats per sec when the tension in the piano strings is slightly increased. The frequency of the piano string before increasing the tension was :

(A) 510 Hz **(B)** 514 Hz **(C)** 516 Hz **(D)** 508 Hz

38. Each of the two strings of length $51.6\,cm$ and $49.1\,cm$ are tensioned separately by $20\,N$ force. Mass per unit length of both the strings is same and equal to $1\,g/m$. When both the strings vibrate simultaneously the number of beats is :

(A) 7 **(B)** 8 **(C)** 3 **(D)** 5

39. A wave in a string has an amplitude of $2\,cm$. The wave travels in the $+ve$ direction of x-axis with a speed of $128\,m/\text{sec}$. and it is noted that 5 complete waves fit in $4\,m$ length of the string. The equation describing the wave is :

(A) $y = (0.02)m\sin(15.7\,x - 2010\,t)$ **(B)** $y = (0.02)m\sin(15.7\,x + 2010\,t)$

(C) $y = (0.02)m\sin(7.85\,x - 1005\,t)$ **(D)** $y = (0.02)m\sin(7.85\,x + 1005\,t)$

40. The wave described by $y = 0.25\sin(10\pi x - 2\pi t)$, where x and y are in meters and t in seconds, is a wave traveling along the :

(A) $+ve\ x$ direction with frequency 1 Hz and wavelength $\lambda = 0.2\,m$.

(B) $-ve\ x$ direction with amplitude 0.25 Hz and wavelength $\lambda = 0.2\,m$.

(C) $-ve\ x$ direction with frequency 1 Hz.

(D) $+ve\ x$ direction with frequency p Hz and wavelength $\lambda = 0.2\,m$.

41. Two vibrating tuning forks produce wave given by $y_1 = 4\sin 500\,\pi t$ and $y_2 = 2\sin 506\,\pi t$. Number of beats produced per minute is :

(A) 360 **(B)** 180 **(C)** 60 **(D)** 3

42. A point source emits sound equally in all directions in non-absorbing medium. Two points P and Q are at distances of $2\,m$ and $3\,m$ respectively from the source. The ratio of the intensities of the waves at P and Q is:

(A) $3:2$ **(B)** $2:3$ **(C)** $9:4$ **(D)** $4:9$

43. A car moving towards a high cliff. The driver sounds a horn of frequency f. The reflected sound heards by the driver has frequency $2f$. If v is the velocity of sound, then the velocity of the car, in the same velocity units, will be :

(A) $v/\sqrt{2}$ **(B)** $v/3$ **(C)** $v/4$ **(D)** $v/2$

44. An observer moves towards a stationary source of sound with a speed 1/5th of the speed of sound. The wavelength and frequency of the source emitted are λ and f respectively. The apparent frequency and wavelength recorded by the observer are respectively :

(A) $1.2f,\ 1.2\lambda$ **(B)** $1.2f,\ \lambda$ **(C)** $f,\ 1.2\lambda$ **(D)** $0.8f,\ 0.8\lambda$

45. A wave travelling in positive X-direction with $a = 0.2\,m$, velocity $= 360\,m/\text{sec}$. and $\lambda = 60\,m$, then correct expression for the wave is :

(A) $y = 0.2\sin\left[2\pi\left(6t + \dfrac{x}{60}\right)\right]$ **(B)** $y = 0.2\sin\left[\pi\left(6t + \dfrac{x}{60}\right)\right]$

(C) $y = 0.2\sin\left[2\pi\left(6t - \dfrac{x}{60}\right)\right]$ **(D)** $y = 0.2\sin\left[\pi\left(6t - \dfrac{x}{60}\right)\right]$

46. Two waves having equation $x_1 = a\sin(\omega t - kx + \phi_1)$, $x_2 = a\sin(\omega t - kx + \phi_2)$. If in the resultant wave the frequency and amplitude remain equal to amplitude of superimposting waves, the phase difference between them is :

(A) $\dfrac{\pi}{6}$ **(B)** $\dfrac{2\pi}{3}$ **(C)** $\dfrac{\pi}{4}$ **(D)** $\dfrac{\pi}{3}$

47. A string is cut into three parts, having fundamental frequencies n_1, n_2, n_3 respectively. Then original fundamental frequency n related by the expression as :

(A) $\frac{1}{n}=\frac{1}{n_1}+\frac{1}{n_2}+\frac{1}{n_3}$ **(B)** $n=n_1 \times n_2 \times n_3$

(C) $n=n_1+n_2+n_3$ **(D)** $n=\frac{n_1+n_2+n_3}{3}$

48. A standing was having 3 nodes and 2 antinodes is formed between two atoms having a distance $1.21 Å$ between them. The wavelength of the standing wave is :

(A) $6.05 Å$ **(B)** $2.42 Å$ **(C)** $1.21 Å$ **(D)** $3.63 Å$

49. A cylindrical tube, open at both ends has fundamental frequency f in air. The tube is dipped vertically in water, so that half of it is in water. The fundamental frequency of air column is now :

(A) $f/2$ **(B)** $3f/4$ **(C)** $2f$ **(D)** f

50. A star, which is emitting radiation at a wavelength of $5000 Å$, is approaching the earth with a velocity of $1.5 \times 10^6 m/s$. The change in wavelength of the radiation as received on the earth is :

(A) 25 Å **(B)** 100 Å **(C)** Zero **(D)** 2.5 Å

INTEGER ANSWER TYPE QUESTIONS

The Answers to the following questions are positive integers of 1/2/3 digits or zero.

51. A closed organ pipe of cross sectional area 10 cm^2 resonates with a tuning fork of frequency 1000 Hz in fundamental tone. The minimum value of water to be drained out so that the pipe again resonates with the same turning fork is(take velocity of wave =300 m/s) (in cm^3)

52. An organ pipe of 3.9 πm long, open at both ends is driven to third harmonic standing wave. If the amplitude of pressure oscillation is 1% of mean atmospheric pressure $\left[p_0=10^5 N/m^2\right]$. The maximum displacement of particle from mean position will be: [Given velocity of sound=300 m/s and density of air $=1.3 kg/m^3$] (in mm to nearest integer)

53. A wire under tension vibrates with a fundamental frequency of 600 Hz. If the length of the wire is doubled, the radius is halved and the wire is made to vibrate under one fourth the tension. Then the fundamental frequency will became (in Hz)

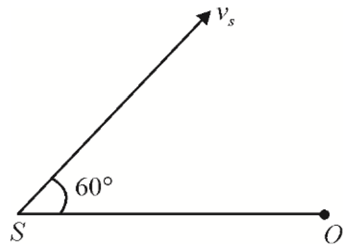
54. A string fixed at both ends oscillates in 5 segments, length 10 m and velocity of wave is 40 m/s. What is the frequency? (in Hz)

55. When two progressive waves $y_1=4 sin(2x-6t)$ and $y_2=3 sin\left(2x-6t-\frac{\pi}{2}\right)$ are superimposed, the amplitude of the resultant wave is:

56. A car sounding its horn at 512 Hz moves towards a high wall at a speed of $20 ms^{-1}$. If the speed of sound is $340 ms^{-1}$, the frequency of the reflected sound heard by the girl sitting in the car will be closest to: (in Hz)

57. A source of sound S emitting waves of frequency 100 Hz and an observer O are located at some distance from each other. The source is moving with a speed of 3.46 m/s at an angle of 60° with the source observer line as shown in the figure. The observer is at rest.

The apparent frequency observed by the observer (velocity of sound in air $330\,ms^{-1}$, is: (in Hz)



58. The fundamental frequency of a closed orgain pipe of length 10 cm is equal to the second overtone of an organ pipe open at both the ends. The length of organ pipe open at both the ends is: (in *cm*)

59. The number of possible natural oscillations of air column in a pipe closed at one end of length 85 cm whose frequencies lie equal or below 1300 Hz are: (Velocity of sound $= 340\,ms^{-1}$)

60. A speeding motorcyclist sees traffic jam ahead him. He slows down to $36\,km\,hour^{-1}$. He find that traffic has eased and a car moving ahead of him at 18 $km\,hour^{-1}$ is honking at a frequency of 1830 Hz. If the speed of sound is 300 m/s, the frequency of the honk as heard by him will be: (in decaHz)

61. A source of unknown frequency gives 4 beats/s when sounded with a source of known frequency 250 Hz. The second harmonic of the source of unknown frequency gives five beats per second, when sounded with a source of frequency 513 Hz. He unknown frequency is: (in Hz)

62. Two waves are represented by the equations $y_1 = a\,sin\left(\omega t + kx + 0.57\right)$ and $y_2 = a\,cos\left(\omega t + kx\right)m,$ where x is in meter and t in sec. The phase difference between them is: (in radian)

63. A tunning fork of frequency 500 Hz makes 4 beats per second with the vibrating string of a piano. The beat frequency decreases to 2 beats per sec when the tension in the piano string is slightly increased. The frequency of the piano string before increasing the tension was: (in Hz)

64. Each of the two strings of length 50 cm and 48.8 cm are tensioned separately by 40 N force. Mass per unit length of both the strings is same and equal to $1\frac{g}{m}$. When both the strings vibrate simultaneously the number of beats is:

65. Two vibrating turning forks produce wave given by $y_1 = 4\,sin\,500\pi t$ and $y_2 = 2\,sin\,506\pi t$. Number of beats produced per minute is: